Dynamic Betti Numbers Are Equivalent to Dynamic Sparse Matrix Rank

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Persistent homology is increasingly computed on data that change over time, yet for general complexes of dimension two or more every known method for updating it after the insertion or deletion of a single simplex takes time polynomial in the number n of simplices in the worst case. We show that this is no accident: for every $p \geq 1$, maintaining the Betti number β_p is equivalent, up to constant factors, to maintaining the rank of a sparse matrix. Precisely, over the two-element field, the following two problems reduce to each other with constant overhead per operation: maintaining the first Betti number of a 2-complex under simplex insertions and deletions, and maintaining the rank of a sparse matrix under entry updates; over every field, the same holds for matrices with entries in $\{0, \pm 1\}$. The key ingredient is a gadget, adapted from a static construction of Edelsbrunner and Parsa, in which switching a row or column off or on costs one simplex update and every entry update costs a constant number. The switches alone imply that, under the Online Matrix–Vector multiplication (OMv) conjecture, no data structure with polynomial preprocessing time can maintain a Betti number β_p, $p \geq 1$, of a subcomplex of a fixed complex with n simplices in amortized time $O(n^{1/2-\varepsilon})$ per update and $O(n^{1-\varepsilon})$ per query, for any $\varepsilon > 0$. This holds over every field, and hence also for persistent Betti numbers under function-value updates and for online zigzag persistence; it survives the restriction to complexes of bounded degree, to 2-complexes linearly embedded in $\mathbb{R}^4$, and to Vietoris–Rips complexes of changing metric spaces and of changing point sets in $\mathbb{R}^{O(\log n)}$, and it applies even to deciding whether the first homology group vanishes. In contrast, all Betti numbers of subcomplexes of any fixed complex linearly embedded in $\mathbb{R}^3$ can be maintained in $O(\log^2 n)$ amortized time per update after polynomial-time preprocessing. Maintaining Betti numbers is thus easy for complexes in $\mathbb{R}^3$, whereas for 2-complexes in $\mathbb{R}^4$ it is as hard as maintaining the rank of a sparse $0/1$ matrix, and over the two-element field equivalent to it.

CCS Concepts: • **Theory of computation** → **Computational geometry**; **Data structures design and analysis**; **Problems, reductions and completeness**; • **Mathematics of computing** → *Algebraic topology*.

Additional Key Words and Phrases: persistent homology, Betti numbers, dynamic data structures, dynamic matrix rank, fine-grained complexity, OMv conjecture, topological data analysis

1 Introduction

Persistent homology summarizes the shape of data by the cycles of a simplicial complex and the scales at which they appear and disappear [24, 25, 62]. In a growing number of applications the complex is not fixed but changes over time. Examples include collective motion and other time-varying point clouds [40, 56], sliding-window embeddings of signals [51], streaming data, where complexes evolve by insertions and edge contractions [39], and gradient-based optimization and learning, where the complex or its filtration changes at every step [8, 35]. Recomputing persistent homology from scratch takes $O(n^\omega)$ time for a complex with n simplices [47], where $\omega < 2.372$ is the exponent of matrix multiplication [2]. So a natural goal is a data structure that absorbs a local change, such as the insertion or deletion of one simplex, in time much smaller than n.

The p-th Betti number β_p of a complex counts its independent p-dimensional holes: β_0 counts connected components, β_1 independent loops, and β_2 enclosed voids. For β_0 the goal of fast updates has been reached. The number β_0 of connected components is maintained by dynamic connectivity in polylogarithmic time per update [36, 38, 60], and Dey, Hou, and Parsa [22] update the persistence of graph filtrations in sublinear time by maintaining minimum spanning forests in dynamic trees. For time series, the banana trees of Cultrera di Montesano et al. [13] support insertions, deletions, cuts, and concatenations in time logarithmic in the length of the series plus the number of changes to the augmented persistence diagram. For general complexes of dimension two or more, the

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update procedures we know of take time polynomial in n per insertion or deletion of a simplex in the worst case. Vines and vineyards [12] process a transposition of two consecutive simplices in $O(n)$ time, but moving one simplex may take up to $n-1$ transpositions; updates of zigzag persistence, which allows deletions as well as insertions [20], warm-started reductions [45], and recomputation are also polynomial in the worst case. Is this a shortcoming of the algorithms, or is maintaining homology inherently hard? We show that the answer is tied to a classical open problem: maintaining Betti numbers is equivalent, up to constant factors, to maintaining the rank of a sparse matrix (an arbitrary one over $\mathbb{F}_2$, and one with entries in $\{0, \pm 1\}$ over other fields). As a consequence, under the Online Matrix–Vector multiplication (OMv) conjecture [33], for no $\varepsilon > 0$ can a data structure with polynomial preprocessing time maintain a Betti number β_p with $p \geq 1$ under insertions and deletions in amortized time $O(n^{1/2-\varepsilon})$ per update while answering queries in time $O(n^{1-\varepsilon})$. In the OMv problem, an algorithm receives a Boolean $\ell \times \ell$ matrix M, then Boolean vectors $v_1, \ldots, v_\ell$ arrive one at a time, and it must output the Boolean product Mv_t before v_{t+1} is revealed; the conjecture states that no algorithm does this in total time $O(\ell^{3-\varepsilon})$ for any $\varepsilon > 0$, that is, polynomially faster than computing every product naively. It is the standard source of tight conditional lower bounds for dynamic problems [33]. Our bounds concern insertions, deletions, and value changes; they say nothing about the cost of a single transposition (Section 2).

The problem. We study the following basic problem, defined formally in Section 3. A fixed simplicial complex $\mathcal{U}$ with n simplices, the *universe*, is given in advance. A data structure maintains a subcomplex $K \subseteq \mathcal{U}$ under the insertion and deletion of single simplices and, on request, reports the p-th Betti number $\beta_p(K) = \dim H_p(K; \mathbb{F})$ over a fixed field $\mathbb{F}$. We call this problem *dynamic Betti numbers*. It is the simplest dynamic question one can ask about homology, because the answer is a single integer. In persistence, a monotone function on the simplices defines a *filtration* by its sublevel sets, the *barcode* records the interval of values during which each homology class lives, and a *persistent Betti number* counts the bars that contain given values. In particular, $\beta_p(K)$ is the persistent Betti number, at value 0, of the function that is 0 on K and 1 elsewhere, so a data structure that maintains a barcode under changes of function values and can count the bars containing a given value also solves dynamic Betti numbers (Lemma 5.9); our lower bounds therefore apply to it.

1.1 Our results

Our main finding is that dynamic Betti numbers are a disguised form of a classical problem in dynamic linear algebra: *dynamic matrix rank*, in which one maintains the rank of a matrix under changes of single entries [30, 54, 57].

Rank-completeness. We call a dynamic problem *rank-complete* if it and dynamic matrix rank reduce to each other with constant overhead per operation and a linear change in instance size (Section 3); over $\mathbb{F}_2$ this refers to the rank of arbitrary matrices, and over other fields to matrices with entries in $\{0, \pm 1\}$. We show that dynamic β_p is rank-complete for every fixed $p \geq 1$; dynamic β_0, which is dynamic connectivity, is not, unless the OMv conjecture fails. One direction is folklore: $\beta_p(K)$ is determined by the ranks of two boundary matrices, and inserting or deleting a simplex changes one column of one of them (Lemma 4.1). Our first contribution is the converse, with constant overhead. We use a slightly richer version of dynamic rank (Problem 3.2). Besides changing an entry, an update may *switch* a whole row or column off, which removes it from the matrix whose rank is queried, or switch it back on. A set of *admissible positions*, fixed in advance, contains every position that may ever hold a nonzero entry, so that the size of an instance can reflect sparsity.

Theorem 1.1 (informal version of Theorem 4.6 and Corollary 4.7). *Let $\mathbb{F}$ be a field.*

(i) *Maintaining the rank of a* $0/1$ *matrix over* $\mathbb{F}$ *with* r *rows,* k *columns, and* m *admissible positions, under entry flips and switches, reduces to maintaining* β_1 *of a 2-complex with* $O(r+k+m)$ *simplices: an entry flip costs seven simplex insertions or deletions, a switch costs one, and a rank query costs one Betti query. For matrices with entries in* $\{0, \pm 1\}$*, an entry update costs at most eight simplex updates.*

(ii) *Conversely, for every fixed* $p \geq 0$*, maintaining* β_p *of a subcomplex of a universe with* n *simplices reduces to maintaining the rank of a matrix with entries in* $\{0, \pm 1\}$ *and* $O(n)$ *admissible positions, with one switch per simplex update.*

Theorem 1.1 shows that over $\mathbb{F}_2$, dynamic β_1 of 2-complexes and dynamic matrix rank are equivalent up to constant factors in time and size; over every field, the same holds for matrices with entries in $\{0, \pm 1\}$, which include all boundary matrices (Section 4.4). Size here is the instance size $N = r + k + m$ of Problem 3.2, not the dimension of the matrix: a dense $s \times s$ matrix becomes a complex with $\Theta(s^2)$ simplices. This measure matters when comparing bounds: the hard instances of our lower bound (Theorem 1.2 below) encode dense $\sqrt{n} \times \sqrt{n}$ matrices, on which the dynamic-rank algorithm of van den Brand, Nanongkai, and Saranurak [57] takes $O(n^{0.7035})$ time per update, whereas on the boundary matrices of an arbitrary complex it takes $O(n^{1.407})$ time (Corollary 5.16); the remaining gap is inherited from dense dynamic rank.

The reduction rests on a local, switchable gadget (Figure 1 and Section 1.2). It follows the pattern of the static reduction of Edelsbrunner and Parsa [26], with rows and columns exchanged: rows become hollow triangles, columns become punctured spheres, and the 2-cycles encode the linear dependencies among the columns, so β_2 equals the nullity of the matrix and β_1 follows from the Euler–Poincaré formula. Edelsbrunner and Parsa only need some triangulation with $O(h)$ simplices of each sphere with h holes, and such a triangulation may change in $\Theta(h)$ simplices when a hole is added or removed; on dense matrices, $h = \Theta(\sqrt{n})$. We add two features. *Switches*, one triangle each, turn rows and columns off and on; they alone already yield the basic OMv lower bound (Theorem 1.2 below). And we assemble every punctured sphere from constant-size pieces, one per admissible position, so that every entry update costs a constant number of simplex updates; this locality yields the equivalence. Switches alone suffice for the basic lower bound and most of its variants, and they could equally be added to the static complex of [26]; constant-cost entry updates are needed for the equivalence itself and for the results that flip entries, namely the worst-case bounds (Theorem 5.13 and Corollary 5.15) and the hardness of deciding whether H_1 vanishes (Theorem 5.30).

Conditional lower bounds. Dynamic rank with switches is hard under the OMv conjecture by an elementary argument: for Boolean vectors u, v, the Boolean product $u^\top M v$ equals 1 exactly when the submatrix $M[\operatorname{supp} u, \operatorname{supp} v]$ formed by the rows in the support $\operatorname{supp} u = \{i : u_i = 1\}$ and the columns in the support $\operatorname{supp} v$ is nonzero, that is, has rank at least one (Lemma 5.1). Because our gadget implements every switch with a single simplex update, this hardness transfers to Betti numbers, deterministically and over every field.

Theorem 1.2 (informal version of Theorem 5.5 and Corollaries 5.7 and 5.10). *Assume the OMv conjecture, and let* $\mathbb{F}$ *be a field,* $p \geq 1$*, and* $\varepsilon > 0$*. No data structure with polynomial preprocessing time maintains* $\beta_p(K; \mathbb{F})$ *of a subcomplex* K *of a given universe of dimension* $\max\{p, 2\}$ *with* n *simplices under simplex insertions and deletions with amortized update time* $O(n^{1/2-\varepsilon})$ *and query time* $O(n^{1-\varepsilon})$*. The same holds for persistent Betti numbers of the sublevel-set filtration of a monotone function on a fixed complex, under updates that change one function value at a time, already for* $\{0, 1\}$*-valued functions.*

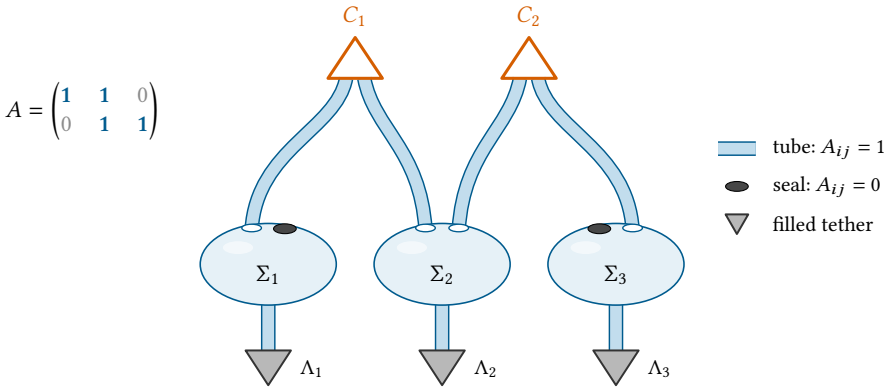


Fig. 1. The gadget $K(A)$ for a 2×3 matrix A, drawn schematically. Row i becomes a hollow triangle, the *row cycle* C_i. Column j becomes a surface Σ_j, drawn as a sphere with one hole per admissible position (here every position is admissible) and one more hole, which is tubed to its *tether* Λ_j. The hole of row i is tubed to C_i if $A_{ij} = 1$ and sealed if $A_{ij} = 0$; the tubes and seals belong to Σ_j. Filling a row cycle switches its row off, and filling a tether, as drawn, switches its column on. Here $\operatorname{rank} A = 2$, and over $\mathbb{F}_2$ the only nonzero 2-cycle consists of the three surfaces and the three filled tethers; it encodes the fact that the three columns of A sum to zero, so $\beta_2(K(A)) = 3 - \operatorname{rank} A = 1$. Figure 4 shows how each surface is assembled from constant-size pieces.

The hard universes form one explicit family; by Theorem 1.3(i) below, the bound does not hold for universes in $\mathbb{R}^3$.

Robustness. The lower bound of Theorem 1.2 survives each of the following restrictions, and it extends to related problems (Section 5).

- *Bounded degree.* Every vertex of the universe lies in a constant number of simplices (Theorem 5.19 and Corollary 5.20).
- *Geometry.* The universe is a fixed 2-complex that is linearly embedded in $\mathbb{R}^4$, that is, its triangles are straight triangles that meet only in common edges and vertices; this holds for β_1 and β_2 (Theorem 5.25).
- *Point clouds and metric spaces.* The complex is a clique complex (the complex of all cliques of a graph) under vertex or edge updates, the Vietoris–Rips complex (all sets of points with pairwise distances at most 1) of a finite metric space whose distances change one at a time, or the Vietoris–Rips complex of a point set in $\mathbb{R}^{O(\log n)}$ under point insertions and deletions; the query asks for β_1 (Theorem 5.29) or, by the same argument, for β_p with any fixed $p \geq 2$, and n is the number of vertices or points.
- *Almost no homology.* The query only asks whether $H_1(K) = 0$, on connected complexes with $\beta_1 = \beta_2 \leq 1$ (Theorem 5.30). So neither algorithms whose running time depends on the size of the answer nor algorithms that approximate β_1 within any factor can beat the bound on all inputs.
- *Zigzag persistence.* A sequence of simplex insertions and deletions is a *zigzag filtration*, and its degree-one barcode records the interval of steps during which each independent loop lives. An online algorithm that reports it for filtrations of length L needs total time $L^{3/2-o(1)}$ (Corollary 5.11).
- *Partially dynamic.* The data structure supports only insertions, or only deletions; here the bound is for worst-case update time (Theorem 5.12).

Relation to dynamic rank. For dynamic rank itself, OMv-hardness over large fields was observed by van den Brand, Nanongkai, and Saranurak [57]. For row and column updates, it follows from [58, Theorem E.5]: OMv-hardness of st-reachability under node updates [33] transfers to bipartite perfect matching under node updates [1], whose size is, with high probability, the rank of the biadjacency matrix with random entries [58, Theorem C.24]. Lemma 5.1 instead gives a direct and deterministic reduction from OuMv (the variant of OMv whose requests are pairs (u, v) of Boolean vectors, answered by the Boolean value $u^\top M v = \bigvee_{i,j} u_i M_{ij} v_j$, not its parity) over every field, using switches or, equivalently, the padding identity (3). On the side of rank, the bounded-degree version of our gadget shows that over every field, dynamic rank under entry updates is OMv-hard already for matrices with entries in $\{0, \pm 1\}$ and at most three admissible positions in every row and column (Corollary 5.21); this also follows without topology from Lemma 5.1 and an elementary dynamic sparsification by partial sums, and static sparsifications of this kind are known [26, 61].

Worst-case bounds and algorithms. Worst-case lower bounds for dynamic rank of dense $\ell \times \ell$ 0/1 matrices under entry flips transfer to Betti numbers through the equivalence, with the exponent halved because the universe has $n = \Theta(\ell^2)$ simplices (Theorem 5.13). Combining this transfer with the hinted variants of OMv proposed by van den Brand et al. [57], in which information about the future query vectors is revealed in advance, we show that over every field, every data structure with polynomial preprocessing time needs worst-case update or query time $n^{\alpha/2-o(1)}$, where α is an exponent determined by rectangular matrix multiplication (Section 5.5), with $\alpha/2 \geq 5/8$ and $\alpha/2 \approx 0.703$ for the current bounds (Corollary 5.15). We keep amortized and worst-case statements separate throughout. In the other direction, the equivalence turns every dynamic-rank algorithm into a dynamic-Betti algorithm. With the randomized dynamic-rank algorithm of van den Brand, Nanongkai, and Saranurak [57] (see also [54]), every Betti number of every complex can be maintained with $O(n^{1.407})$ field operations per update in the worst case, which is faster than recomputation (Corollary 5.16).

Where maintenance becomes easy. Our lower bound is about cycles of dimension at least one. It cannot extend to β_0, because maintaining β_0 is dynamic connectivity. It does not extend to complexes in three-dimensional space either: there, a simple combination of Alexander duality with fully dynamic connectivity maintains all Betti numbers, whereas four dimensions already suffice for hardness.

Theorem 1.3 (informal version of Corollary 6.6 and Theorem 5.25).

(i) For every fixed simplicial complex $\mathcal{U}$ with n simplices that is linearly embedded in $\mathbb{R}^3$, with rational vertex coordinates, all Betti numbers of a subcomplex $K \subseteq \mathcal{U}$ can be maintained in $O(\log^2 n)$ amortized time per simplex insertion or deletion, with constant query time, after preprocessing in time polynomial in the input size.

(ii) Under the OMv conjecture, the lower bound of Theorem 1.2 for β_1 and for β_2 holds even when the universe is a 2-complex that is linearly embedded in $\mathbb{R}^4$.

In part (ii), the reduction from dynamic rank of 0/1 matrices holds verbatim for universes linearly embedded in $\mathbb{R}^4$, so over $\mathbb{F}_2$ maintaining β_1 of such 2-complexes is equivalent to dynamic matrix rank (Corollary 5.26). Part (i) rests on a fully dynamic algorithm for subcomplexes of a triangulated 3-sphere (Theorem 6.4), which combines a combinatorial form of Alexander duality with dynamic connectivity. It is the fully dynamic analogue of the incremental algorithm of Delfinado and Edelsbrunner [16], and it applies directly to every subcomplex of a fixed triangulation of a 3-ball, for example the Delaunay triangulation of a fixed point set in $\mathbb{R}^3$ and all its alpha complexes (Corollary 6.5); point insertions and deletions, which may change $\Theta(n)$ Delaunay simplices, are not covered. Duality also explains where the line falls: for a subcomplex of a triangulated d-sphere,

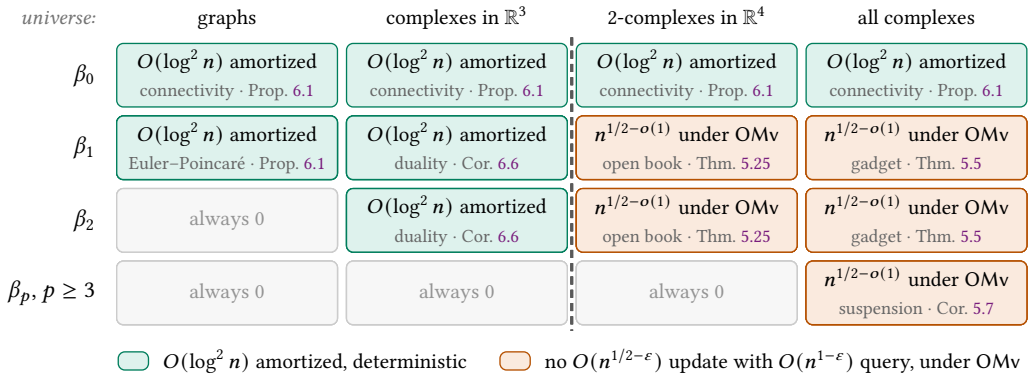


Fig. 2. Maintaining one Betti number of a subcomplex K of a fixed universe with n simplices under single-simplex insertions and deletions; the columns give the class of the universe. Green: deterministic data structures with $O(\log^2 n)$ amortized update time and constant query time, after polynomial preprocessing. Orange: assuming the OMv conjecture, for every $\varepsilon > 0$ and over every field, no data structure with polynomial preprocessing time has amortized update time $O(n^{1/2-\varepsilon})$ and query time $O(n^{1-\varepsilon})$; over $\mathbb{F}_2$, each of these problems is equivalent, up to constant factors, to dynamic matrix rank (Corollary 4.7, and Corollary 5.26 for $\mathbb{R}^4$), and over every field the problems in the last column are equivalent to dynamic rank of $\{0, \pm 1\}$-matrices. The second line of a cell names the method: the *gadget* of Section 4; the *open book*, an explicit linear form of the embedding of Edelsbrunner and Parsa [26] applied to the gadget (Section 5.7); *suspension*, which shifts Betti numbers up by one dimension (Lemma 4.8); and Alexander *duality* (Section 6.2). The dashed line marks the dichotomy between three and four dimensions; Table 1 gives the precise bounds and further settings.

connectivity of the complement determines β_d and β_{d-1}, which, together with β_0 and the Euler–Poincaré formula, determine all Betti numbers when $d = 3$ but leave β_1 and β_2 tied by a single relation when $d = 4$; these are exactly the Betti numbers of part (ii). Theorem 1.3 is the dynamic analogue of the static picture drawn by Edelsbrunner and Parsa [26]: computing the Betti numbers of a complex in $\mathbb{R}^3$ takes near-linear time [16, 18], while for complexes in $\mathbb{R}^4$ it is as hard as computing the rank of a matrix. Finally, sparse structure helps even without geometry. If the *Hasse diagram* of the universe, the graph that joins every simplex to its facets, is given with a tree decomposition of width t, then every Betti number β_p with fixed p can be maintained with $O((t+1)^3 \log n \log(t+2))$ field operations per update in the worst case (Corollary 6.12); a simpler structure handles boundary matrices of small *width*, a parameter akin to bandwidth (Theorem 6.9 and Corollary 6.10). Figure 2 and Table 1 summarize the landscape.

1.2 Technical overview

From homology to rank. For a complex K and a field $\mathbb{F}$, the rank–nullity theorem gives

$$\beta_p(K) = n_p(K) - \operatorname{rank} \partial_p - \operatorname{rank} \partial_{p+1},$$

where $n_p(K)$ is the number of p-simplices and ∂_p is the p-th boundary matrix. Inserting a p-simplex adds one column to ∂_p; it does not change ∂_{p+1}, because no $(p+1)$-simplex containing it is present yet. So a dynamic-rank structure for the boundary matrices of the universe, with absent simplices represented by zero columns, maintains every Betti number.

From rank to homology. Given a matrix $A \in \{0,1\}^{r \times k}$, we build a 2-complex $K(A)$ as follows (Figure 1 and Definition 4.2). Row i becomes a *row cycle* C_i: three edges forming a hollow triangle. Column j becomes a sphere with one hole per row, triangulated from constant-size pieces called

Table 1. The bounds of Figure 2 in detail, together with further settings. Updates are single-simplex insertions and deletions unless a row says otherwise. "Amortized" and "worst-case" refer to update time; the algorithms have query time $O(1)$ and polynomial preprocessing time, and the last two count field operations. Each lower bound holds for every $\varepsilon > 0$, over every field unless stated otherwise, and for every data structure with polynomial preprocessing time and query time $O(n^{1-\varepsilon})$; the zigzag row instead bounds the total time of an online algorithm. Each lower bound assumes the OMv conjecture, except the one marked † in the last row, which assumes the uMv-hinted uMv conjecture of van den Brand, Nanongkai, and Saranurak [57] and bounds the maximum of the update and query times; here $\alpha/2 \geq 5/8$, and $\alpha/2 \approx 0.703$ for the current bounds on rectangular matrix multiplication.

Algorithms		
β_0, any universe	$O(\log^2 n)$ amortized	Prop. 6.1
All β_p, any fixed complex in $\mathbb{R}^3$ (polynomial preprocessing)	$O(\log^2 n)$ amortized	Cor. 6.6
β_p for fixed p, given a tree decomposition of the Hasse diagram of width t	$O((t+1)^3 \log n \log(t+2))$ worst-case	Cor. 6.12
All β_p, any universe (randomized)	$O(n^{1.407})$ worst-case	Cor. 5.16
Conditional lower bounds		
β_p for any $p \geq 1$	$n^{1/2-o(1)}$ amortized	Cor. 5.7
… on universes of bounded degree	$n^{1/2-o(1)}$ amortized	Sec. 5.6
… for $p \leq 2$, on universes linearly embedded in $\mathbb{R}^4$	$n^{1/2-o(1)}$ amortized	Thm. 5.25
… insertions only, or deletions only	$n^{1/2-o(1)}$ worst-case	Thm. 5.12
Deciding $H_1(K) = 0$, for connected K with $\beta_1 = \beta_2 \leq 1$	$n^{1/2-o(1)}$ amortized	Thm. 5.30
Persistent Betti numbers under function-value updates	$n^{1/2-o(1)}$ amortized	Cor. 5.10
β_p of clique and Vietoris–Rips complexes (metric spaces, points in $\mathbb{R}^{O(\log n)}$)	$n^{1/2-o(1)}$ amortized	Sec. 5.8
Online zigzag persistence in degree one, length L	$L^{3/2-o(1)}$ total	Cor. 5.11
β_p for any $p \geq 1$	$n^{\alpha/2-o(1)}$ worst-case†	Cor. 5.15

sockets. The hole of row i is sealed by a single triangle if $A_{ij} = 0$, and joined to C_i by a tube of six triangles if $A_{ij} = 1$. The column therefore becomes a connected surface Σ_j whose boundary consists of the row cycles C_i with $A_{ij} = 1$. The key fact, Lemma 4.4, is that a 2-cycle has the same coefficient x_j on every triangle of Σ_j, because adjacent triangles must cancel along their common edge. A 2-chain with coefficient x_j on every triangle of Σ_j has boundary $\sum_i \big(\sum_j A_{ij}x_j\big)C_i$, so it is a 2-cycle exactly when $Ax = 0$. Hence the 2-cycles correspond exactly to the vectors x with $Ax = 0$, and $\beta_2(K(A)) = k - \operatorname{rank} A$. Flipping A_{ij} exchanges the seal for the tube, which changes seven triangles.

Switches. Think of the columns of A as unknowns and of the rows as equations. A 2-cycle puts a coefficient x_j on Σ_j, and for the boundary to cancel on C_i these coefficients must satisfy the equation $\sum_j A_{ij}x_j = 0$ of row i. If we insert the triangle that fills C_i, a 2-cycle can cancel its boundary on C_i with that triangle instead, and the equation of row i disappears: filling C_i switches row i *off*. Columns work the other way around. Each Σ_j gets one more boundary component, a private hollow triangle Λ_j, the *tether*, that no other surface touches. While Λ_j is hollow, nothing can cancel the boundary of Σ_j there, so $x_j = 0$ in every 2-cycle; filling Λ_j switches column j *on*. Each switch inserts or deletes a single triangle. Now take $A = M \in \{0,1\}^{\ell\times\ell}$ and Boolean vectors $u, v \in \{0,1\}^\ell$ with supports $\operatorname{supp} u = \{i : u_i = 1\}$ and $\operatorname{supp} v = \{j : v_j = 1\}$. If exactly the rows in $\operatorname{supp} u$ and the columns in $\operatorname{supp} v$ are on, then $\beta_2 = |\operatorname{supp} v| - \operatorname{rank} M[\operatorname{supp} u, \operatorname{supp} v]$. This is

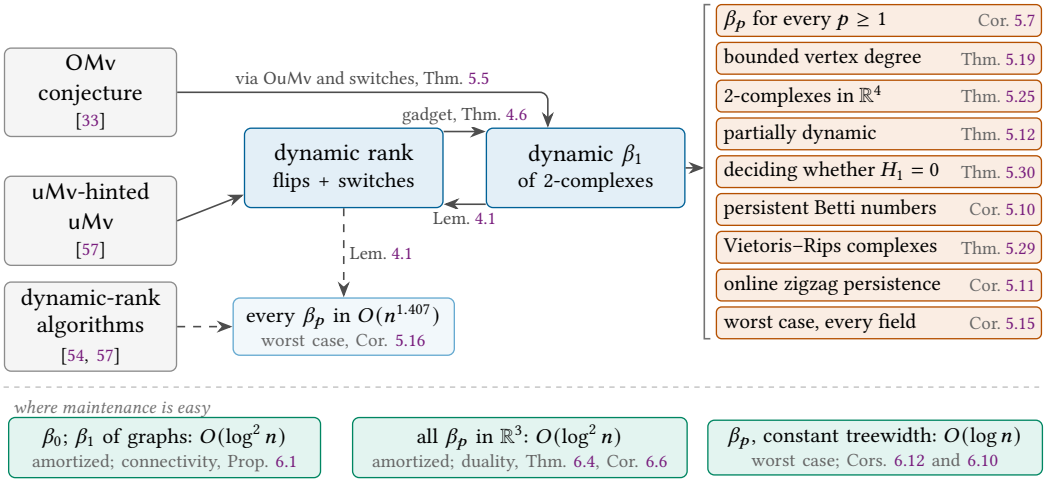


Fig. 3. Roadmap of the results. Solid arrows point in the direction in which hardness propagates; the two arrows between the core problems are the two halves of the equivalence (Theorem 4.6), and the OMv lower bound is proved directly for Betti numbers, using the switches of the gadget. Dashed arrows transfer algorithms from dynamic rank to dynamic Betti numbers. The orange boxes are the variants of the lower bound in Section 5, listed in the order of Table 1; the pale box "every β_p in $O(n^{1.407})$" is the general algorithm obtained from dynamic rank (Corollary 5.16); the green boxes are the algorithms of Section 6, where the treewidth is that of the Hasse diagram and the bandwidth bound is Corollary 6.10. The arrow from the hinted conjecture yields worst-case bounds only.

smaller than $|\operatorname{supp} v|$ exactly when $M[\operatorname{supp} u, \operatorname{supp} v] \neq 0$, that is, exactly when the Boolean product $u^\top M v = \bigvee_{i,j} u_i M_{ij} v_j$ equals 1; a 0/1 matrix is nonzero over every field if it is nonzero at all, so this works over every field. Answering ℓ such requests online is the OuMv problem, which needs total time $\ell^{3-o(1)}$ under the OMv conjecture, even after polynomial preprocessing of M [33]. Our universe has $n = \Theta(\ell^2)$ simplices, and moving from one request to the next takes $O(\ell) = O(\sqrt{n})$ switches and one query. A data structure with update time $n^{1/2-\varepsilon}$ and query time $n^{1-\varepsilon}$ would thus answer all ℓ requests in time $\ell \cdot (\ell \cdot \ell^{1-2\varepsilon} + \ell^{2-2\varepsilon}) = O(\ell^{3-2\varepsilon})$, contradicting the conjecture; this is where the exponents $1/2$ and 1 in Theorem 1.2 come from.

Almost no homology. The complexes above contain the whole 1-skeleton of the universe, so their first Betti number is typically of order n, and one might hope for algorithms whose cost depends on the homology rather than on the size of the complex. To rule this out (Theorem 5.30), we cone off the support graph of the matrix, which kills its cycles without changing the rank information. We then encode an OuMv request in a bordered matrix whose rank deficiency is a single bit, so that $\beta_1 = \beta_2 \in \{0, 1\}$, and a standard random isolation step turns this bit into the Boolean answer.

Easy cases. For a proper subcomplex $K \subsetneq T$ of a triangulated 3-sphere T, Alexander duality gives $\beta_2(K) = c - 1$, where c is the number of connected components of the complement of K in the sphere. These components correspond to the components of a dual graph whose nodes are the tetrahedra of T not in K and whose edges are the triangles of T not in K. A simplex update changes this graph by $O(1)$ nodes or edges. So β_0 and β_2 both reduce to dynamic connectivity, and the Euler–Poincaré formula gives β_1.

1.3 Organization

Figure 3 shows how the results depend on each other. Section 2 discusses related work. Section 3 fixes notation and defines the dynamic problems and the reductions we use. Section 4 presents the gadget and proves the equivalence. Section 5 derives the conditional lower bounds and their variants, including the embedding of the hard universes in $\mathbb{R}^4$. Section 6 gives the algorithms for β_0, for subcomplexes of the 3-sphere and of any fixed geometric complex in $\mathbb{R}^3$, and for universes of small width or bounded treewidth. Section 7 lists open problems.

2 Related work

Updating persistence. Cohen-Steiner, Edelsbrunner, and Morozov [12] introduced vines and vineyards. They maintain a reduced boundary matrix while two adjacent simplices of the filtration are transposed, in time linear in the number of simplices per transposition. A continuously changing function is processed as a sequence of such transpositions. Dey and Hou [20] carried this approach over to zigzag filtrations. They identified eight atomic operations, six of which can be executed as efficiently as their non-zigzag counterparts. Luo and Nelson [45] additionally handle insertions and deletions of cells and batches of changes. The cost of their scheme scales with the number of elementary changes. Piekenbrock and Perea [52] replace the long sequences of transpositions that arise when a simplex moves to a new position in the filtration by coarser moves. Under insertions and elementary collapses, persistence is maintained with annotations [5, 17] and by the streaming algorithm for towers of Kerber and Schreiber [39], and Maria and Oudot [46] compute zigzag persistence online, one insertion or deletion at a time. In the worst case, every one of these methods spends time polynomial in the size of the complex per elementary change.

Sublinear updates are known in two settings. For graphs, Dey, Hou, and Parsa [22] perform a transposition in ordinary graph persistence in $O(\log L)$ time for a filtration of length L, and they update several kinds of bars of graph zigzag persistence in sublinear time. For time series, the banana trees of Cultrera di Montesano, Edelsbrunner, Henzinger, and Ost [13] support insertions, deletions, cuts, and concatenations in time logarithmic in the length of the series plus the number of changes of the augmented persistence diagram. In both settings the homology that changes is that of a graph, and graph homology reduces to connectivity. Our results indicate that such bounds do not extend to insertions and deletions in general 2-complexes unless dynamic rank of sparse matrices admits them too: under the OMv conjecture, amortized update time $n^{1/2-\varepsilon}$ with query time $n^{1-\varepsilon}$ is impossible there, whereas complexes in $\mathbb{R}^3$ remain easy (Section 6.2). A transposition is a different operation from the insertions, deletions, and value changes we consider. Our lower bounds do not apply to it, and the complexity of vineyards remains open (Section 7).

Static complexity of Betti numbers and persistence. The standard persistence algorithm [25, 62] reduces the boundary matrix by Gaussian elimination and takes cubic time in the worst case. Milosavljević, Morozov, and Škraba [47] showed that persistence, and even zigzag persistence, can be computed in matrix multiplication time. Chen and Kerber [10] gave an output-sensitive algorithm. Over finite fields, Wiedemann's method [37, 59] computes the rank of an $n \times n$ matrix with $O(n)$ nonzero entries, and hence every Betti number of a complex of fixed dimension, in time $O(n^{2+o(1)})$, as Edelsbrunner and Parsa note [26]; Casacuberta and Kyng [9] also compute a basis of the null space of such a matrix in expected time $O(n^{2.2131})$. In the other direction, Edelsbrunner and Parsa [26] proved that, over $\mathbb{F}_2$ and over prime fields, computing the Betti numbers of a 2-complex with N simplices has the same complexity as computing the rank of an $N \times N$ 0/1 matrix with N nonzero entries. In their construction every column of the matrix becomes a cycle, and every row becomes a sphere with holes, triangulated with $O(h)$ simplices if it has h holes, and glued to the cycles of its nonzero entries. They also sparsify the matrix to at most three nonzero entries

per row and column [26, Theorem 4.1], a special case of the sparsification of Yuster [61], and they embed their complex in $\mathbb{R}^4$ by placing its cycles in a plane and every punctured sphere in its own 3-dimensional half-space from a pencil through that plane, even as a subcomplex of a triangulation of the whole space computed in linear time. Our gadget follows the same pattern, with the roles of rows and columns exchanged. What is new is the switches, the decomposition of each punctured sphere into constant-size sockets, which makes every entry update local, and the orientation analysis, which gives the static reduction over every field as well; our embedding of the hard universes in Section 5.7 uses their pencil construction. This contrasts with complexes in $\mathbb{R}^3$, whose Betti numbers Delfinado and Edelsbrunner [16] compute in near-linear time; Dey and Guha [18] compute homology groups and their generators for arbitrary complexes in $\mathbb{R}^3$. The contrast between the 3-sphere and $\mathbb{R}^4$ in Sections 6.2 and 5.7 is the dynamic counterpart of this static picture. Recent work extends static hardness in other directions: to sufficiently fine additive and to constant-factor multiplicative approximations of Vietoris–Rips and Čech barcodes [41], and to the average case, via planted cliques, for estimating Betti numbers and deciding whether homology vanishes [55]. Closest to our equivalence on the static side, Ding, Kyng, Probst Gutenberg, and Zhang [23], building on Kyng and Zhang [43], showed that solving linear equations in boundary operators and combinatorial Laplacians of 2-complexes is as hard as solving general sparse linear equations, by a reduction that preserves sparsity up to polylogarithmic factors. Our gadget plays the analogous role for rank and Betti numbers; what is new is that the encoding preserves the instance size up to constant factors and is local, so that every change of an entry and every switch of a row or column costs a constant number of simplex updates. Embeddings in $\mathbb{R}^3$ also help for Laplacian systems, which can be solved in nearly linear time, parameterized by the first Betti number, for subcomplexes of collapsible complexes embedded in $\mathbb{R}^3$ [3], whereas for general 2-complexes they are as hard as general sparse linear equations [23]. To our knowledge, our results are the first conditional lower bounds for maintaining homology in dimension at least one under updates.

Conditional lower bounds for dynamic problems. Henzinger, Krinninger, Nanongkai, and Saranurak [33] introduced the OMv conjecture. They derived from it tight lower bounds for many dynamic problems: amortized bounds for fully dynamic algorithms and worst-case bounds for partially dynamic ones. Abboud and Vassilevska Williams [1] obtained lower bounds for dynamic problems from other popular conjectures. OMv-hardness of dynamic matrix rank over large fields was known. As van den Brand, Nanongkai, and Saranurak observe [58, Theorem E.5], for row and column updates it follows from the OMv lower bound for st-reachability [33] via bipartite matching [1]. Over every field, including $\mathbb{F}_2$, a direct and deterministic reduction from OuMv to dynamic rank follows from the observation that $u^\top M v = 1$ exactly when $M[\operatorname{supp} u, \operatorname{supp} v] \neq 0$, together with switches or the padding identity (3) (Lemma 5.1). Composing it with our equivalence yields the lower bounds for Betti numbers, with a constant number of simplex updates per coordinate of a query vector. Van den Brand et al. also proposed hinted variants of OMv. These imply worst-case lower bounds for dynamic matrix inverse, determinant, and rank under entry updates that match their algorithms [58, Corollary 5.15], and we discuss what they imply for Betti numbers in Section 5.5.

Dynamic linear algebra. Sankowski maintained the inverse and the determinant of a matrix under entry updates in subquadratic time [53]. Sankowski also reduced dynamic rank to dynamic determinant with randomization [54]. Frandsen and Frandsen [30] studied dynamic matrix rank directly. Van den Brand, Nanongkai, and Saranurak [57] improved the update time for entry updates to $O(s^{1.407})$ in the worst case, for $s \times s$ matrices; for rank the algorithm is randomized. All these bounds are stated for dense matrices of a given dimension. Our equivalence identifies dynamic Betti numbers with dynamic rank of sparse matrices, measured by the instance size $N = r + k + m$

of Problem 3.2. So an improvement for sparse matrices in this measure would transfer to Betti numbers, and conversely.

Connectivity, low dimension, and bounded treewidth. The zeroth Betti number is the number of connected components. It can be maintained in $O(\log^2 n)$ amortized time by the deterministic algorithm of Holm, de Lichtenberg, and Thorup [36], improved by Wulff-Nilsen [60]. Kapron, King, and Mountjoy [38] achieve polylogarithmic worst-case time with a Monte Carlo algorithm, and Chuzhoy et al. [11] achieve deterministic worst-case time $n^{o(1)}$. The amortized bounds are optimal up to a logarithmic factor: Pătraşcu and Demaine [50] proved an $\Omega(\log n)$ cell-probe lower bound per operation for dynamic connectivity, and it transfers to β_0, and hence to our algorithms for complexes in $\mathbb{R}^3$. Eppstein [27] maintains a graph embedded on a surface, together with generators of the fundamental group of the surface, in polylogarithmic amortized time per update; this is the closest precedent for our combination of duality and dynamic connectivity. Delfinado and Edelsbrunner [16] process a filtration of a triangulated 3-sphere with two union–find structures, one on the complex and one, run backwards, on the dual graph of its complement. Our algorithm for the 3-sphere replaces both by fully dynamic connectivity, which is what allows arbitrary insertions and deletions. Given a tree decomposition of width t of the bipartite graph of the nonzero entries, Fomin, Lokshtanov, Saurabh, Pilipczuk, and Wrochna [28] compute the rank of an $s \times s$ matrix with $O(t^3 s)$ arithmetic operations. Our dynamic structures store small summaries of groups of columns in a balanced hierarchy (Theorem 6.7); with a balanced tree decomposition [4], they show that bounded treewidth of the Hasse diagram suffices for polylogarithmic worst-case updates (Corollary 6.12).

Output-sensitive and Morse-theoretic methods. Several methods aim for a cost governed by the homology rather than by the size of the complex. Annotations [5, 17] maintain a homology basis under elementary inclusions and collapses. Discrete Morse theory [29] and its use for persistence [48] replace a complex by a smaller chain complex on its *critical cells*, the cells left unmatched by an acyclic matching between cells and their facets; their number is at least the total Betti number. The hard instances of Theorem 5.30 are connected and have $\beta_1 = \beta_2 \le 1$, so no method whose cost is governed by the homology of the complex, rather than its size, can escape our lower bounds. This argument does not cover methods whose cost depends on the number of critical cells of a given matching, which can be much larger than the total Betti number.

3 Preliminaries

Complexes and homology. A (finite, abstract) *simplicial complex* K is a finite family of nonempty finite sets, called *simplices*, that is closed under taking nonempty subsets. A simplex with $p+1$ elements is a *p-simplex*; for $p \ge 1$ its *facets* are its p-element subsets, and it is a *cofacet* of each of them; a vertex has no facets. We call 0-, 1-, 2-, and 3-simplices *vertices*, *edges*, *triangles*, and *tetrahedra*, and the *1-skeleton* $K^{(1)}$ of K consists of its vertices and edges. We write K_p for the set of p-simplices, $n_p(K) = |K_p|$, and $\dim K$ for the largest p with $K_p \neq \emptyset$; a *d-complex* is a simplicial complex of dimension at most d. A *subcomplex* of K is a subset of K that is itself a simplicial complex. We use standard facts from homology theory [24, 32, 49], which we recall briefly. Fix a field $\mathbb{F}$. The p-chains $C_p(K)$ form the $\mathbb{F}$-vector space with basis K_p, where each simplex carries a fixed orientation, that is, an ordering of its vertices up to even permutations. The boundary map $\partial_p \colon C_p(K) \to C_{p-1}(K)$ is the linear map with $\partial_p[v_0, \dots, v_p] = \sum_{i=0}^{p} (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_p]$, where the hat marks an omitted vertex. A simplex whose vertices are listed in another order denotes $\pm$ the basis element: $[v_{\pi(0)}, \dots, v_{\pi(p)}] = \operatorname{sgn}(\pi)[v_0, \dots, v_p]$, so for instance $[x, y] = -[y, x]$. An oriented triangle $[x, y, z]$ induces on its edges the orientations $[x, y]$, $[y, z]$, $[z, x]$ that occur in its boundary, and two triangles with a common edge induce *opposite* orientations on it if these differ by the sign -1.

Over $\mathbb{F}_2$, orientations and signs play no role. Since $\partial_p \circ \partial_{p+1} = 0$, the *$p$-boundaries*, the elements of $\operatorname{im} \partial_{p+1}$, lie in the space $Z_p(K) = \ker \partial_p$ of *p-cycles*. The *p-th homology group* is $H_p(K;\mathbb{F}) = Z_p(K)/\operatorname{im} \partial_{p+1}$, and the *$p$-th Betti number* is $\beta_p(K) = \dim H_p(K;\mathbb{F}) = \dim Z_p(K) - \operatorname{rank} \partial_{p+1}$. Writing ∂_p also for its matrix with respect to the simplex bases, the rank–nullity theorem gives

$$\beta_p(K) \;=\; n_p(K) - \operatorname{rank} \partial_p - \operatorname{rank} \partial_{p+1} \qquad (\text{with } \operatorname{rank} \partial_0 = 0). \tag{1}$$

The *Euler characteristic* is $\chi(K) = \sum_p (-1)^p n_p(K)$, and the Euler–Poincaré formula states that $\chi(K) = \sum_p (-1)^p \beta_p(K)$ over every field. Over $\mathbb{F}_2$, a 2-cycle is simply a set of triangles in which every edge lies in an even number of triangles; the gadget of Section 4 rests on this picture. If $\dim K \le 2$, then K has no 3-simplices, so $\partial_3 = 0$ and $H_2(K) = Z_2(K)$; hence

$$\beta_2(K) = \dim Z_2(K) \qquad \text{and} \qquad \beta_1(K) = \beta_0(K) + \beta_2(K) - \chi(K). \tag{2}$$

For a matrix A with rows indexed by $[r] = \{1, \dots, r\}$ and columns by $[k]$, and for $R \subseteq [r]$ and $J \subseteq [k]$, we write $A[R, J]$ for the submatrix on rows R and columns J, and $\operatorname{supp}(A)$ for the set of positions of its nonzero entries. We write $X \triangle Y$ for the symmetric difference of two sets, and $\operatorname{supp} u = \{i : u_i \neq 0\}$ for the support of a vector u.

Dynamic problems. All our problems have the same shape. An instance is fixed in advance and may be preprocessed. Then an online sequence of *updates* and *queries* arrives, and each query must be answered before the next operation is revealed. A data structure for instances of size n has preprocessing time t_{pre}, *amortized* update time u, and query time q if its preprocessing takes time at most t_{pre} and, afterwards, every sequence of t updates interleaved with s queries takes total time $O((n+t)\cdot u + s \cdot q)$; it has *worst-case* update time u and query time q if every single update takes time $O(u)$ and every single query takes time $O(q)$. Worst-case bounds imply amortized bounds, and the term $n \cdot u$ prevents a data structure from postponing its preprocessing into the updates. Running times refer to the word RAM with words of $\Theta(\log n)$ bits, the model in which the OMv conjecture is stated, so our lower bounds over every field, including $\mathbb{Q}$ and $\mathbb{R}$, apply to word-RAM data structures. Algorithms that count field operations at unit cost (Corollary 5.16 and Section 6.3) are stated in the algebraic model; over a prime field of polynomial size, a field operation takes $O(1)$ word operations.

Problem 3.1 (Dynamic Betti numbers). Fix a field $\mathbb{F}$ and an integer $p \ge 0$. The instance is a simplicial complex $\mathcal{U}$, the *universe*, with $n = |\mathcal{U}|$ simplices. The data structure maintains a subcomplex $K \subseteq \mathcal{U}$, initially empty, under the operations

- Insert(σ), for $\sigma \in \mathcal{U} \setminus K$ all of whose facets lie in K, which replaces K by $K \cup \{\sigma\}$;
- Delete(σ), for $\sigma \in K$ none of whose cofacets lies in K, which replaces K by $K \setminus \{\sigma\}$;
- Query, which returns $\beta_p(K;\mathbb{F})$.

The preconditions guarantee that K is a subcomplex after every operation. We write *dynamic β_p over $\mathbb{F}$* for Problem 3.1 with parameters p and $\mathbb{F}$, and *dynamic β_p of d-complexes* for its restriction to universes of dimension at most d; the size of an instance is $n = |\mathcal{U}|$. We call Insert and Delete *simplex updates*. Knowing the universe in advance can only help an algorithm, so our lower bounds also hold when the universe is revealed online. Our algorithms also assume a fixed universe, for example a fixed triangulation of the 3-sphere (Section 6).

Our version of dynamic rank has two features that the standard version lacks. A set P fixes in advance the positions that may ever hold a nonzero entry, so that the size of an instance can reflect sparsity. And a *switch* includes or excludes a whole row or column at unit cost.

Problem 3.2 (Dynamic rank). Fix a field $\mathbb{F}$. The instance consists of integers r, k and a set $P \subseteq [r] \times [k]$ of *admissible positions*, with $m = |P|$. The data structure maintains a matrix $A \in \mathbb{F}^{r \times k}$

with $\operatorname{supp}(A) \subseteq P$, a set $R \subseteq [r]$ of *active rows*, and a set $J \subseteq [k]$ of *active columns*. Initially $A = 0$, $R = [r]$, and $J = [k]$. The operations are:

- Set(i, j, x), for $(i, j) \in P$ and $x \in \mathbb{F}$, which sets $A_{ij} = x$;
- Toggle(row i), which adds i to R if $i \notin R$ and removes it otherwise, and Toggle(column j), which does the same for j and J;
- Query, which returns $\operatorname{rank} A[R, J]$.

We call Set an *entry update* and Toggle a *switch*. The size of an instance is $N = r + k + m$. Dynamic rank of an $s \times s$ matrix under entry updates [30, 54, 57] is the special case $r = k = s$ and $P = [s] \times [s]$ without switches; its size is $N = 2s + s^2$. For a set $S \subseteq \mathbb{F}$ with $0 \in S$, *dynamic rank of S-matrices* is the special case in which every update Set(i, j, x) has $x \in S$; we use $S = \{0, 1\}$ and $S = \{0, \pm 1\}$. For a 0/1 matrix, an *entry flip* is the update Set$(i, j, 1 - A_{ij})$. Switches do not make the problem harder: if X_R is the diagonal $r \times r$ matrix with $(X_R)_{ii} = 1$ for $i \notin R$ and 0 otherwise, and Y_J is defined in the same way, then

$$\operatorname{rank} \begin{pmatrix} A & X_R \\ Y_J & 0 \end{pmatrix} = \operatorname{rank} A[R, J] + (r - |R|) + (k - |J|), \tag{3}$$

so a switch becomes a single entry update of a padded matrix of linear size; we call (3) the *padding identity*. We keep switches as primitive operations because our gadget implements each of them with a single simplex update.

Reductions. We say that a dynamic problem Π *reduces to* a dynamic problem Π' *with overhead* (a, b) if there is a polynomial-time map from instances of Π of size N to instances of Π' of size $O(N)$ with the following properties. During preprocessing, the reduction may perform $O(N)$ updates of Π' to create its initial state. Afterwards, every update of Π is carried out by at most a updates of Π' plus $O(a)$ extra time, and every query of Π is answered by at most b queries of Π' plus $O(b)$ extra time. If a and b are constants, a data structure for Π' with update time $u(N)$ and query time $q(N)$ yields one for Π with update time $O(u(O(N)))$ and query time $O(q(O(N)))$, both in the amortized and in the worst-case sense; the preprocessing time grows by $O(N \cdot u(O(N)))$ plus a polynomial. Reductions compose: overheads (a, b) and (a', b') give overhead (aa', bb'), sizes stay linear, and preprocessing stays a linear number of updates. An update names a simplex or an admissible position by its index in an enumeration fixed during preprocessing. All reductions in this paper are deterministic, except that the proof of Theorem 5.30 uses random probes, the proof of Corollary 5.15 draws random masks, and the proof of Theorem 5.29 draws a random embedding during preprocessing; the OMv conjecture, and the uMv-hinted uMv conjecture as we read it (Section 5.5), cover randomized algorithms, so the lower bounds are unaffected.

The OMv conjecture. In the *Online Matrix–Vector multiplication* problem (OMv), over the Boolean semiring, an algorithm first receives a Boolean matrix $M \in \{0, 1\}^{\ell \times \ell}$. Then vectors $v_1, \ldots, v_\ell \in \{0, 1\}^\ell$ arrive one at a time, and the algorithm must output the Boolean product Mv_t before it sees v_{t+1}. The OMv *conjecture* states that for every $\varepsilon > 0$, no algorithm solves OMv in total time $O(\ell^{3-\varepsilon})$, including all time spent before the first vector arrives, with error probability at most $1/3$ [34, Conjecture 1.1]. The fastest known algorithm takes time $\ell^3 / 2^{\Omega(\sqrt{\log \ell})}$ [44], which does not refute the conjecture. In the related OuMv problem, each request is a pair (u_t, v_t) of Boolean vectors, and the answer is the Boolean value $u_t^\top M v_t = \bigvee_{i,j} u_{t,i} M_{ij} v_{t,j}$. Henzinger et al. [34, Theorem 2.7] showed that the OMv conjecture implies that no algorithm that preprocesses M in polynomial time solves OuMv with ℓ requests in computation time $O(\ell^{3-\varepsilon})$, the time spent after preprocessing, with error probability at most $1/3$. All our lower bounds use this form. Note that $u^\top M v = 1$ if and only if the submatrix $M[\operatorname{supp} u, \operatorname{supp} v]$ is nonzero; this observation drives our reduction.

4 Dynamic Betti numbers are equivalent to dynamic rank

This section proves that dynamic Betti numbers and dynamic rank of matrices with entries in $\{0, \pm 1\}$ reduce to each other with constant overhead and a linear change in size (Theorem 4.6 and Corollary 4.7); over $\mathbb{F}_2$ and over $\mathbb{F}_3$, every matrix has entries in $\{0, \pm 1\}$. Section 4.1 gives the easy direction, from Betti numbers to rank. The other direction rests on the gadget of Section 4.2. Every row of the matrix becomes a hollow triangle, every column becomes a punctured sphere assembled from constant-size pieces, and each entry decides whether a hole of the sphere of its column is closed off or joined by a tube to the triangle of its row (Figure 1). Section 4.3 proves that the 2-cycles of the resulting complex correspond exactly to the linear dependencies among the active columns, restricted to the active rows, and Section 4.4 assembles the reductions.

4.1 From Betti numbers to rank

Let $\mathcal{U}$ be a universe and $p \geq 0$. For $q \geq 1$ let B_q be the boundary matrix of $\mathcal{U}$ in dimension q, with rows indexed by $\mathcal{U}_{q-1}$ and columns by $\mathcal{U}_q$. For a subcomplex $K \subseteq \mathcal{U}$ let B_q^K be the matrix obtained from B_q by replacing the columns of the q-simplices not in K with zero columns.

Lemma 4.1. *For every subcomplex $K \subseteq \mathcal{U}$ and every $p \geq 0$,*

$$\beta_p(K) = n_p(K) - \operatorname{rank} \begin{pmatrix} B_p^K & 0 \\ 0 & B_{p+1}^K \end{pmatrix},$$

where B_0^K is the empty matrix. Inserting or deleting a q-simplex with $q \geq 1$ changes one column of B_q^K, and no other column of B_p^K or B_{p+1}^K; inserting or deleting a vertex changes only $n_0(K)$.

Proof. The columns of B_q^K indexed by K_q form the matrix of ∂_q on K, extended by rows for the $(q-1)$-simplices of $\mathcal{U}$ that are not in K. These extra rows are zero in those columns, because every facet of a simplex of K lies in K. Adding zero rows and zero columns does not change the rank, so $\operatorname{rank} B_q^K = \operatorname{rank} \partial_q$, and the formula follows from (1). The column of a q-simplex σ is zero in B_q^K if $\sigma \notin K$ and equals its column in B_q otherwise, so updating σ changes exactly that column. □

A column change of B_q^K is a single switch, or at most $q+1$ entry updates. We call *dynamic rank under switches* the variant of Problem 3.2 whose instance also contains a matrix $A_0 \in \mathbb{F}^{r \times k}$, with $P = \operatorname{supp}(A_0)$, and in which $A = A_0$ throughout, so that the only updates are switches; its size is again $N = r + k + m$. So for fixed p, dynamic β_p reduces to dynamic rank under switches with overhead $(1, 1)$ and to dynamic rank with overhead $(p+2, 1)$: one data structure for the block matrix of Lemma 4.1 and a counter for $n_p(K)$ suffice. Combined with the fastest known dynamic-rank algorithm, this gives a randomized data structure with worst-case update time $O(n^{1.407})$ for every Betti number (Corollary 5.16 in Section 5.5).

4.2 The gadget

We now go the other way. Fix a 0/1 matrix A with r rows and k columns. We build a 2-complex whose 2-cycles encode the kernel of A. The construction uses four building blocks, each attached to *triangle cycles*. A triangle cycle $a = (a_0, a_1, a_2)$ consists of three vertices and the three edges a_0a_1, a_1a_2, a_2a_0; we call the triangle $a_0a_1a_2$ its *filling*. Indices of vertices of a triangle cycle are taken modulo 3, and we write uv for the edge $\{u, v\}$ and uvw for the triangle $\{u, v, w\}$.

- The *cone* on a with apex o is $\{o\, a_t\, a_{t+1} : t \in \mathbb{Z}_3\}$, a disk with boundary a (three triangles).
- The *cylinder* between disjoint triangle cycles a and b is $\operatorname{Cyl}(a, b) = \{a_t\, a_{t+1}\, b_t,\ a_{t+1}\, b_t\, b_{t+1} : t \in \mathbb{Z}_3\}$, an annulus with boundary $a \sqcup b$ (six triangles).

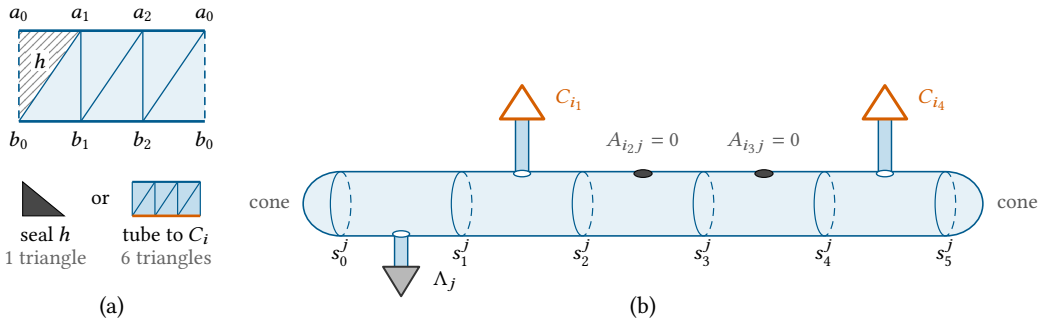


Fig. 4. The pieces of the gadget (Definition 4.2). (a) A socket: the punctured cylinder, that is, the cylinder $\mathrm{Cyl}(a, b)$ between the triangle cycles a and b, cut open along the edge a_0b_0 (dashed; the two copies are identified), with the triangle $h = a_0a_1b_0$ removed (hatched), and its two possible attachments. The hole h is either sealed by the single triangle h or tubed to a row cycle C_i by the six triangles of $\mathrm{Cyl}(h, C_i)$, so flipping one entry of A costs $1 + 6 = 7$ triangle updates (Lemma 4.5). (b) The surface Σ_j formed by the triangles of column j: from left to right, a cone, the socket tubed to the tether Λ_j, one socket per admissible position of column j (here four, in rows $i_1 < i_2 < i_3 < i_4$, with $A_{i_1 j} = A_{i_4 j} = 1$ and $A_{i_2 j} = A_{i_3 j} = 0$), and a cone, glued in a chain along the triangle cycles $s_0^j, \ldots, s_5^j$.

- The *punctured cylinder* $\mathrm{Cyl}^\circ(a, b) = \mathrm{Cyl}(a, b) \setminus \{h\}$ is the cylinder with the triangle $h = a_0a_1b_0$ removed (five triangles). We call the triangle cycle (a_0, a_1, b_0) its *hole*.
- The *tube* from the hole h to a triangle cycle c is $\mathrm{Cyl}(h, c)$ (six triangles).

A *socket* between a and b consists of the punctured cylinder $\mathrm{Cyl}^\circ(a, b)$ and one or two *attachments* at its hole: the *seal*, which is the triangle h, and the *tube* $\mathrm{Cyl}(h, c)$ to a triangle cycle c disjoint from a and b. In every complex $K(A; R, J)$ below, exactly one attachment of each socket is present. With the seal, the socket is the annulus $\mathrm{Cyl}(a, b)$, and we call it *sealed*; with the tube, it is a *pair of pants*, a sphere with three holes, with boundary $a \sqcup b \sqcup c$ (see the proof of Lemma 4.3), and we call it *tubed*. Figure 1 shows the whole construction schematically, and Figure 4 shows a socket and one column surface.

Definition 4.2 (The gadget). Let $P \subseteq [r] \times [k]$ be a set of admissible positions, and for $j \in [k]$ let $S_j = \{i : (i, j) \in P\}$, listed in increasing order as $i_1 < \cdots < i_{d_j}$. The complex $\mathcal{U}(P)$ consists of the following simplices, all on distinct vertices unless stated otherwise.

- For each row $i \in [r]$, a triangle cycle C_i, the *row cycle*, and its filling D_i.
- For each column $j \in [k]$, a triangle cycle Λ_j, the *tether*, and its filling E_j.
- For each column j, triangle cycles $s_0^j, s_1^j, \ldots, s_{d_j+1}^j$, cones on s_0^j and on $s_{d_j+1}^j$, a socket between s_0^j and s_1^j whose only attachment is the tube to Λ_j, and, for every $t \in [d_j]$, a socket between s_t^j and s_{t+1}^j with both attachments, the seal and the tube to C_{i_t}, which we call the *socket of position* (i_t, j).
- All vertices and edges of the simplices above.

For a 0/1 matrix A with $\mathrm{supp}(A) \subseteq P$ and sets $R \subseteq [r]$, $J \subseteq [k]$, the complex $K(A; R, J) \subseteq \mathcal{U}(P)$ consists of all vertices and edges of $\mathcal{U}(P)$, the cones, the punctured cylinders, the tubes to the tethers, and

- for every $(i, j) \in P$, the seal of the socket of position (i, j) if $A_{ij} = 0$ and its tube to C_i if $A_{ij} = 1$;
- the filling D_i for every row $i \notin R$, and the filling E_j for every column $j \in J$.

So row i is *active* when its row cycle is hollow, that is, when its filling D_i is absent, and column j is *active* when its tether is filled. We write $K(A) = K(A; [r], [k])$ for the complex in which all rows and all columns are active. Row cycles and punctured spheres already appear in the static reduction [26, Theorem 3.1]; the constant-size sockets and the switches make every matrix operation local (Lemma 4.5). We write Σ_j for the set of triangles of $K(A; R, J)$ that belong to column j: its two cones, its punctured cylinders, its seals and tubes. Only admissible positions get sockets; a socket at a position that is always zero would always be sealed and would contribute nothing. The size of $\mathcal{U}(P)$ is $O(r + k + |P|)$, and $\mathcal{U}(P)$ can be constructed in linear time.

Lemma 4.3. *Let A be a $0/1$ matrix with $\mathrm{supp}(A) \subseteq P$, let $R \subseteq [r]$ and $J \subseteq [k]$, and take Σ_j in $K(A; R, J)$. For every $j \in [k]$, the triangles of Σ_j form a connected triangulated surface of genus zero, namely a sphere with $1 + |\{i : A_{ij} = 1\}|$ holes, whose boundary is the disjoint union of the tether Λ_j and the row cycles C_i with $A_{ij} = 1$. Every edge of $K(A; R, J)$ that is not on a row cycle or a tether lies in either no triangle or exactly two triangles of $K(A; R, J)$, and in the latter case both triangles belong to the same set Σ_j. Any two triangles of Σ_j are joined by a sequence of triangles of Σ_j in which consecutive triangles share an edge that is not on a row cycle or a tether.*

Proof. Each cone is a disk, and a sealed socket is the annulus $\mathrm{Cyl}(a, b)$. A tubed socket between a and b with tube to c is a pair of pants, which we check directly. The punctured cylinder alone is not a surface, because it is pinched at b_0; the tube repairs this. In the tubed socket, the edges a_1b_0 and b_0a_0 of the hole lie in one triangle of $\mathrm{Cyl}^\circ(a, b)$ and in one triangle of the tube, the edge a_0a_1 lies only in the tube triangle $a_0a_1c_0$, and the *link* of every vertex, the graph formed by the edges opposite to it in the triangles that contain it, is a single path; for instance, the link of b_0 is the path $b_1\, a_1\, c_1\, c_2\, a_0\, b_2$. The edges in only one triangle form the three disjoint cycles a, b, and c, and the socket has 9 vertices, 21 edges, and 11 triangles, so its Euler characteristic is -1; hence it is a sphere with three holes. The pieces of column j are glued in a chain along the complete triangle cycles $s^j_0, \ldots, s^j_{d_j+1}$, so Σ_j is a connected surface: a disk, followed by pairs of pants and annuli, followed by a disk. Its boundary consists of the far ends of its tubes, namely Λ_j and the row cycles C_i with $A_{ij} = 1$. Each cone and each socket is connected through its interior edges, by direct inspection of the triangle lists, and consecutive pieces share the three edges of a cycle s^j_t, which are interior edges of Σ_j; this gives the last claim. All vertices of column j other than those of the row cycles are private to column j. Hence an edge that lies on no row cycle and no tether has at least one private endpoint, so all triangles that contain it belong to one column j, and it is an interior edge of the surface Σ_j or an edge in no triangle of K. An interior edge of a triangulated surface lies in exactly two of its triangles. The fillings D_i and E_j use only edges of row cycles and tethers. □

4.3 The crux: two-cycles are column dependencies

Over $\mathbb{F}_2$ we may ignore orientations. For other fields we orient the triangles explicitly. For a triangle cycle x let $[x] = x_0x_1 + x_1x_2 + x_2x_0$ denote the 1-cycle that traverses it in index order. We orient the triangles of $\mathrm{Cyl}(a, b)$ as $a_t\, a_{t+1}\, b_t$ and $a_{t+1}\, b_{t+1}\, b_t$, for $t \in \mathbb{Z}_3$. A direct check shows that at every edge shared by two of these triangles the two induced orientations are opposite, and that $\partial\,\mathrm{Cyl}(a, b) = [a] - [b]$. The hole $h = a_0a_1b_0$ is one of these oriented triangles, so $\partial\,\mathrm{Cyl}^\circ(a, b) = [a] - [b] - [h]$, and the same rule gives $\partial\,\mathrm{Cyl}(h, c) = [h] - [c]$ for a tube. Writing o for the apex and a for the coned cycle, we orient the cone on $a = s^j_0$ as $o\, a_{t+1}\, a_t$, so that its boundary is $-[s^j_0]$, and the cone on the last cycle $a = s^j_{d_j+1}$ as $o\, a_t\, a_{t+1}$, so that its boundary is $+[s^j_{d_j+1}]$. Summing over the pieces of column j, the contributions of the cycles s^j_t and of the holes cancel, and the sum $[\Sigma_j]$ of the oriented triangles of Σ_j satisfies $\partial[\Sigma_j] = -[\Lambda_j] - \sum_{i:\,A_{ij}=1}[C_i]$. Since the pieces induce opposite orientations on every cycle along which they are glued, this orientation is coherent: the

two triangles at any interior edge of Σ_j induce opposite orientations on it. We replace $[\Sigma_j]$ by its negative, and we orient the fillings so that $\partial D_i = [C_i]$ and $\partial E_j = [\Lambda_j]$. Then

$$\partial[\Sigma_j] = [\Lambda_j] + \sum_{i:\, A_{ij}=1} [C_i]. \tag{4}$$

All signs in (4) are equal; this is what makes the construction work over every field.

Lemma 4.4 (Crux). *Let $\mathbb{F}$ be a field, let A be a $0/1$ matrix with $\operatorname{supp}(A) \subseteq P$, and let $R \subseteq [r]$ and $J \subseteq [k]$. Then $K = K(A; R, J)$ satisfies*

$$\beta_2(K;\mathbb{F}) \;=\; \dim Z_2(K;\mathbb{F}) \;=\; |J| - \operatorname{rank}_{\mathbb{F}} A[R, J].$$

Proof. The first equality holds by (2), since $\dim K \le 2$. Let z be a 2-cycle of K. Consider an edge e that lies in two triangles τ, τ' of some Σ_j (Lemma 4.3). No other triangle contains e, and τ, τ' induce opposite orientations on e, so the coefficient of e in ∂z is the difference of the coefficients of τ and τ' in z. Since $\partial z = 0$, these coefficients agree. By the last claim of Lemma 4.3, z has the same coefficient x_j on every triangle of Σ_j. Every other triangle of K is one of the fillings D_i ($i \notin R$) or E_j ($j \in J$). Therefore

$$z = \sum_{j\in[k]} x_j[\Sigma_j] + \sum_{i\notin R} y_i D_i + \sum_{j\in J} w_j E_j$$

for some scalars $x_j, y_i, w_j \in \mathbb{F}$. By (4),

$$\begin{aligned}\partial z = &\sum_{j\in J} (x_j + w_j)\,[\Lambda_j] + \sum_{j\notin J} x_j\,[\Lambda_j] \\ &+ \sum_{i\in R} \Big(\sum_j A_{ij}x_j\Big)[C_i] + \sum_{i\notin R} \Big(\sum_j A_{ij}x_j + y_i\Big)[C_i].\end{aligned}$$

The row cycles and the tethers have pairwise disjoint edge sets, so $\partial z = 0$ holds if and only if all coefficients vanish. For $j \notin J$ this forces $x_j = 0$; for $j \in J$ it determines $w_j = -x_j$. For $i \notin R$ it determines y_i, and for $i \in R$ it says $\sum_{j\in J} A_{ij}x_j = 0$. So $z \mapsto (x_j)_{j\in J}$ is a linear bijection from $Z_2(K)$ onto the kernel of $A[R, J]$, whose dimension is $|J| - \operatorname{rank}_{\mathbb{F}} A[R, J]$. □

Because $K(A; R, J)$ always contains the whole 1-skeleton of $\mathcal{U}(P)$, the number $\beta_0(K)$ and the numbers of vertices and edges do not change. By (2),

$$\beta_1(K) = \beta_1(\mathcal{U}^{(1)}) + \beta_2(K) - n_2(K), \tag{5}$$

where $\mathcal{U}^{(1)}$ is the 1-skeleton of $\mathcal{U}(P)$. Here $\beta_1(\mathcal{U}^{(1)}) = \beta_0(\mathcal{U}^{(1)}) - n_0(\mathcal{U}) + n_1(\mathcal{U})$ is the number of independent cycles of the graph $\mathcal{U}^{(1)}$, a constant of the universe; intuitively, each triangle of K either fills one of these cycles or creates a void. So β_1 and β_2 determine each other once the number $n_2(K)$ of triangles is known.

Lemma 4.5 (Locality). *Let $K = K(A; R, J)$.*

(i) If A' is a $0/1$ matrix that differs from A in exactly one entry $(i, j) \in P$, then $K(A'; R, J)$ differs from K in exactly 7 triangles.

(ii) If $R' = R\triangle\{i\}$ and $J' = J\triangle\{j\}$, then $K(A; R', J)$ and $K(A; R, J')$ each differ from K in exactly one triangle.

In both cases the change can be carried out by single-triangle insertions and deletions in any order, and every intermediate family of simplices is a subcomplex of $\mathcal{U}(P)$.

Proof. Flipping A_{ij} exchanges the seal of the socket of position (i, j) (one triangle) for its tube (six triangles), or vice versa: one triangle is deleted and six are inserted, or the reverse. Switching row i or column j inserts or deletes the filling D_i or E_j. All edges and vertices of $\mathcal{U}(P)$ are always present, so every set of triangles together with them is a subcomplex. □

4.4 The equivalence

Theorem 4.6 (Rank-completeness). *Let $\mathbb{F}$ be a field.*

(i) For every fixed $p \geq 0$, dynamic β_p over $\mathbb{F}$ reduces to dynamic rank under switches over $\mathbb{F}$ of a $\{0, \pm 1\}$-matrix with overhead $(1, 1)$, and to dynamic rank over $\mathbb{F}$ of $\{0, \pm 1\}$-matrices with overhead $(p + 2, 1)$.

(ii) Dynamic rank over $\mathbb{F}$ of $0/1$ matrices reduces to dynamic β_2 over $\mathbb{F}$ of 2-complexes, and to dynamic β_1 over $\mathbb{F}$ of 2-complexes, with overhead $(7, 1)$. Every entry flip becomes 7 simplex updates and every switch becomes one simplex update.

(iii) For every fixed $p \geq 1$, dynamic β_1 over $\mathbb{F}$ of 2-complexes reduces to dynamic β_p over $\mathbb{F}$ of $(p + 1)$-complexes with overhead $(3^{p-1}, 1)$, and for every fixed $p \geq 2$, dynamic β_2 over $\mathbb{F}$ of 2-complexes reduces to dynamic β_p over $\mathbb{F}$ of p-complexes with overhead $(3^{p-2}, 1)$.

(iv) Dynamic rank over $\mathbb{F}$ of $\{0, \pm 1\}$-matrices reduces to dynamic β_2 over $\mathbb{F}$ of 2-complexes, and to dynamic β_1 over $\mathbb{F}$ of 2-complexes, with overhead $(8, 1)$.

Since boundary matrices have entries in $\{0, \pm 1\}$, Theorem 4.6 has the following consequence.

Corollary 4.7. *Let $\mathbb{F}$ be a field, and let $p \geq 1$ and $d \geq \max\{p, 2\}$ be fixed. Dynamic rank over $\mathbb{F}$ of $\{0, \pm 1\}$-matrices, dynamic β_1 over $\mathbb{F}$ of 2-complexes, and dynamic β_p over $\mathbb{F}$ of complexes of dimension at most d reduce to one another with constant overhead. Over $\mathbb{F}_2$ and over $\mathbb{F}_3$, this includes dynamic rank of all matrices.*

Part (iii) uses the suspension. The *suspension* of a complex K with two new vertices a, b is

$$SK = K \cup \{\{a\}, \{b\}\} \cup \{\sigma \cup \{a\},\ \sigma \cup \{b\} : \sigma \in K\}.$$

Lemma 4.8. *For every nonempty subcomplex K of a universe $\mathcal{U}$, every field $\mathbb{F}$, and every $p \geq 0$, we have $\widetilde{H}_{p+1}(SK; \mathbb{F}) \cong \widetilde{H}_p(K; \mathbb{F})$; moreover, $\beta_{p+1}(SK) = \beta_p(K)$ for every subcomplex K, empty or not, and every $p \geq 1$. Inserting or deleting a simplex σ of K corresponds to inserting or deleting the three simplices σ, $\sigma \cup \{a\}$, $\sigma \cup \{b\}$ of SK, in an order in which every intermediate family is a subcomplex of $S\mathcal{U}$.*

Proof. If $K = \emptyset$, then SK consists of two points, and $\beta_{p+1}(SK) = 0 = \beta_p(K)$ for $p \geq 1$. Otherwise SK is the union of the cones $aK = \{\{a\}\} \cup K \cup \{\sigma \cup \{a\} : \sigma \in K\}$ and bK, which are contractible, and $aK \cap bK = K$. The Mayer–Vietoris sequence in reduced homology [32, Section 2.2] therefore gives $\widetilde{H}_{p+1}(SK) \cong \widetilde{H}_p(K)$ for every $p \geq 0$, and reduced and unreduced homology agree in positive degrees. For insertion, insert σ first and then its two cones; for deletion, use the reverse order. Every intermediate family is then a subcomplex of $S\mathcal{U}$. □

Proof of Theorem 4.6. Part (i) is Lemma 4.1. The rank instance is the block matrix of Lemma 4.1; its size is $O(n)$ for fixed p. In the version with switches, the instance is the block matrix itself, preprocessing switches off all its columns, and afterwards only switches are used. In the version with entry updates, the admissible positions are the at most $(p + 1)\, n_p(\mathcal{U}) + (p + 2)\, n_{p+1}(\mathcal{U})$ nonzero positions of the block matrix, and all entries are initially zero. Inserting or deleting a q-simplex for $q \in \{p, p + 1\}$ switches one column, or updates its at most $q + 1$ nonzero entries; the query subtracts the answer from a counter for $n_p(K)$.

For part (ii), we map an instance (r, k, P) to the universe $\mathcal{U}(P)$ of Definition 4.2. In preprocessing we insert the simplices of $K(0; [r], [k])$ in order of increasing dimension. We maintain the invariant $K = K(A; R, J)$, together with counters for $|J|$ and $n_2(K)$. By Lemma 4.5, a flip is carried out by 7 simplex updates and a switch by one. By Lemma 4.4, $\operatorname{rank} A[R, J] = |J| - \beta_2(K)$, and by (5), $\beta_2(K) = \beta_1(K) - \beta_1(\mathcal{U}^{(1)}) + n_2(K)$, so one query of either Betti number answers a rank query. The universe has $O(r + k + |P|)$ simplices.

For part (iii) the universe is the $(p-1)$-fold suspension $S^{p-1}\mathcal{U}$ of $\mathcal{U}$ (the suspension applied $p-1$ times, with two new vertices each time), which has fewer than $3^{p-1}(n+1)$ simplices. In preprocessing we insert the $3^{p-1}-1$ simplices of the $(p-1)$-fold suspension of the empty complex, in order of increasing dimension. Afterwards each update of K becomes 3^{p-1} updates, by Lemma 4.8 applied $p-1$ times. The reduction from β_2 of 2-complexes to β_p of p-complexes is the same with the $(p-2)$-fold suspension.

For part (iv), let $\mathcal{U}^{\pm}(P)$ consist of $\mathcal{U}(P)$ and, for every $(i, j) \in P$, the *reversed tube* $\operatorname{Cyl}(h, \bar{C}_i)$ with its edges, where h is the hole of the socket of (i, j) and $\bar{C}_i = (c_0, c_2, c_1)$ traverses $C_i = (c_0, c_1, c_2)$ backwards. For $A \in \{0, \pm 1\}^{r \times k}$ with $\operatorname{supp}(A) \subseteq P$, the complex $K^{\pm}(A; R, J)$ is defined as in Definition 4.2, with all vertices and edges of $\mathcal{U}^{\pm}(P)$, and with the reversed tube instead of the tube at every position with $A_{ij} = -1$. A socket with a reversed tube is a socket tubed to the triangle cycle $\bar{C}_i$, so Lemma 4.3 holds with $A_{ij} = 1$ replaced by $A_{ij} \neq 0$. Betti numbers do not depend on orientations, so we may choose them separately for each complex: every triangle of $K^{\pm}(A; R, J)$ is oriented as the piece of $K^{\pm}(A; R, J)$ that contains it orients it. Since $\partial \operatorname{Cyl}(h, \bar{C}_i) = [h] + [C_i]$, the computation before (4) gives $\partial[\Sigma_j] = [\Lambda_j] + \sum_i A_{ij}[C_i]$, and the proof of Lemma 4.4 gives $\beta_2(K^{\pm}(A; R, J); \mathbb{F}) = |J| - \operatorname{rank}_{\mathbb{F}} A[R, J]$. Changing an entry between 0 and ± 1 exchanges the seal and a tube, which takes 7 triangle updates; changing it between 1 and -1 deletes 4 triangles and inserts 4, because the two tubes share the two triangles $h_0h_1c_0$ and $h_2c_1c_2$. The universe still has $O(r + k + |P|)$ simplices, and the rest of the proof is as for part (ii). □

Proof of Corollary 4.7. By Theorem 4.6(i), dynamic β_p reduces to dynamic rank of the block matrix of Lemma 4.1, whose entries lie in $\{0, \pm 1\}$. By part (iv), dynamic rank of $\{0, \pm 1\}$-matrices reduces to dynamic β_1 and to dynamic β_2 of 2-complexes. For $p \leq 2$ we are done, since $d \geq 2$; for $p \geq 3$, part (iii) reduces dynamic β_2 of 2-complexes to dynamic β_p of p-complexes, and $p \leq d$. Over $\mathbb{F}_2$ and over $\mathbb{F}_3$, every matrix has entries in $\{0, \pm 1\}$. □

Two consequences are worth stating separately. First, composing (ii) and (iv) with the switch-only version of (i) for $p = 2$ shows that switches alone are as powerful as entry updates.

Corollary 4.9. *Let $\mathbb{F}$ be a field. Dynamic rank over $\mathbb{F}$ of $0/1$ matrices reduces with overhead $(7, 1)$, and dynamic rank over $\mathbb{F}$ of $\{0, \pm 1\}$-matrices reduces with overhead $(8, 1)$, to dynamic rank under switches over $\mathbb{F}$ of a $\{0, \pm 1\}$-matrix with $O(N)$ nonzero entries, using column switches only. Over $\mathbb{F}_2$ and over $\mathbb{F}_3$, this applies to all matrices.*

Second, if nothing is known in advance about the positions of the nonzero entries, we may take $P = [r] \times [k]$; then the universe has $\Theta(rk)$ simplices. This is the setting of the lower bounds in Section 5.

5 Conditional lower bounds

In this section we prove lower bounds for dynamic Betti numbers that are conditional on the OMv conjecture. Section 5.1 shows that the gadget of Section 4 answers one OuMv request with $O(\ell)$ simplex updates and a single query. Section 5.2 derives amortized lower bounds for β_p for every $p \geq 1$, and Section 5.3 extends them to persistent Betti numbers of sublevel sets. Section 5.4 restates the bounds as a lower bound on the total time of online zigzag persistence. Section 5.5 proves

worst-case lower bounds for data structures that support only insertions or only deletions, and it transfers worst-case lower bounds from dynamic rank. Section 5.6 shows that the bounds hold on universes of bounded degree, Section 5.7 on 2-complexes that are linearly embedded in $\mathbb{R}^4$, Section 5.8 extends them to clique complexes and Vietoris–Rips complexes, and Section 5.9 to complexes with almost no homology. Section 5.10 discusses what the bounds do not cover.

5.1 One request of OuMv

Throughout this section, *dynamic β_p over $\mathbb{F}$* denotes Problem 3.1 with parameters p and $\mathbb{F}$, and n denotes the number of simplices of the universe. For a vector $u \in \{0,1\}^\ell$ we write $\operatorname{supp} u = \{i \in [\ell] : u_i = 1\}$. A matrix with no rows or no columns has rank 0.

The hypothesis. We use the consequence of the OMv conjecture recalled in Section 3 [34, Theorem 2.7]. If the conjecture holds, then for every constant $\delta > 0$ no algorithm solves OuMv for matrices $M \in \{0,1\}^{\ell\times\ell}$ and ℓ requests with preprocessing time polynomial in ℓ, computation time $O(\ell^{3-\delta})$, and error probability at most $1/3$. Here the *computation time* is the total time spent after preprocessing.

Dynamic rank with switches. Switches make dynamic rank OMv-hard by an elementary argument, which we state first.

Lemma 5.1. *Let $\mathbb{F}$ be a field and $\varepsilon > 0$. If the OMv conjecture holds, then no data structure for Problem 3.2 over $\mathbb{F}$ on the instances with $r = k = \ell$ and $P = [\ell] \times [\ell]$, $\ell \geq 1$, has polynomial preprocessing time, amortized update time $O(N^{1/2-\varepsilon})$, and query time $O(N^{1-\varepsilon})$, where $N = 2\ell + \ell^2$ is the instance size. This holds even if every entry update sets an entry to 0 or 1 and all updates after the first ℓ^2 are switches.*

Proof. Given an OuMv instance $M \in \{0,1\}^{\ell\times\ell}$, we set $A = M$ with at most ℓ^2 entry updates; afterwards $R = J = [\ell]$. For a request (u, v), at most 2ℓ switches make $R = \operatorname{supp} u$ and $J = \operatorname{supp} v$, and one query returns $\operatorname{rank} A[R, J]$. Since $A[R, J]$ is a 0/1 matrix, its rank over $\mathbb{F}$ is positive if and only if it has a nonzero entry, that is, if and only if $u^\top M v = 1$. By the definition of amortized time, after the polynomial preprocessing of the data structure, the at most $3\ell^2$ updates and the ℓ interleaved queries take time $O((N + 3\ell^2)N^{1/2-\varepsilon} + \ell N^{1-\varepsilon}) = O(\ell^{3-2\varepsilon})$ in total, and reading the requests takes $O(\ell^2)$ time; the total $O(\ell^{3-2\varepsilon} + \ell^2) = O(\ell^{3-\delta})$, with $\delta = \min\{2\varepsilon, 1\}$, contradicts the consequence of the OMv conjecture stated above. □

Together with Theorem 4.6(ii), which maps every switch to one simplex update and every rank query to one Betti query, Lemma 5.1 already implies the lower bound for β_1 and β_2 in Theorem 5.5 below. We nevertheless give a direct proof through the following abstraction of the gadget, because the same argument also yields the worst-case bounds for partially dynamic data structures (Theorem 5.12), in which switching a row off and switching a column on are insertions, and the bounds for universes of bounded degree (Theorem 5.19).

Encodings. Our reduction uses only three properties of the gadget, which we isolate in the following definition. We apply it to the gadget of Section 4 and, in Section 5.6, to a variant of bounded degree.

Definition 5.2 (Encoding). Let $\mathbb{F}$ be a field, let ℓ be a positive integer, and let $M \in \{0,1\}^{\ell\times\ell}$. An *encoding of M over $\mathbb{F}$* consists of a 2-dimensional universe $\mathcal{U}$ and a subcomplex $K_{R,J} \subseteq \mathcal{U}$ for every pair of sets $R, J \subseteq [\ell]$, with the following properties.

(E1) Every complex $K_{R,J}$ contains all vertices and all edges of $\mathcal{U}$.
(E2) $\beta_2(K_{R,J}; \mathbb{F}) = |J| - \operatorname{rank}_{\mathbb{F}} M[R, J]$ for all $R, J \subseteq [\ell]$.

(E3) There are distinct triangles $D_1, \dots, D_\ell, E_1, \dots, E_\ell$ of $\mathcal{U}$, none of them in $K_{[\ell],\emptyset}$, such that $K_{R,J} = K_{[\ell],\emptyset} \cup \{D_i : i \notin R\} \cup \{E_j : j \in J\}$ for all $R, J \subseteq [\ell]$.

Lemma 5.3. *For every field $\mathbb{F}$, every $\ell \geq 1$, and every $M \in \{0,1\}^{\ell \times \ell}$, the universe $\mathcal{U}([\ell] \times [\ell])$ of Definition 4.2, together with the complexes $K_{R,J} = K(M; R, J)$, is an encoding of M over $\mathbb{F}$, in which the triangles of (E3) are the fillings D_i and E_j. This universe has $\Theta(\ell^2)$ simplices and can be constructed in time $O(\ell^2)$.*

Proof. Property (E1) holds by Definition 4.2, and property (E2) is Lemma 4.4 with $A = M$ and $P = [\ell] \times [\ell]$. By Definition 4.2, the complex $K(M; R, J)$ consists of $K(M; [\ell], \emptyset)$ together with the fillings D_i for $i \notin R$ and E_j for $j \in J$, none of which lies in $K(M; [\ell], \emptyset)$; this is (E3). The size and the construction time are stated in Section 4. □

Lemma 5.4. *Let $\mathcal{U}$, together with the complexes $K_{R,J}$, be an encoding of $M \in \{0,1\}^{\ell \times \ell}$ over $\mathbb{F}$.*

(i) For all $u, v \in \{0,1\}^\ell$, we have $u^\top M v = 1$ if and only if $\beta_2(K_{\operatorname{supp} u, \operatorname{supp} v}; \mathbb{F}) < |\operatorname{supp} v|$.

(ii) For all $R, J \subseteq [\ell]$,

$$\beta_2(K_{R,J}) = \beta_1(K_{R,J}) - \beta_1(\mathcal{U}^{(1)}) + n_2(K_{R,J}),$$

where $\mathcal{U}^{(1)}$ is the 1-skeleton of $\mathcal{U}$, so that $\beta_1(\mathcal{U}^{(1)}) = \beta_0(\mathcal{U}^{(1)}) - n_0(\mathcal{U}) + n_1(\mathcal{U})$.

(iii) For all $R, J, R', J' \subseteq [\ell]$, the $|R \triangle R'| + |J \triangle J'|$ insertions and deletions of the triangles D_i, $i \in R \triangle R'$, and E_j, $j \in J \triangle J'$, in any order, transform $K_{R,J}$ into $K_{R',J'}$, and every intermediate family of simplices is a subcomplex of $\mathcal{U}$. If $R' \subseteq R$ and $J \subseteq J'$, the sequence consists of insertions only; if $R \subseteq R'$ and $J' \subseteq J$, it consists of deletions only.

Proof. (i) Let $R = \operatorname{supp} u$ and $J = \operatorname{supp} v$; Figure 5 illustrates this part for the gadget. By property (E2), the inequality $\beta_2(K_{R,J}) < |J|$ holds if and only if $\operatorname{rank}_\mathbb{F} M[R, J] \geq 1$, that is, if and only if the submatrix $M[R, J]$ has a nonzero entry. Since the entries of M are 0 and 1, and $1 \neq 0$ in $\mathbb{F}$, this happens if and only if $M_{ij} = 1$ for some $i \in R$ and $j \in J$. This is the condition $u^\top M v = 1$.

(ii) Let $K = K_{R,J}$. By (E1), K has the same vertices and edges as $\mathcal{U}$. Hence $n_0(K) = n_0(\mathcal{U})$ and $n_1(K) = n_1(\mathcal{U})$, and $\beta_0(K) = \beta_0(\mathcal{U}^{(1)})$ because β_0 depends only on vertices and edges. Since $\dim K \leq 2$, we have $\chi(K) = n_0(K) - n_1(K) + n_2(K)$, and (2) gives

$$\beta_1(K) = \beta_0(\mathcal{U}^{(1)}) + \beta_2(K) - n_0(\mathcal{U}) + n_1(\mathcal{U}) - n_2(K),$$

which is the claim.

(iii) By (E3), $K_{R',J'}$ arises from $K_{R,J}$ by inserting D_i for $i \in R \setminus R'$ and E_j for $j \in J' \setminus J$, and by deleting D_i for $i \in R' \setminus R$ and E_j for $j \in J \setminus J'$. Every intermediate family consists of all vertices and edges of $\mathcal{U}$, by (E1), and some of its triangles. Since $\mathcal{U}$ is 2-dimensional, such a family is a subcomplex of $\mathcal{U}$, every inserted triangle has its edges present, and no deleted triangle has a cofacet; so every update satisfies the preconditions of Problem 3.1. If $R' \subseteq R$ and $J \subseteq J'$, all these updates are insertions; if $R \subseteq R'$ and $J' \subseteq J$, all are deletions. □

5.2 Amortized lower bounds

Theorem 5.5. *Let $\mathbb{F}$ be a field, let $p \in \{1, 2\}$, and let $\varepsilon > 0$. If the OMv conjecture holds, then no data structure for dynamic β_p over $\mathbb{F}$ on 2-dimensional universes has polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$.*

Proof. Suppose that $\mathcal{D}$ is such a data structure. We use it to solve OuMv for matrices $M \in \{0,1\}^{\ell \times \ell}$ and ℓ requests. Let $\mathcal{U}$, together with the complexes $K_{R,J}$, be the encoding of M over $\mathbb{F}$ from Lemma 5.3, and let $n = |\mathcal{U}| = \Theta(\ell^2)$.

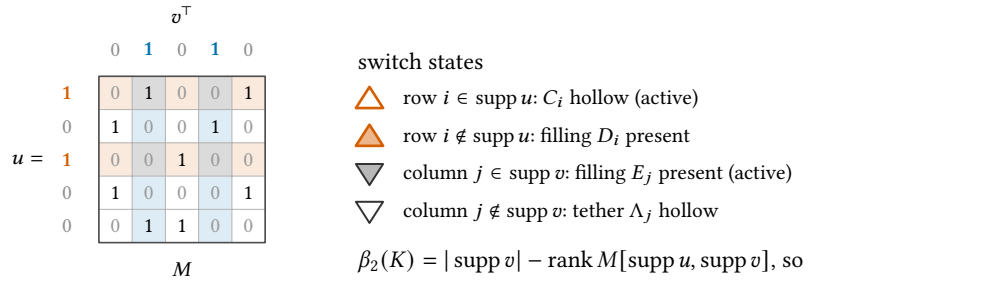


Fig. 5. Answering one OuMv request (u, v) with the gadget $K(M; R, J)$; see Lemma 5.4(i). Exactly the rows in $R = \operatorname{supp} u$ and the columns in $J = \operatorname{supp} v$ are switched on, so the active part of M is the submatrix $M[R, J]$ (dark cells). Here $M[R, J]$ contains a 1, so $\operatorname{rank} M[R, J] = 1$, $\beta_2 = |J| - 1 < |J|$, and the Boolean product $u^\top M v$ is 1. Moving from one request to the next takes at most 2ℓ switches, each of which inserts or deletes a single triangle.

Preprocessing. We construct $\mathcal{U}$, compute the number $\beta_1(\mathcal{U}^{(1)})$ of Lemma 5.4(ii), and initialize $\mathcal{D}$ with the universe $\mathcal{U}$. Then we insert the simplices of $K_{[\ell],\emptyset}$ into $\mathcal{D}$ in order of increasing dimension, so that every insertion is valid. We maintain sets $R, J \subseteq [\ell]$ such that $\mathcal{D}$ stores $K_{R,J}$, initially $R = [\ell]$ and $J = \emptyset$, together with the number $n_2(K_{R,J})$ of its triangles.

Requests. For a request (u, v), let $R' = \operatorname{supp} u$ and $J' = \operatorname{supp} v$. We perform the at most $|R \triangle R'| + |J \triangle J'| \leq 2\ell$ updates of Lemma 5.4(iii), which transform $K_{R,J}$ into $K_{R',J'}$, and we set $R = R'$ and $J = J'$. Then we query $\mathcal{D}$. If $p = 2$, the answer is $\beta_2(K_{R,J})$; if $p = 1$, we compute $\beta_2(K_{R,J})$ from the answer by Lemma 5.4(ii). We output 1 if $\beta_2(K_{R,J}) < |J|$, and 0 otherwise. By Lemma 5.4(i), the output is $u^\top M v$.

Running time. In total, $\mathcal{D}$ processes at most $n + 2\ell^2 = O(\ell^2)$ updates: at most n insertions that create $K_{[\ell],\emptyset}$ and at most 2ℓ updates per request. By the definition of amortized time (Section 3), after the preprocessing of $\mathcal{D}$ these updates together with the ℓ interleaved queries take total time $O((2n + 2\ell^2) \cdot n^{1/2-\varepsilon} + \ell \cdot n^{1-\varepsilon}) = O(\ell^{3-2\varepsilon})$. Reading a request, computing $R \triangle R'$ and $J \triangle J'$, and maintaining the counter take $O(\ell)$ time per request. The preprocessing of $\mathcal{D}$ takes time polynomial in n, and hence in ℓ. So the preprocessing time is polynomial in ℓ, and the computation time is $O(\ell^{3-2\varepsilon} + \ell^2) = O(\ell^{3-\delta})$ for $\delta = \min\{2\varepsilon, 1\} > 0$. This contradicts the consequence of the OMv conjecture stated in Section 5.1.

We note for later use that the argument runs $\mathcal{D}$ only on the universes $\mathcal{U}([\ell] \times [\ell])$, $\ell \geq 1$, and that it uses these universes only through properties (E1)–(E3), the bound $n = O(\ell^2)$, and the polynomial construction time. □

Since worst-case bounds imply amortized bounds (Section 3), Theorem 5.5 also excludes worst-case update time $O(n^{1/2-\varepsilon})$ together with query time $O(n^{1-\varepsilon})$. The reduction is deterministic. The requests of an OuMv instance may depend on the answers to earlier requests, for example when the instance is produced by the reduction from OMv in [34]; but as long as all earlier answers are correct, the requests, and hence the operations that the reduction performs on $\mathcal{D}$, are determined by the input. Hence the theorem also applies to randomized data structures that, for every fixed sequence of operations, answer all queries correctly with probability at least 2/3: we apply this guarantee to the sequence that arises when all answers are correct. It also applies to data structures whose time bounds hold only in expectation: we run k independent copies, abort a copy once its running time exceeds 12 times its expected bound, and output the answer given by more than

$k/2$ copies; by Markov's inequality, each copy answers all requests correctly within this time with probability at least $2/3 - 1/12 = 7/12 > 1/2$, so for a large enough constant k more than $k/2$ copies do so with probability at least $2/3$. In the operation sequences of the proof, all vertices and edges of the universe are inserted during preprocessing and are never deleted, and afterwards only triangles are inserted and deleted. So the hardness does not come from changes of the 1-skeleton; moreover, after preprocessing the maintained complex has $\Theta(n)$ simplices, so the bound also holds with n replaced by the size of the current complex.

Output-sensitive data structures. Inserting or deleting a q-simplex changes exactly one Betti number, namely β_q or β_{q-1}, and changes it by exactly one. Indeed, inserting a q-simplex adds a column to ∂_q, which changes $\operatorname{rank}\partial_q$ by zero or one, and it does not change $\operatorname{rank}\partial_{q+1}$, because no coface of the new simplex is present; by (1), either β_q increases by one or β_{q-1} decreases by one, and deletions are symmetric. So the number of changes of the Betti numbers is exactly one per update, and Theorem 5.5 also excludes output-sensitive data structures whose update time is $O(f(n) + k \cdot g(n))$ with $f(n) + g(n) = O(n^{1/2-\varepsilon})$, where k is the number of changes, together with query time $O(n^{1-\varepsilon})$. This contrasts with time series, where the banana trees of Cultrera di Montesano et al. [13] absorb a local change in time $O(\log n + k)$, with k the number of changes of the persistence diagram.

Higher Betti numbers. For $p \geq 3$ we iterate the suspension of Section 4.4. For a universe $\mathcal{U}$ and $m \geq 0$, the *m-fold suspension* is defined by $S^0\mathcal{U} = \mathcal{U}$ and $S^m\mathcal{U} = S(S^{m-1}\mathcal{U})$, where each suspension uses two new vertices. For a subcomplex $K \subseteq \mathcal{U}$ we define $S^m K$ in the same way, with the same new vertices as for $\mathcal{U}$. For $m \geq 1$, the complex $S^m\emptyset$ is not empty; for instance, $S\emptyset$ consists of the two new vertices.

Lemma 5.6. *Let $\mathcal{U}$ be a nonempty universe with n simplices, and let $m \geq 0$.*

(i) For every subcomplex $K \subseteq \mathcal{U}$, the complex $S^m K$ is a subcomplex of $S^m\mathcal{U}$, and $\beta_{q+m}(S^m K; \mathbb{F}) = \beta_q(K; \mathbb{F})$ for every field $\mathbb{F}$ and every $q \geq 1$.

(ii) $\dim S^m\mathcal{U} = \dim \mathcal{U} + m$, $|S^m\mathcal{U}| < 3^m(n+1)$, and $|S^m\emptyset| < 3^m$.

(iii) If K' arises from a subcomplex $K \subseteq \mathcal{U}$ by one simplex insertion (deletion), then some sequence of 3^m simplex insertions (deletions) transforms $S^m K$ into $S^m K'$, and every intermediate family is a subcomplex of $S^m\mathcal{U}$.

(iv) For fixed m and every $q \geq 1$, dynamic β_q over $\mathbb{F}$ on $\mathcal{U}$ reduces to dynamic β_{q+m} over $\mathbb{F}$ on $S^m\mathcal{U}$ with overhead $(3^m, 1)$, and the reduction carries out insertions by insertions and deletions by deletions.

Proof. We prove (i)–(iii) by induction on m; the case $m = 0$ is trivial. Let $m \geq 1$, let $L = S^{m-1}K$ and $V = S^{m-1}\mathcal{U}$, so that $S^m K = SL$ and $S^m\mathcal{U} = SV$.

(i) By induction $L \subseteq V$, and the suspension preserves inclusion, so $SL \subseteq SV$. By Lemma 4.8 with $p = q + m - 1 \geq 1$, and by induction, $\beta_{q+m}(SL) = \beta_{q+m-1}(L) = \beta_q(K)$.

(ii) Since $V \supseteq \mathcal{U}$ is not empty, $\dim SV = \dim V + 1$. The suspension of a complex X has $|X| + 2 + 2|X| = 3|X| + 2$ simplices. By induction $|V| \leq 3^{m-1}(n+1) - 1$, so $|SV| \leq 3^m(n+1) - 1$. The same computation, starting from $|\emptyset| = 0$, gives $|S^m\emptyset| \leq 3^m - 1$.

(iii) By induction, a sequence of 3^{m-1} updates of the same kind transforms L into $S^{m-1}K'$ through subcomplexes of V. By Lemma 4.8, each of these updates is carried out by three updates of the same kind on the suspension, through subcomplexes of SV.

(iv) The universe $S^m\mathcal{U}$ can be computed in polynomial time, and $|S^m\mathcal{U}| = O(n)$ by (ii). During preprocessing we insert the $3^m - 1$ simplices of $S^m\emptyset$ in order of increasing dimension. Afterwards the data structure for $S^m\mathcal{U}$ stores $S^m K$. Every update of K is carried out by the 3^m updates of (iii), and a query for $\beta_q(K)$ is answered by a query for $\beta_{q+m}(S^m K)$, by (i). □

Corollary 5.7. *Let $\mathbb{F}$ be a field, let $p \geq 1$ be an integer, and let $\varepsilon > 0$. If the OMv conjecture holds, then no data structure for dynamic β_p over $\mathbb{F}$ on universes of dimension $\max\{p, 2\}$ has polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$.*

Proof. For $p \leq 2$ this is Theorem 5.5. Let $p \geq 3$, let $m = p - 2$, and suppose that $\mathcal{D}$ is such a data structure. For $\mathcal{U} = \mathcal{U}([\ell] \times [\ell])$, the universe $S^m\mathcal{U}$ has dimension p and $O(n)$ simplices by Lemma 5.6(ii). By Lemma 5.6(iv) and the paragraph on reductions in Section 3, running $\mathcal{D}$ on $S^m\mathcal{U}$ yields a data structure for dynamic β_2 over $\mathbb{F}$ on the universes $\mathcal{U}([\ell] \times [\ell])$ with polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$. By the proof of Theorem 5.5, no such data structure exists. □

The dimension $\max\{p, 2\}$ in Corollary 5.7 cannot be lowered. If $\dim \mathcal{U} < p$, then $\beta_p(K) = 0$ for every subcomplex $K \subseteq \mathcal{U}$. If $\dim \mathcal{U} \leq 1$, then $\beta_1(K) = n_1(K) - n_0(K) + \beta_0(K)$ by (2), and $\beta_0(K)$ is the number of connected components of the graph K, which a dynamic connectivity structure maintains in $O(\log^2 n)$ amortized time per update [36]. The same structure maintains β_0 of every complex (Section 6). So the lower bounds concern exactly the Betti numbers β_p with $p \geq 1$ on universes of dimension at least $\max\{p, 2\}$.

5.3 Persistent Betti numbers

We now consider sublevel-set filtrations of functions whose values change. A function $f : \mathcal{U} \to \mathbb{R}$ on a universe is *monotone* if $f(\tau) \leq f(\sigma)$ whenever $\tau \subseteq \sigma$. For such f and $t \in \mathbb{R}$, the *sublevel set* $\mathcal{U}_t = \{\sigma \in \mathcal{U} : f(\sigma) \leq t\}$ is a subcomplex of $\mathcal{U}$, because the faces of a simplex $\sigma \in \mathcal{U}_t$ have values at most $f(\sigma) \leq t$.

Problem 5.8 (Sublevel-set persistence). Fix a field $\mathbb{F}$ and an integer $p \geq 0$. The instance is a universe $\mathcal{U}$ with n simplices and a monotone function $f_0 : \mathcal{U} \to \mathbb{R}$. The data structure maintains a monotone function $f : \mathcal{U} \to \mathbb{R}$, initially f_0, under the operations

- Set(σ, x), for $\sigma \in \mathcal{U}$ and a rational number x, given in binary, such that the function obtained from f by changing its value at σ to x is monotone, which makes this change;
- Query(s, t), for real numbers $s \leq t$, which returns the *persistent Betti number* $\beta_p^{s,t}(f)$, the rank of the linear map $H_p(\mathcal{U}_s; \mathbb{F}) \to H_p(\mathcal{U}_t; \mathbb{F})$ induced by the inclusion $\mathcal{U}_s \subseteq \mathcal{U}_t$.

We call Set a *value update*. By the structure theorem for persistence modules [62], the vector spaces $H_p(\mathcal{U}_t; \mathbb{F})$, $t \in \mathbb{R}$, together with the maps induced by inclusions, decompose into intervals, which form the degree-p *barcode* of f. The persistent Betti number $\beta_p^{s,t}(f)$ is the number of intervals of the barcode that contain both s and t [24, 25].

Lemma 5.9. *Let $\mathbb{F}$ be a field and let $p \geq 0$. Dynamic β_p over $\mathbb{F}$ on a universe $\mathcal{U}$ reduces to Problem 5.8 over $\mathbb{F}$ on the same universe with overhead $(1, 1)$. In the reduction, f takes values in $\{0, 1\}$, and every query is Query$(0, 0)$.*

Proof. For a subcomplex $K \subseteq \mathcal{U}$, let $f_K(\sigma) = 0$ if $\sigma \in K$ and $f_K(\sigma) = 1$ otherwise. The function f_K is monotone: if $\tau \subseteq \sigma$ and $f_K(\sigma) = 0$, then $\sigma \in K$, so $\tau \in K$ and $f_K(\tau) = 0$; if $f_K(\sigma) = 1$, then $f_K(\tau) \leq f_K(\sigma)$ trivially. Its sublevel set at 0 is K. We let $f_0 = f_\emptyset$, the constant function 1, and we maintain the invariant $f = f_K$. An update Insert(σ) produces the subcomplex $K \cup \{\sigma\}$, so Set$(\sigma, 0)$ is a valid value update that restores the invariant. Similarly, Delete(σ) is carried out by Set$(\sigma, 1)$. A query is answered by Query$(0, 0)$, which returns the rank of the identity map of $H_p(K; \mathbb{F})$, that is, $\beta_p(K; \mathbb{F})$. □

Corollary 5.10. *Let $\mathbb{F}$ be a field, let $p \geq 1$, and let $\varepsilon > 0$. If the OMv conjecture holds, then no data structure for Problem 5.8 over $\mathbb{F}$ on universes of dimension $\max\{p, 2\}$ has polynomial preprocessing*

time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$. This holds even when f takes values in $\{0,1\}$ and every query is $\textsc{Query}(0,0)$.

Proof. By Lemma 5.9 and the paragraph on reductions in Section 3, such a data structure yields a data structure for dynamic β_p over $\mathbb{F}$ on the same universes with polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$. The reduction uses only values in $\{0,1\}$ and only queries $\textsc{Query}(0,0)$, so a data structure that supports only these suffices. This contradicts Corollary 5.7. □

Corollary 5.10 applies to every data structure that maintains the barcode under value updates and counts, within the stated time bounds, the intervals that contain two given numbers. We ask for this count rather than for the barcode itself, because the barcode can have $\Theta(n)$ intervals and listing them takes linear time. Explicit maintenance of the barcode is excluded as well. In the reduction, f takes only the values 0 and 1, so the barcode of H_p consists of $\beta_p(K) - \rho$ copies of $[0,1)$, ρ copies of $[0,\infty)$, and $\beta_p(\mathcal{U}) - \rho$ copies of $[1,\infty)$, where $K = \mathcal{U}_0$ and ρ is the rank of the map $H_p(K) \to H_p(\mathcal{U})$. A value update changes K by one simplex, which changes $\beta_p(K)$ by at most one and, since H_p of the larger complex is spanned by the image of H_p of the smaller one and at most one further class, also changes ρ by at most one. So every value update changes $O(1)$ intervals, and Corollary 5.10 applies to every data structure that maintains the barcode explicitly, even an output-sensitive one. Corollary 5.10 also holds for injective functions, so data structures that assume distinct values gain nothing. Fix an injective map $\mathrm{id}\colon \mathcal{U} \to [n]$, let $d = \dim \mathcal{U}$, and let $g(\sigma) = (\dim \sigma + \mathrm{id}(\sigma)/(n+1))/(2d+2) \in [0, \frac{1}{2})$. In the proof of Lemma 5.9, replace f_K by the function that is $g(\sigma)$ on $\sigma \in K$ and $1 + g(\sigma)$ on $\sigma \notin K$, and $\textsc{Query}(0,0)$ by $\textsc{Query}(\frac{1}{2}, \frac{1}{2})$. This function is injective, it is monotone because g increases strictly from a simplex to each of its cofaces, its sublevel set at $\frac{1}{2}$ is K, and a simplex update of K is still a single value update. In the operation sequences obtained from the proof of Theorem 5.5 through Lemma 5.9, all vertices and edges have value 0 after preprocessing, and only the values of triangles change.

5.4 Online zigzag persistence

A sequence of simplex insertions and deletions is the same as a simplex-wise *zigzag filtration*

$$\emptyset = K_0 \leftrightarrow K_1 \leftrightarrow \cdots \leftrightarrow K_L,$$

in which every K_t arises from K_{t-1} by inserting or deleting one simplex. Zigzag persistence [6] assigns to such a filtration a barcode: a multiset of intervals of indices. The number of intervals of the barcode of H_p that contain the index t is $\beta_p(K_t)$. Algorithms for zigzag persistence go back to Carlsson, de Silva, and Morozov [7]; Maria and Oudot [46] update a zigzag barcode under one simplex insertion or removal at a time, in cubic total time in the worst case. Offline, zigzag persistence can be computed in matrix multiplication time [47], and in practice by a reduction to standard persistence [21]; for graphs, near-linear time suffices [19, 22]. We call an algorithm *online* if it receives the steps one at a time and, after step t, has reported all intervals that begin at t or end at $t-1$. Such an algorithm knows how many intervals of the barcode of H_p contain t, so it maintains $\beta_p(K_t)$ with $O(1)$ extra work per step.

Corollary 5.11 (Online zigzag persistence). *Assume the OMv conjecture, and let $\mathbb{F}$ be a field and $\varepsilon > 0$. No online algorithm computes the barcode of H_1 over $\mathbb{F}$ of zigzag filtrations of length L of subcomplexes of a 2-dimensional universe in total time $O(L^{3/2-\varepsilon})$, even with polynomial preprocessing of the universe. In particular, the amortized time per step cannot be $O(L^{1/2-\varepsilon})$.*

Proof. The reduction in the proof of Theorem 5.5 produces, for an OuMv instance of dimension ℓ, a universe with $n = \Theta(\ell^2)$ simplices and a sequence of $L = \Theta(\ell^2)$ simplex insertions and deletions,

interleaved with ℓ queries of β_1. The steps of each round are determined by the request (u, v) of that round, and the answer to the round is needed before the next request arrives; an online zigzag algorithm provides it in time. An online zigzag algorithm answers each query in $O(1)$ time from its count of intervals containing the current index. If its total time were $O(L^{3/2-\varepsilon}) = O(\ell^{3-2\varepsilon})$, then OuMv would be solvable in time $O(\ell^{3-2\varepsilon} + \ell^2) = O(\ell^{3-\delta})$, with $\delta = \min\{2\varepsilon, 1\}$, after polynomial preprocessing, contradicting the OMv conjecture. □

By Theorem 5.30, the same bound holds for filtrations in which every complex at which the reduction reads off an answer is connected and has $\beta_1 = \beta_2 \le 1$. Offline, the whole barcode can be computed in time $O(L^\omega)$ [47], which exceeds $L^{3/2}$ because $\omega \ge 2$, so Corollary 5.11 does not separate the online from the offline setting. What it rules out is an online algorithm with amortized time $L^{1/2-\varepsilon}$ per step; for graphs, in contrast, the number of intervals containing the current index is maintained online in polylogarithmic amortized time per step by dynamic connectivity (Proposition 6.1).

5.5 Worst-case lower bounds

Partially dynamic data structures. *Incremental* dynamic β_p over $\mathbb{F}$ is the variant of Problem 3.1 in which the instance also contains a subcomplex $K_0 \subseteq \mathcal{U}$, the data structure starts with $K = K_0$, and the only updates are insertions. *Decremental* dynamic β_p over $\mathbb{F}$ is defined in the same way, with deletions instead of insertions. In both variants the data structure may preprocess K_0 together with $\mathcal{U}$.

Theorem 5.12. *Let $\mathbb{F}$ be a field, let $p \ge 1$, and let $\varepsilon > 0$. If the OMv conjecture holds, then no data structure for incremental dynamic β_p over $\mathbb{F}$, and no data structure for decremental dynamic β_p over $\mathbb{F}$, on universes of dimension $\max\{p, 2\}$ has polynomial preprocessing time, worst-case update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$.*

Proof. First let $p \in \{1, 2\}$. We use the rollback technique [1, 33]. Suppose that $\mathcal{D}$ is an incremental data structure with the stated bounds. Let $M \in \{0,1\}^{\ell\times\ell}$ be the matrix of an instance of OuMv, and let $\mathcal{U}$, together with the complexes $K_{R,J}$, be the encoding of M over $\mathbb{F}$ from Lemma 5.3.

Rollback. We run $\mathcal{D}$ under a simulator that records, for every write to a memory cell, the address and the previous content of the cell in a log. Replaying a log backwards restores every cell written since the log was started. Afterwards the memory of $\mathcal{D}$, and hence its state, is the same as when the log was started. The simulation and the log slow $\mathcal{D}$ down by a constant factor. An operation writes at most as many cells as it takes steps, so an update and its rollback take worst-case time $O(n^{1/2-\varepsilon})$, and a query and its rollback take worst-case time $O(n^{1-\varepsilon})$.

Preprocessing. We construct $\mathcal{U}$, compute $\beta_1(\mathcal{U}^{(1)})$, and initialize $\mathcal{D}$ with the universe $\mathcal{U}$ and the initial complex $K_0 = K_{[\ell],\emptyset}$. This takes time polynomial in ℓ.

Requests. For a request (u, v), let $R' = \operatorname{supp} u$ and $J' = \operatorname{supp} v$. Since $R' \subseteq [\ell]$ and $\emptyset \subseteq J'$, Lemma 5.4(iii) gives a sequence of at most 2ℓ triangle insertions that transforms $K_{[\ell],\emptyset}$ into $K_{R',J'}$. We start a new log, perform these insertions, and query $\mathcal{D}$. If $p = 1$, we compute $\beta_2(K_{R',J'})$ from the answer by Lemma 5.4(ii), where $n_2(K_{R',J'})$ is $n_2(K_0)$ plus the number of inserted triangles. We output 1 if $\beta_2(K_{R',J'}) < |J'|$ and 0 otherwise, which is $u^\top M v$ by Lemma 5.4(i). Finally we roll back the log. Then $\mathcal{D}$ is again in the state it had after preprocessing, and it stores $K_{[\ell],\emptyset}$. Between two rollbacks, $\mathcal{D}$ receives a sequence of insertions that starts from K_0, so its guarantees on correctness and running time apply.

Running time. A request takes time $O(\ell \cdot n^{1/2-\varepsilon} + n^{1-\varepsilon} + \ell) = O(\ell^{2-2\varepsilon} + \ell)$. So the computation time is $O(\ell^{3-2\varepsilon} + \ell^2) = O(\ell^{3-\delta})$ for $\delta = \min\{2\varepsilon, 1\} > 0$, and the preprocessing time is polynomial in ℓ. This contradicts the consequence of the OMv conjecture stated in Section 5.1.

For a decremental data structure we use $K_0 = K_{\emptyset,[\ell]}$ instead. Since $\emptyset \subseteq R'$ and $J' \subseteq [\ell]$, Lemma 5.4(iii) now gives at most 2ℓ triangle deletions, and the rest of the argument is unchanged. As for Theorem 5.5, both arguments use only properties (E1)–(E3), the bound $n = O(\ell^2)$, and the polynomial construction time of the encodings.

Now let $p \geq 3$ and $m = p - 2$. The reduction of Lemma 5.6(iv) maps an initial complex K_0 to $S^m K_0$, insertions to insertions, and deletions to deletions. By Lemma 5.6(ii), $S^m \mathcal{U}([\ell] \times [\ell])$ has dimension p and $O(n)$ simplices. Hence a data structure as in the theorem for β_p yields an incremental, respectively decremental, data structure for β_2 on the universes $\mathcal{U}([\ell] \times [\ell])$ with polynomial preprocessing time, worst-case update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$, which we have just excluded. □

The rollback argument needs worst-case bounds. An amortized data structure may perform an expensive operation, such as a rebuild, whose cost is charged to many earlier updates. After a rollback, the next request can trigger the same expensive operation again, so the total time is no longer bounded by the number of updates times the amortized update time. Theorem 5.12 also applies to randomized data structures that answer each query correctly with probability at least $1 - 1/n$. Indeed, as long as all earlier answers are correct, each answer is produced by a run of $\mathcal{D}$ on a sequence of operations that is determined by the input, namely the preprocessing followed by the operations of one request. By a union bound, all ℓ answers are correct with probability at least $1 - \ell/n \geq 2/3$ for all sufficiently large ℓ, since $n = \Theta(\ell^2)$.

Fully dynamic data structures. For fully dynamic data structures with worst-case bounds, Theorem 5.5 gives the lower bound discussed after its proof. Stronger worst-case lower bounds follow from lower bounds for dynamic rank. Recall that an entry flip replaces an entry A_{ij} of a 0/1 matrix by $1 - A_{ij}$.

Theorem 5.13. *Let $\mathbb{F}$ be a field, let $p \geq 1$ be an integer, and let $\alpha > 0$. Suppose that for every $\varepsilon > 0$ no data structure maintains* $\operatorname{rank}_{\mathbb{F}} A$ *for a matrix $A \in \{0,1\}^{\ell \times \ell}$, initially zero, under entry flips, with polynomial preprocessing time and worst-case time $O(\ell^{\alpha-\varepsilon})$ per flip, where the time of a flip includes reporting the rank after the flip. Then for every $\varepsilon > 0$ no data structure for dynamic β_p over $\mathbb{F}$ on universes of dimension $\max\{p, 2\}$ has polynomial preprocessing time, worst-case update time $O(n^{\alpha/2-\varepsilon})$, and query time $O(n^{\alpha/2-\varepsilon})$.*

Proof. Suppose that $\mathcal{D}$ is such a data structure. First let $p \in \{1, 2\}$. Maintaining $\operatorname{rank}_{\mathbb{F}} A$ under entry flips is the special case of Problem 3.2 with $r = k = \ell$, $P = [\ell] \times [\ell]$, 0/1 entries, and no switches; its instance size is $\Theta(\ell^2)$. By Theorem 4.6(ii), this problem reduces to dynamic β_p over $\mathbb{F}$ on the 2-dimensional universe $\mathcal{U}([\ell] \times [\ell])$, which has $n = \Theta(\ell^2)$ simplices, with constant overhead. The reduction makes $O(n)$ insertions during preprocessing, which take time $O(n \cdot n^{\alpha/2-\varepsilon})$, polynomial in ℓ. After a flip we report the rank with one query. So $\mathcal{D}$ yields a data structure for the rank problem with polynomial preprocessing time and worst-case time $O(n^{\alpha/2-\varepsilon}) = O(\ell^{\alpha-2\varepsilon})$ per flip, including the report of the rank. This contradicts the hypothesis with 2ε in place of ε.

For $p \geq 3$, Lemma 5.6(iv) with $m = p - 2$ turns $\mathcal{D}$ into a data structure for dynamic β_2 over $\mathbb{F}$ on the universes $\mathcal{U}([\ell] \times [\ell])$ with polynomial preprocessing time and worst-case update and query time $O(n^{\alpha/2-\varepsilon})$, because $S^m \mathcal{U}([\ell] \times [\ell])$ has dimension p and $O(n)$ simplices. The argument for $p = 2$ applies to this data structure. □

We use the following Schur-complement identity here and in Section 5.9.

Lemma 5.14. *Let $Y \in \mathbb{F}^{m \times m}$ be invertible and $x, w \in \mathbb{F}^m$. Then*

$$\operatorname{rank} \begin{pmatrix} Y & w \\ x^\top & 0 \end{pmatrix} = m + \left[x^\top Y^{-1} w \neq 0\right],$$

where $[\,\cdot\,]$ *is* 1 *if the condition in brackets holds and* 0 *otherwise.*

Proof. The matrix equals

$$\begin{pmatrix} I & 0 \\ x^\top Y^{-1} & 1 \end{pmatrix} \begin{pmatrix} Y & w \\ 0 & -x^\top Y^{-1} w \end{pmatrix}.$$

The first factor is invertible, and the second is block triangular with an invertible diagonal block Y. □

The hypothesis of Theorem 5.13 follows from a hinted variant of the OMv conjecture. Van den Brand, Nanongkai, and Saranurak [58, Conjecture 5.12] conjecture the hardness of the *uMv-hinted uMv* problem [58, Definition 5.11]. In the form used in the proof of [58, Theorem 5.13], this problem maintains a product $M_1M_2M_3M_4M_5$ of five Boolean matrices of dimension at most s: the matrices M_1, M_3, M_5 are given in advance, M_2 and M_4 start at zero and receive element updates, and a single query asks for one entry of the product. Under the conjecture, every data structure for it with polynomial preprocessing time needs worst-case time $s^{\alpha-o(1)}$ per update or per query; we call this consequence the *five-matrix bound*. Its exponent α is a function of the exponents of rectangular matrix multiplication. It satisfies $\alpha \geq 5/4$, with equality if $\omega = 2$, and $\alpha \approx 1.407$ with the current bounds on these exponents, which also give the exponent 1.407 of the dynamic rank algorithm of [57] (Corollary 5.16 below). As for the OMv conjecture, we read the conjecture as a statement about randomized algorithms with error probability at most 1/3; the lower bound of van den Brand et al. for dynamic rank, discussed after the next proof, relies on this reading as well.

Corollary 5.15 (Worst-case lower bound). *Let* $\mathbb{F}$ *be a field, let* $p \geq 1$*, and let* $\varepsilon > 0$*. If the uMv-hinted uMv conjecture [58, Conjecture 5.12] holds, then no data structure for dynamic* β_p *over* $\mathbb{F}$ *on universes of dimension* $\max\{p, 2\}$ *has polynomial preprocessing time, worst-case update time* $O(n^{\alpha/2-\varepsilon})$*, and query time* $O(n^{\alpha/2-\varepsilon})$*. Here* $\alpha \geq 5/4$ *is the exponent of the five-matrix bound; with the current bounds on rectangular matrix multiplication,* $\alpha/2 \approx 0.703$*.*

Proof. By Theorem 5.13, it suffices to rule out a data structure $\mathcal{R}$ with the following properties. It maintains $\operatorname{rank}_{\mathbb{F}} A$ of a matrix $A \in \{0,1\}^{\ell \times \ell}$ under entry flips. It has polynomial preprocessing time. It takes worst-case time $O(\ell^{\alpha-\varepsilon})$ per flip, including the report of the rank. By running a constant number of independent copies and taking majority votes, we may assume that $\mathcal{R}$ errs with probability at most $1/1000$ on every fixed operation sequence.

Suppose that $\mathcal{R}$ exists, and consider an instance of the five-matrix product problem above. During preprocessing we draw independent, uniformly random 0/1 matrices Ξ_1, Ξ_3, Ξ_5 of the shapes of M_1, M_3, M_5, and we replace M_t by the entrywise product $M'_t = M_t \circ \Xi_t$ for $t \in \{1, 3, 5\}$; we set $M'_2 = M_2$ and $M'_4 = M_4$. Following [58, Theorem C.1], we place $M'_1, \ldots, M'_5$ on the block superdiagonal of a block matrix X whose diagonal blocks are identity matrices and whose other blocks are zero. Then X is a 0/1 matrix of dimension $m = O(s)$. It is upper unitriangular, hence invertible over every field. The block of X^{-1} in the first block row and the last block column is $-P$, where $P = M'_1M'_2M'_3M'_4M'_5$ is computed in $\mathbb{F}$. For a pair (a, b), call a sequence (i_1, i_2, i_3, i_4) of indices a *path* if $(M_1)_{ai_1}(M_2)_{i_1i_2}(M_3)_{i_2i_3}(M_4)_{i_3i_4}(M_5)_{i_4b} = 1$. Then $P_{ab} = \sum (\Xi_1)_{ai_1}(\Xi_3)_{i_2i_3}(\Xi_5)_{i_4b}$, summed over all paths, and distinct paths contribute distinct monomials in the entries of the masks. If entry (a, b) of the Boolean product is 0, there is no path, and $P_{ab} = 0$. Otherwise P_{ab} is a nonzero multilinear polynomial of degree three in the entries of the masks, so $P_{ab} \neq 0$ with probability at least $1/8$, by the argument in the proof of Lemma 5.32.

Let a' be the row of X that corresponds to row a of the first block, and let b' be the column that corresponds to column b of the last block. By Lemma 5.14 with $Y = X$, $x = e_{a'}$, and $w = e_{b'}$,

$$\operatorname{rank}\begin{pmatrix} X & e_{b'} \\ e_{a'}^{\top} & 0 \end{pmatrix} = m + \left[(X^{-1})_{a'b'} \neq 0\right] = m + \left[P_{ab} \neq 0\right].$$

We run nine independent copies of this construction, each with its own masks and its own instance of $\mathcal{R}$ on the bordered 0/1 matrix of dimension $\ell = m + 1$, whose border row and border column are initially zero. In preprocessing we flip the entries of each matrix X into place; there are $O(s^2)$ of them, so this takes polynomial time. An element update of M_2 or M_4 is one flip in every copy. A query for entry (a, b) sets, in every copy, the single nonzero entries of the border row and the border column, reads the rank, and resets them, which takes four flips; we answer 1 if and only if some copy reports rank $m + 1$. If entry (a, b) of the Boolean product is 0, this answer is wrong only if some copy of $\mathcal{R}$ errs; otherwise it is wrong only if all nine masks fail or some copy of $\mathcal{R}$ errs. So the error probability is at most $(7/8)^9 + 9/1000 < 1/3$. The problem is thus solved with polynomial preprocessing, worst-case time $O(\ell^{\alpha-\varepsilon}) = O(s^{\alpha-\varepsilon})$ per update and per query, and error probability less than $1/3$; by the proof of the five-matrix bound, this contradicts the uMv-hinted uMv conjecture. □

Van den Brand, Nanongkai, and Saranurak also state a lower bound for dynamic rank directly [58, Corollary 5.15]. Their rank instances, however, come from a reduction through bipartite matching that substitutes random elements of a large field for the entries, so they are not 0/1 matrices, and our gadget encodes only matrices with entries in $\{0, \pm 1\}$ (Theorem 4.6). This is why we go back to the five-matrix product, whose block matrix X has 0/1 entries. Over a field of characteristic zero the masks are not needed, because P_{ab} for Ξ_1, Ξ_3, Ξ_5 all-ones counts the paths, so that reduction is deterministic; in positive characteristic the number of paths can vanish, and the masks repair this. Corollary 5.16 below gives the randomized worst-case upper bound $O(n^{1.407})$. With the current bounds on matrix multiplication, the exponents $\alpha/2 \approx 0.7035$ and $\alpha \approx 1.407$ differ by exactly the factor two incurred by encoding a dense $\ell \times \ell$ matrix with $\Theta(\ell^2)$ simplices.

In the other direction, dynamic-rank algorithms yield dynamic-Betti algorithms.

Corollary 5.16. *Let $\mathbb{F}$ be a field and let $p \geq 0$ be an integer. There is a randomized (Monte Carlo) data structure for dynamic β_p over $\mathbb{F}$ that, for every fixed operation sequence of polynomial length, answers all queries correctly with high probability, with preprocessing time polynomial in n, worst-case update time $O(n^{1.407})$, and query time $O(1)$, where operations in $\mathbb{F}$, or, if $\mathbb{F}$ is small, in an extension field of $\mathbb{F}$ of size n^c for a large enough constant c, count as unit cost.*

Proof. The block matrix of Lemma 4.1 has at most n rows and at most n columns, and a simplex update changes at most $p + 2$ of its entries (Theorem 4.6(i)). Van den Brand, Nanongkai, and Saranurak [57] maintain the rank of an $\ell \times \ell$ matrix under entry updates in $O(\ell^{1.407})$ worst-case time per update, with a randomized (Monte Carlo) algorithm that obtains the rank from a dynamic determinant through a reduction of Sankowski [54]; over a small field the algorithm works in an extension field; a field of size n^c, for a large enough constant c, makes the error probability of every query polynomially small, so that all answers of a polynomially long operation sequence are correct with high probability. We run it on the block matrix, padded with zero rows and columns to size $n \times n$, and keep a counter for $n_p(K)$. Over $\mathbb{Q}$, the unit-cost assumption can be removed: the block matrix has entries in $\{0, \pm 1\}$, so by Hadamard's inequality each of its nonzero minors has absolute value at most $n^{n/2}$ and hence at most $O(n \log n)$ prime divisors. For a prime q drawn uniformly from the primes with $c \log n$ bits, for a large enough constant c, the rank modulo q therefore equals the rank over $\mathbb{Q}$ for all of the polynomially many matrices of a fixed operation sequence, with high

probability, and arithmetic modulo q takes constant time on a word RAM with words of $O(\log n)$ bits. □

For fully dynamic β_p, $p \geq 1$, the best upper bound that we know is thus the randomized $O(n^{1.407})$ worst-case bound of Corollary 5.16. Our lower bounds are $n^{1/2-o(1)}$ for amortized, and hence also for worst-case, update time from the OMv conjecture, for every data structure with query time $O(n^{1-\varepsilon})$ (Theorem 5.5), and $n^{\alpha/2-o(1)}$, with $\alpha/2 \geq 5/8$ and $\alpha/2 \approx 0.703$ for the current bounds on matrix multiplication, for worst-case update or query time over every field from the uMv-hinted uMv conjecture (Corollary 5.15). Over $\mathbb{F}_2$, Theorem 4.6 shows that closing the gap is equivalent to closing the corresponding gap for dynamic rank, with time measured in the instance size $N = r + k + m$ of Problem 3.2.

5.6 Bounded degree

In the universe $\mathcal{U}([\ell] \times [\ell])$, every row cycle C_i meets the tubes of ℓ sockets, so its vertices lie in $\Theta(\ell)$ simplices. We now replace each row cycle by a chain of copies, one for every column (Figure 6). The proof of Lemma 4.4 carries over by a telescoping argument. We call the triangle cycles $s_0^j, \ldots, s_{\ell+1}^j$ of column j in Definition 4.2 the *running cycles* of column j; consecutive running cycles are joined by a socket.

Definition 5.17 (Bounded-degree gadget). Let $\ell \geq 1$. The universe $\mathcal{U}^\flat = \mathcal{U}^\flat(\ell)$ is obtained from $\mathcal{U}([\ell] \times [\ell])$ of Definition 4.2 by the following changes; all new vertices are distinct.

- Each row cycle C_i is replaced by ℓ triangle cycles $C_i^1, \ldots, C_i^\ell$, the *copies* of C_i, and D_i is the filling of C_i^1.
- For every $(i, j) \in [\ell] \times [\ell]$, the tube of the socket of position (i, j) is the tube to C_i^j instead of C_i.
- For every $i \in [\ell]$ and $j \in [\ell - 1]$, the annulus $Q_i^j = \mathrm{Cyl}(C_i^j, C_i^{j+1})$ is added, together with its vertices and edges.

For $A \in \{0,1\}^{\ell \times \ell}$ and $R, J \subseteq [\ell]$, the complex $K^\flat(A; R, J) \subseteq \mathcal{U}^\flat$ consists of all vertices and edges of $\mathcal{U}^\flat$, the cones, the punctured cylinders, the tubes to the tethers, all annuli Q_i^j, the seal of the socket of (i, j) if $A_{ij} = 0$ and its tube to C_i^j if $A_{ij} = 1$, the filling D_i for every $i \notin R$, and the filling E_j for every $j \in J$.

Lemma 5.18. *Let $\mathbb{F}$ be a field, let $\ell \geq 1$, let $A \in \{0,1\}^{\ell \times \ell}$, and let $R, J \subseteq [\ell]$. Then*

$$\beta_2\big(K^\flat(A; R, J); \mathbb{F}\big) = |J| - \operatorname{rank}_{\mathbb{F}} A[R, J].$$

Proof. Let $K = K^\flat(A; R, J)$, and let Σ_j be the set of triangles of K that belong to column j, that is, its cones, punctured cylinders, seals, and tubes.

Incidences. Call a vertex *private* to column j if it lies on a running cycle s_t^j or on the tether Λ_j, or if it is the apex of a cone of column j. As in the proof of Lemma 4.3, Σ_j is a connected triangulated surface whose boundary is the disjoint union of Λ_j and the copies C_i^j with $A_{ij} = 1$. As there, every edge of K that has an endpoint private to column j and does not lie on Λ_j lies in no triangle of K or in exactly two triangles of K, both in Σ_j. Every other edge of K lies on a tether or joins two vertices of copies. An edge that joins two vertices of copies but does not lie on a copy joins C_i^j and C_i^{j+1} for some i and j, because only the annuli contain such edges, and it lies in exactly two triangles of K, both in Q_i^j. Every triangle of K lies in some Σ_j or in some Q_i^j, or it is a filling D_i with $i \notin R$ or E_j with $j \in J$.

Orientations. We orient all triangles as in Section 4.3, and we orient each annulus Q_i^j like every cylinder there. The computation in Section 4.3 does not depend on the cycles to which the tubes

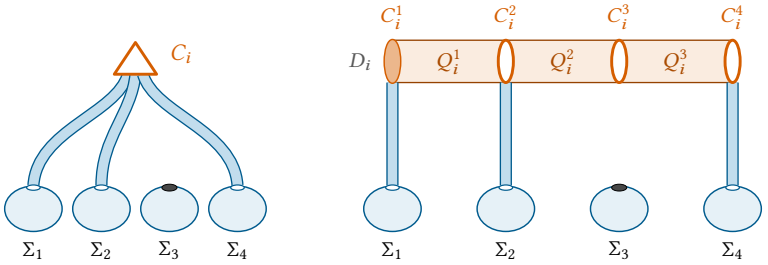


Fig. 6. Bounding the degree (Definition 5.17), shown for one row i with $A_{i1} = A_{i2} = A_{i4} = 1$ and $A_{i3} = 0$. (a) In the gadget of Definition 4.2, all ℓ sockets of row i have a tube to the row cycle C_i in the universe (only the tubes of the 1-entries are drawn), so every vertex of C_i lies in $\Theta(\ell)$ simplices. (b) In the bounded-degree gadget (here $\ell = 4$), C_i is replaced by a chain of copies $C_i^1, \dots, C_i^\ell$ joined by annuli Q_i^j; the socket of position (i, j) is tubed to its own copy C_i^j, and the row switch D_i fills the first copy. Now every vertex of a copy lies in $O(1)$ simplices and every edge of a copy in at most three triangles, and the identity $\beta_2 = |J| - \operatorname{rank} A[R, J]$ survives by telescoping along the chain (Lemma 5.18).

lead. It therefore yields a coherently oriented chain $[\Sigma_j]$ with $\partial[\Sigma_j] = [\Lambda_j] + \sum_{i:\, A_{ij}=1} [C_i^j]$, and a coherently oriented chain $[Q_i^j]$ with $\partial[Q_i^j] = [C_i^j] - [C_i^{j+1}]$. We orient the fillings so that $\partial D_i = [C_i^1]$ and $\partial E_j = [\Lambda_j]$.

Cycles. Let z be a 2-cycle of K. If an edge lies in exactly two triangles t, t' of K, both in Σ_j or both in Q_i^j, then t and t' induce opposite orientations on it, so its coefficient in ∂z is, up to sign, the difference of the coefficients of t and t' in z; hence these coefficients agree. The surface Σ_j is connected through such edges, by the last claim of Lemma 4.3, and so is Q_i^j, whose six triangles form a cyclic sequence in which consecutive triangles share an edge between C_i^j and C_i^{j+1}. Therefore

$$z = \sum_{j \in [\ell]} x_j [\Sigma_j] + \sum_{i \in [\ell]} \sum_{j \in [\ell-1]} y_{ij} [Q_i^j] + \sum_{i \notin R} a_i D_i + \sum_{j \in J} b_j E_j$$

for some scalars $x_j, y_{ij}, a_i, b_j \in \mathbb{F}$. We set $a_i = 0$ for $i \in R$, $b_j = 0$ for $j \notin J$, and $y_{i0} = y_{i\ell} = 0$ for all i. Then

$$\partial z = \sum_{j \in [\ell]} (x_j + b_j)\, [\Lambda_j] + \sum_{i \in [\ell]} \big(A_{i1} x_1 + y_{i1} + a_i\big)\, [C_i^1] + \sum_{i \in [\ell]} \sum_{j=2}^{\ell} \big(A_{ij} x_j + y_{ij} - y_{i,j-1}\big)\, [C_i^j].$$

The copies and the tethers have pairwise disjoint edge sets, so $\partial z = 0$ holds if and only if all coefficients vanish.

Solving. The coefficients of the tethers vanish if and only if $x_j = 0$ for $j \notin J$ and $b_j = -x_j$ for $j \in J$. Fix a row i. The sum of the coefficients of $C_i^1, \dots, C_i^\ell$ is $\sum_j A_{ij} x_j + a_i$, because the terms y_{ij} cancel. So if all coefficients vanish, then $\sum_{j \in J} A_{ij} x_j = 0$ for $i \in R$, and $a_i = -\sum_j A_{ij} x_j$ for $i \notin R$. Moreover, $y_{ij} = -a_i - \sum_{j' \le j} A_{ij'} x_{j'}$ for $j \in [\ell - 1]$. Conversely, let $x \in \mathbb{F}^\ell$ satisfy $x_j = 0$ for $j \notin J$ and $\sum_{j \in J} A_{ij} x_j = 0$ for $i \in R$. If we define a_i, b_j, and y_{ij} by these formulas, then the coefficients of Λ_j and of $C_i^1, \dots, C_i^{\ell-1}$ vanish by construction, and the coefficient of C_i^ℓ vanishes because the sum of all coefficients of row i is $\sum_j A_{ij} x_j + a_i = 0$. Hence $z \mapsto (x_j)_{j \in J}$ is a linear bijection from $Z_2(K)$ onto the kernel of $A[R, J]$. Since $\dim K = 2$, we get $\beta_2(K) = \dim Z_2(K) = |J| - \operatorname{rank}_{\mathbb{F}} A[R, J]$. □

Theorem 5.19. *Let $\mathbb{F}$ be a field, let $p \in \{1,2\}$, and let $\varepsilon > 0$. Let $\mathcal{B}$ be the class of 2-dimensional universes in which every vertex lies in at most 37 simplices and every edge lies in at most three triangles. If the OMv conjecture holds, then the following statements hold for universes in $\mathcal{B}$.*

(i) No data structure for dynamic β_p over $\mathbb{F}$ has polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$.

(ii) No data structure for incremental dynamic β_p over $\mathbb{F}$, and no data structure for decremental dynamic β_p over $\mathbb{F}$, has polynomial preprocessing time, worst-case update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$.

(iii) No data structure for Problem 5.8 over $\mathbb{F}$ has polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$.

Proof. For $\ell \geq 1$ and $M \in \{0,1\}^{\ell\times\ell}$, the universe $\mathcal{U}^\flat(\ell)$ together with the complexes $K_{R,J} = K^\flat(M;R,J)$ is an encoding of M over $\mathbb{F}$. Indeed, (E1) holds by Definition 5.17, (E2) is Lemma 5.18, and (E3) holds with the fillings D_i and E_j, as in $\mathcal{U}([\ell]\times[\ell])$. The universe $\mathcal{U}^\flat(\ell)$ has $\Theta(\ell^2)$ simplices, since each position (i,j) contributes one socket, one copy, and at most one annulus, each of constant size, and it can be constructed in time $O(\ell^2)$. The proofs of Theorem 5.5 and of Theorem 5.12 for $p \in \{1,2\}$ use only these properties, so they apply to these encodings and prove (i) and (ii) for the universes $\mathcal{U}^\flat(\ell)$. Part (iii) follows from (i) and Lemma 5.9, which does not change the universe. It remains to show that $\mathcal{U}^\flat(\ell) \in \mathcal{B}$.

Triangles at a vertex. A vertex of a running cycle s_t^j lies in at most three triangles of each of the at most two cylinders or cones that contain s_t^j, where a seal counts as a triangle of its cylinder. It also lies in three triangles of each of the at most two tubes whose hole contains it. A vertex of a copy C_i^j lies in three triangles of the tube to C_i^j, in three triangles of each of the at most two annuli Q_i^{j-1} and Q_i^j, and possibly in D_i. A vertex of a tether lies in three triangles of its tube and in its filling, and the apex of a cone lies in three triangles. So every vertex lies in at most 12 triangles. Every edge of $\mathcal{U}^\flat(\ell)$ lies in a triangle, so a vertex in t triangles lies in at most $2t$ edges. Hence every vertex lies in at most $1 + 24 + 12 = 37$ simplices.

Triangles at an edge. An edge of a running cycle lies in at most one triangle of each of the two cylinders or cones that contain the cycle and, if it is an edge of a hole, in one triangle of the tube of that hole. An edge that joins two consecutive running cycles lies in at most two triangles of their cylinder and, if it is an edge of a hole, in one triangle of the tube. An edge of a copy C_i^j lies in one triangle of the tube to C_i^j, in one triangle of each annulus Q_i^{j-1} and Q_i^j that exists, and, if $j = 1$, in D_i; since Q_i^0 does not exist, these are at most three triangles. An edge of a tether lies in one triangle of its tube and in its filling. Every remaining edge joins the apex of a cone to its base, or a hole to the cycle at the far end of its tube, or two consecutive copies, and lies in exactly two triangles. So every edge lies in at most three triangles. □

Entry flips are also local in $\mathcal{U}^\flat(\ell)$: flipping A_{ij} exchanges the seal of the socket of position (i,j) with its tube to C_i^j, which takes a constant number of triangle updates, and Lemma 5.18 holds for every matrix A. Hence the reduction in the proof of Theorem 4.6(ii), with Lemma 5.18 in place of Lemma 4.4, also works on the universes $\mathcal{U}^\flat(\ell)$, and so does the proof of Theorem 5.13 for $p \in \{1,2\}$. Since the boundary matrices of these universes are sparse, the bounds also transfer back to dynamic rank of sparse matrices.

Corollary 5.20. *For every fixed $p \geq 3$, the bounds of Theorem 5.5 hold for dynamic β_p over every field $\mathbb{F}$ on universes of dimension p in which every vertex lies in at most κ_p simplices, where κ_p depends only on p.*

Proof. Suspension does not preserve bounded degree, because each of its two apexes lies in $\Theta(n)$ simplices. We use a product with a sphere instead. Let $S = \partial\Delta^{p-1}$ be the boundary of a $(p-1)$-simplex, a triangulated $(p-2)$-sphere with a constant number of simplices. Fix a total order on the vertices of the bounded-degree universe $\mathcal{U}^\flat$ of Theorem 5.19 and on the vertices of S, and let $\mathcal{U}^\flat \times S$ be the standard (staircase) triangulation of the product of the two ordered complexes [49]. Its simplices are the chains of the product of the two vertex orders whose projections to the factors are simplices. Restricted to the simplices whose first projection lies in a subcomplex $K \subseteq \mathcal{U}^\flat$, it is a triangulation of $\|K\| \times \|S\|$, which we denote $K \times S$. Every vertex of $\mathcal{U}^\flat \times S$ lies in a number of simplices bounded in terms of p and the maximum number of simplices at a vertex of $\mathcal{U}^\flat$, and a simplex update of K corresponds to inserting or deleting the boundedly many simplices of $K \times S$ that project onto it, in order of increasing dimension for insertions and decreasing dimension for deletions. By the Künneth formula over a field [32, Section 3.B], $H_p(K \times S) \cong \bigoplus_{i+j=p} H_i(K) \otimes H_j(S)$. Since $\dim K = 2 < p$ and $H_j(S) \neq 0$ only for $j \in \{0, p-2\}$, we get $\beta_p(K \times S) = \beta_2(K)\,\beta_{p-2}(S) = \beta_2(K)$. So the reduction of Theorem 5.19 carries over with a constant factor more updates and a constant factor larger universe. □

Corollary 5.21. *Let $\mathbb{F}$ be a field and let $\varepsilon > 0$. If the OMv conjecture holds, then no data structure for Problem 3.2 over $\mathbb{F}$ has polynomial preprocessing time, amortized update time $O(N^{1/2-\varepsilon})$, and query time $O(N^{1-\varepsilon})$, where N is the instance size. This holds even for instances in which every row and every column contains at most three admissible positions, and for operation sequences without switches in which every entry update $\textsc{Set}(i, j, x)$ has $x \in \{-1, 0, 1\}$.*

Proof. Suppose that $\mathcal{D}$ is such a data structure. We build a data structure for dynamic β_2 over $\mathbb{F}$ on the universes $\mathcal{U}^\flat(\ell)$. Let B be the boundary matrix of $\mathcal{U}^\flat(\ell)$ in dimension 2, with rows indexed by the edges and columns indexed by the triangles, for the orientations used in Lemma 5.18. Its entries lie in $\{-1, 0, 1\}$. Every column of B has three nonzero entries, and every row has at most three, as shown in the proof of Theorem 5.19. The rank instance has $r = n_1(\mathcal{U}^\flat)$ rows, $k = n_2(\mathcal{U}^\flat)$ columns, and the admissible set $P = \operatorname{supp}(B)$, so its size is $N = O(n)$. We use no switches, so all rows and columns stay active, and a rank query returns the rank of the whole matrix. We maintain the matrix B^K obtained from B by replacing the columns of the triangles not in K with zero columns. Initially $K = \emptyset$ and $B^K = 0$. Inserting a triangle sets the three entries of its column to their values in B, deleting a triangle sets them to 0, and inserting or deleting a vertex or an edge does not change B^K. Since $\mathcal{U}^\flat$ has no 3-simplices, Lemma 4.1 gives $\beta_2(K) = n_2(K) - \operatorname{rank} B^K$, so a query is answered with one rank query and a counter for $n_2(K)$. The resulting data structure has polynomial preprocessing time, amortized update time $O(n^{1/2-\varepsilon})$, and query time $O(n^{1-\varepsilon})$. By the proof of Theorem 5.19(i), no such data structure exists on the universes $\mathcal{U}^\flat(\ell)$. □

Corollary 5.21 also follows without topology. Replace the equation of row i by the chain $y_{i,j} - y_{i,j-1} - A_{ij}x_j^{(i)} = 0$ for $j \in [\ell]$, with $y_{i,0} = 0$, tie the copies of each column by $x_j^{(i)} - x_j^{(i-1)} = 0$, and realize the row switch of i by the single-entry row $y_{i,\ell} = 0$ and the column switch of j by the single-entry row $x_j^{(1)} = 0$. Every row and every column then has at most three nonzero entries, all in $\{0, \pm 1\}$, the kernel is isomorphic to that of $A[R, J]$, and every operation changes $O(1)$ entries; together with Lemma 5.1, this proves Corollary 5.21. The gadget of Definition 5.17 yields, in addition, the bounded-degree complexes of Theorem 5.19.

Three admissible positions per row and column is the smallest number for which Corollary 5.21 can hold. Statically, Edelsbrunner and Parsa [26, Theorem 4.1], in a special case of a sparsification of Yuster [61], reduce the rank of a 0/1 matrix to the rank of a 0/1 matrix with at most three nonzero entries in every row and every column, by applying their construction twice; Corollary 5.21 is a

dynamic counterpart, and two admissible positions per column already make the problem easy. They also remark that their complexes, in which every edge lies in at most three triangles, are already as hard as sparse rank [26, remark after Theorem 4.1], the static analogue of the class $\mathcal{B}$ of Theorem 5.19.

Proposition 5.22. *Let $\mathbb{F}$ be a field. If every column, or every row, has at most two admissible positions, then Problem 3.2 over $\mathbb{F}$ restricted to entries in $\{-1, 0, 1\}$ can be solved with $O(\log^2 N)$ amortized time per operation.*

Proof. Let $c(\cdot)$ denote the number of connected components of a graph. First suppose that every column has at most two admissible positions. Let Γ be the multigraph on the row set $[r]$ with an edge between the two nonzero rows of every active column that has two nonzero entries, a half-edge at the nonzero row of every active column that has one, and a half-edge at every inactive row. Call an edge *positive* if its two entries have opposite signs or $\mathbb{F}$ has characteristic 2, and *negative* otherwise. Then $\operatorname{rank} A[R, J] = \operatorname{rank}\big(A[[r], J] \;\big|\; I_{[r]\setminus R}\big) - (r - |R|)$, and the left kernel of the matrix on the right consists of the vectors $y \in \mathbb{F}^r$ with $y_i = y_{i'}$ on positive edges, $y_i = -y_{i'}$ on negative edges, and $y_i = 0$ at half-edges. Its dimension is the number of *balanced* components of Γ, those without half-edges in which every cycle has an even number of negative edges. In the double cover $\widetilde{\Gamma}$, with nodes i^+, i^-, where a positive edge ii' lifts to $i^+i'^+$ and $i^-i'^-$, a negative edge to $i^+i'^-$ and $i^-i'^+$, and a half-edge at i to i^+i^-, a component of Γ lifts to two components if it is balanced and to one otherwise. Hence $\operatorname{rank} A[R, J] = |R| - c(\widetilde{\Gamma}) + c(\Gamma)$. Every operation of Problem 3.2 changes $O(1)$ edges of Γ and $\widetilde{\Gamma}$, so dynamic connectivity [36] maintains the rank in $O(\log^2 N)$ amortized time per operation; since $\operatorname{rank} A = \operatorname{rank} A^\top$, the same holds if every row has at most two admissible positions. □

This is the matrix counterpart of the contrast between β_1 of graphs, whose boundary matrices have two nonzero entries per column, and β_1 of 2-complexes.

Over $\mathbb{F}_2$, the matrices in Corollary 5.21 are 0/1 matrices.

5.7 Universes embedded in $\mathbb{R}^4$

Edelsbrunner and Parsa [26] observed that their static reduction produces 2-complexes that embed in $\mathbb{R}^4$, whereas Betti numbers of complexes in $\mathbb{R}^3$ can be computed in near-linear time [16, 18]. We show that the dynamic lower bound also holds in $\mathbb{R}^4$: the universes $\mathcal{U}(P)$ of Definition 4.2 have linear embeddings in $\mathbb{R}^4$, with all row cycles in a common 2-plane and each column surface in its own *page*, a closed half of a 3-dimensional affine subspace bounded by that plane (Figure 7). Section 6.2 shows that no such lower bound can hold in $\mathbb{R}^3$. This open-book layout is that of Edelsbrunner and Parsa [26, Section 3], who place the cycles of their complex in a 2-plane of $\mathbb{R}^4$ and every punctured sphere in its own 3-dimensional subspace, taken from a pencil of subspaces through that plane. What is new here is that each page carries a column surface assembled from sockets, with both attachments of every socket, the tether, and its filling, at explicit rational coordinates.

A *geometric simplex* in $\mathbb{R}^d$ is the convex hull of a set of affinely independent points, its *vertices*; its *faces* are the convex hulls of the subsets of its vertices, including the empty face. A *geometric simplicial complex* in $\mathbb{R}^d$ is a finite set X of geometric simplices that contains all faces of its members and in which any two simplices intersect in a common face. Its underlying space $\|X\|$ is the union of its simplices. A *linear embedding* of a simplicial complex K in $\mathbb{R}^d$ is an injective map f from the vertices of K to $\mathbb{R}^d$ such that, for every $\sigma \in K$, the points $f(\sigma)$ are affinely independent, and the simplices $\operatorname{conv} f(\sigma)$, $\sigma \in K$, form a geometric simplicial complex. For two geometric simplicial complexes X and Y, the set $X \cap Y$ of their common simplices is a geometric simplicial complex.

Lemma 5.23 (Gluing). *Let X and Y be geometric simplicial complexes in $\mathbb{R}^d$ such that*

$$\|X\| \cap \|Y\| = \|X \cap Y\|.$$

Then $X \cup Y$ is a geometric simplicial complex.

Proof. The set $X \cup Y$ contains all faces of its members. Let $\sigma \in X$ and $\tau \in Y$. Since $\sigma \cap \tau \subseteq \|X\| \cap \|Y\| = \|X \cap Y\|$, every point of $\sigma \cap \tau$ lies in some $\rho \in X \cap Y$. For such ρ, the set $\sigma \cap \rho$ is a common face of σ and ρ, because both lie in X, and $\tau \cap \rho$ is a common face of τ and ρ, because both lie in Y. Both are faces of ρ, so both lie in $X \cap Y$, and their intersection $\sigma \cap \tau \cap \rho$ is a common face of the two. Hence $\sigma \cap \tau \cap \rho$ is a face of σ and a face of τ. So $\sigma \cap \tau$ is a union of common faces of σ and τ. Let V be the set of vertices of these faces. Then $\sigma \cap \tau \subseteq \operatorname{conv} V$, and $\operatorname{conv} V \subseteq \sigma \cap \tau$ because $\sigma \cap \tau$ is convex and contains V. So $\sigma \cap \tau = \operatorname{conv} V$, which is a common face of σ and τ, because V consists of common vertices. □

One column. Fix $P \subseteq [r] \times [k]$ and a column $j \in [k]$, and let $i_1 < \cdots < i_d$ be the elements of S_j, where $d = d_j$. We place the simplices of $\mathcal{U}(P)$ that belong to column j in $\mathbb{R}^3$, with coordinates (x, y, z). We write the vertices of the triangle cycle s_t^j as $s_{t,0}, s_{t,1}, s_{t,2}$, and the vertices of a triangle cycle C as $C_{(0)}, C_{(1)}, C_{(2)}$. Let $\delta_0 = (0, 1)$, $\delta_1 = (1, 1)$, $\delta_2 = (\frac{1}{2}, 2)$, and $\eta_0 = (0, 0)$, $\eta_1 = (0, 1)$, $\eta_2 = (1, 0)$. For $t \in \{0, \dots, d+1\}$ and $u \in \mathbb{Z}_3$, we place

$$s_{t,u} = (t, \delta_u), \qquad (\Lambda_j)_{(u)} = \left(\tfrac{1}{2}\eta_u, \tfrac{1}{2}\right), \qquad (C_i)_{(u)} = \left((i, 0) + \tfrac{1}{2}\eta_u,\ 0\right) \ \ (i \in [r]),$$

and the apexes of the cones on s_0^j and on s_{d+1}^j at $(-1, \frac{1}{2}, \frac{3}{2})$ and $(d+2, \frac{1}{2}, \frac{3}{2})$. For $0 \le t \le d$, the hole of the socket between s_t^j and s_{t+1}^j is the triangle cycle $h_t = (s_{t,0}, s_{t,1}, s_{t+1,0})$, and its vertices are the points $((t, 0) + \eta_u, 1)$, $u \in \mathbb{Z}_3$. Let W be the set of the following geometric simplices and all their faces: the six triangles of $\operatorname{Cyl}(s_t^j, s_{t+1}^j)$ for $0 \le t \le d$, the two cones, the tube $\operatorname{Cyl}(h_0, \Lambda_j)$, the tubes $\operatorname{Cyl}(h_t, C_{i_t})$ for $t \in [d]$, and the filling E_j of Λ_j. Apart from the seal of h_0, which is not in $\mathcal{U}(P)$, these are exactly the simplices of $\mathcal{U}(P)$ that contain a vertex of column j, together with all their faces.

Lemma 5.24. *The set W is a geometric simplicial complex in the half-space $\{z \ge 0\}$, and $\|W\| \cap \{z = 0\} = \bigcup_{t \in [d]} \|C_{i_t}\|$.*

Proof. We assemble W from three parts with Lemma 5.23.

The cylinders and cones. Let $\Delta \subset \mathbb{R}^2$ be the triangle with vertices $\delta_0, \delta_1, \delta_2$, write $[\delta_u, \delta_{u+1}]$ for the segment between δ_u and δ_{u+1}, and let $R_u = [0, d+1] \times [\delta_u, \delta_{u+1}]$, $u \in \mathbb{Z}_3$, be the lateral faces of the prism $[0, d+1] \times \Delta$. The triangles $s_{t,u}s_{t,u+1}s_{t+1,u}$ and $s_{t,u+1}s_{t+1,u}s_{t+1,u+1}$ of the cylinders cut the rectangle $[t, t+1] \times [\delta_u, \delta_{u+1}]$ along a diagonal. So, for fixed u, the cylinder triangles in R_u triangulate R_u. Two distinct faces R_u meet in a segment $[0, d+1] \times \{\delta\}$, where δ is a common vertex of their edges of Δ, and both triangulations subdivide this segment at the integer points. By Lemma 5.23, applied twice, the cylinder triangles and their faces form a geometric simplicial complex Y_0. The cone on s_0^j consists of three facets of the tetrahedron spanned by $(-1, \frac{1}{2}, \frac{3}{2})$ and s_0^j, so it is a geometric simplicial complex. It lies in $\{x \le 0\}$, the complex Y_0 lies in $\{x \ge 0\}$, and both meet the plane $\{x = 0\}$ exactly in the boundary of the triangle spanned by s_0^j. By Lemma 5.23, and by the symmetric argument for the cone on s_{d+1}^j, the cylinders and the cones form a geometric simplicial complex Y. It lies in $\{1 \le z \le 2\}$, and the edges of every hole h_t belong to Y.

The tubes. Let Δ' be the triangle with vertices η_0, η_1, η_2; its projection to the x-axis is $[0, 1]$. For $t \in [d]$ let F_t be the convex hull of h_t and C_{i_t}. Its bottom face, spanned by C_{i_t}, is a copy of its top face, spanned by h_t, scaled by $\frac{1}{2}$ and translated, with the vertices $(h_t)_{(u)}$ and $(C_{i_t})_{(u)}$ corresponding. So F_t is a frustum of a triangular pyramid. Its lateral facets are the three trapezoids spanned by

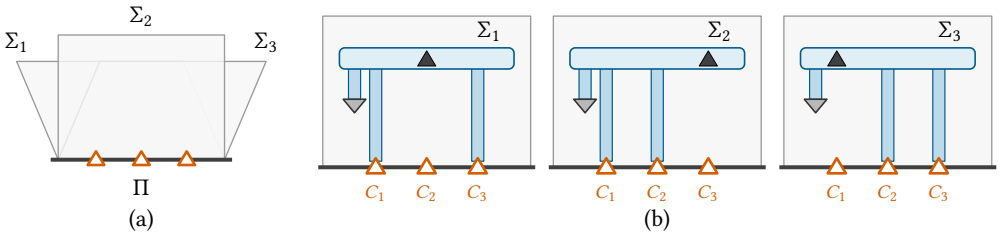


Fig. 7. The open-book embedding of Theorem 5.25, an explicit linear version of the pencil-of-3-planes embedding of Edelsbrunner and Parsa [26], drawn schematically for the 3×3 matrix A with columns $(1, 0, 1)$, $(1, 1, 0)$, and $(0, 1, 1)$. (a) The row cycles C_1, C_2, C_3 lie in the binding plane Π, and each column surface Σ_j lies in its own page, a closed half of a 3-dimensional affine subspace of $\mathbb{R}^4$ bounded by Π; distinct pages meet only in Π. (b) Each page is drawn flat, with its bottom edge representing Π. In the page of Σ_j, a tube joins the hole of the socket of position (i, j) to C_i if $A_{ij} = 1$, and a seal (black) closes the hole if $A_{ij} = 0$; a short tube leads to the private tether Λ_j (gray). So different surfaces meet only along the row cycles. Here $P = [3] \times [3]$; in the universe $\mathcal{U}(P)$, the hole of every socket has both its seal and its tube.

$(h_t)_{(u)}, (h_t)_{(u+1)}, (C_{i_t})_{(u+1)}, (C_{i_t})_{(u)}$, and two of them meet exactly in a common lateral edge. The six triangles of $\mathrm{Cyl}(h_t, C_{i_t})$ are obtained by cutting each trapezoid along its diagonal from $(h_t)_{(u+1)}$ to $(C_{i_t})_{(u)}$, so they form a geometric simplicial complex. A point of F_t at height $z \in [0, 1]$ is a convex combination $zp + (1 - z)q$ with p in the top face and q in the bottom face, so the horizontal section of F_t at height z is

$$\tfrac{1+z}{2}\,\Delta' + \big(x_t(z), 0\big), \qquad \text{where } x_t(z) = zt + (1 - z)\,i_t. \tag{6}$$

In the same way, the convex hull F_0 of h_0 and Λ_j is a frustum whose section at height $z \in [\frac{1}{2}, 1]$ is $z\Delta'$, and $\mathrm{Cyl}(h_0, \Lambda_j)$ is a geometric simplicial complex.

The tubes meet only at shared vertices of holes. For $z < 1$ the section (6) projects to the x-interval $[x_t(z), x_t(z) + \frac{1+z}{2}]$, whose length is less than one. For $1 \le t < t' \le d$ we have $t' - t \ge 1$ and $i_{t'} - i_t \ge 1$, so $x_{t'}(z) - x_t(z) = z(t' - t) + (1 - z)(i_{t'} - i_t) \ge 1$. Hence F_t and $F_{t'}$ can meet only at height $z = 1$, where they meet in $\operatorname{conv} h_t \cap \operatorname{conv} h_{t'}$. This is the point $s_{t+1,0}$ if $t' = t + 1$, and empty otherwise. Similarly, the section of F_0 at height $z \in [\frac{1}{2}, 1)$ projects into $[0, z]$, while $x_t(z) \ge 1$ for $t \ge 1$. So F_0 meets F_t only in the point $s_{1,0}$, and only if $t = 1$. These points are common vertices of the tubes involved. By Lemma 5.23, applied once for each tube, the tubes and their faces form a geometric simplicial complex Z.

Gluing. The complex Y lies in $\{z \ge 1\}$, and Z lies in $\{z \le 1\}$. Each tube meets the plane $\{z = 1\}$ exactly in the boundary of the triangle spanned by its hole, and the edges of this boundary belong to Y and to Z. So $\|Y\| \cap \|Z\| = \|Y \cap Z\|$, and $Y \cup Z$ is a geometric simplicial complex by Lemma 5.23. The filling E_j is the bottom face of F_0 at height $\frac{1}{2}$. It does not meet Y, it meets $\mathrm{Cyl}(h_0, \Lambda_j)$ exactly in $\|\Lambda_j\|$, and it does not meet F_t for $t \ge 1$, because E_j projects into $[0, \frac{1}{2}]$ while the section of F_t at height $\frac{1}{2}$ projects into $[1, \infty)$. A final application of Lemma 5.23 shows that W is a geometric simplicial complex. All its vertices have $z \ge 0$, so W lies in $\{z \ge 0\}$. Only the tubes $\mathrm{Cyl}(h_t, C_{i_t})$ with $t \in [d]$ reach height 0, and they meet $\{z = 0\}$ exactly in $\|C_{i_t}\|$. □

The open book. For $j \in [k]$ let $\varphi_j \colon \mathbb{R}^3 \to \mathbb{R}^4$ be the linear map $\varphi_j(x, y, z) = (x, y, z, jz)$, and let $\Pi = \{(x, y, 0, 0) : x, y \in \mathbb{R}\}$. Each φ_j is injective and fixes the plane $\{z = 0\}$ in the sense that $\varphi_j(x, y, 0) = (x, y, 0, 0)$. If $j \ne j'$ and $\varphi_j(x, y, z) = \varphi_{j'}(x', y', z')$, then $x = x'$, $y = y'$, $z = z'$, and

$jz = j'z$, so $z = 0$. Hence the images of distinct maps φ_j meet exactly in Π. Figure 7 shows the resulting embedding.

Theorem 5.25.

(i) For all $r, k \geq 1$ and every $P \subseteq [r] \times [k]$, the complex $\mathcal{U}(P)$ of Definition 4.2 has a linear embedding in $\mathbb{R}^4$ with rational coordinates, and such an embedding can be computed in $O(r + k + |P|)$ time.

(ii) Assume the OMv conjecture, and let $\mathbb{F}$ be a field, $p \in \{1, 2\}$, and $\varepsilon > 0$. No data structure with polynomial preprocessing time solves Problem 3.1 for β_p over $\mathbb{F}$ with amortized update time $O(n^{1/2-\varepsilon})$ and query time $O(n^{1-\varepsilon})$, even if the universe is a 2-dimensional geometric simplicial complex in $\mathbb{R}^4$ that is given with the coordinates of its vertices.

Proof. (i) For every column $j \in [k]$, let W_j be the complex of Lemma 5.24, and let W'_j be W_j without the seal of h_0. Removing a triangle that is a face of no other simplex keeps a geometric simplicial complex, so W'_j is one. Since φ_j is linear and injective, $\varphi_j(W'_j)$ is a geometric simplicial complex in $\mathbb{R}^4$. The position of a vertex of a row cycle does not depend on the column, because $\varphi_j(x, y, 0) = (x, y, 0, 0)$ for every j. We place the filling D_i of C_i in Π, for every $i \in [r]$. The triangles D_i are pairwise disjoint, because D_i projects to the x-interval $[i, i + \frac{1}{2}]$. Let X be the union of the complexes $\varphi_j(W'_j)$, $j \in [k]$, and of the triangles D_i, $i \in [r]$, with their faces.

We show by Lemma 5.23 that X is a geometric simplicial complex, adding one column at a time and then the triangles D_i. By Lemma 5.24, the complex $\varphi_j(W'_j)$ meets Π exactly in $\bigcup_{i \in S_j} \|C_i\|$. The images of distinct maps φ_j meet only in Π. So, for $j \geq 2$, the union of the first $j - 1$ columns meets $\varphi_j(W'_j)$ exactly in $\bigcup \|C_i\|$, over the rows $i \in S_j$ that also lie in $S_{j'}$ for some $j' < j$. This set is the underlying space of the common simplices, namely the vertices and edges of these row cycles. Likewise, the union of all columns meets the triangles D_i exactly in the row cycles that occur in some column, and these consist of common simplices. So X is a geometric simplicial complex.

The vertex positions are injective: the vertices of column j that are not on a row cycle have $z \geq \frac{1}{2}$, so they are mapped into the image of φ_j minus Π, and distinct vertices of W_j have distinct positions. The simplices of X are the images of the simplices of $\mathcal{U}(P)$: those that contain a vertex of column j come from W'_j, and the remaining ones are the row cycles and their fillings. So the vertex positions form a linear embedding of $\mathcal{U}(P)$. All coordinates are rational, and each is computed in $O(1)$ time; $\mathcal{U}(P)$ has $O(r + k + |P|)$ simplices.

(ii) The proof of Theorem 5.5 uses only universes of the form $\mathcal{U}([\ell] \times [\ell])$ (Lemma 5.3). By (i), the coordinates of a linear embedding of such a universe in $\mathbb{R}^4$ can be computed during preprocessing, in linear time, and added to the instance. So the proof of Theorem 5.5 applies without change. □

Corollary 5.26. *Let $\mathbb{F}$ be a field. Dynamic rank over $\mathbb{F}$ of 0/1 matrices reduces with overhead $(7, 1)$ to dynamic β_1, and to dynamic β_2, over $\mathbb{F}$ on 2-complexes given with a linear embedding in $\mathbb{R}^4$. Over $\mathbb{F}_2$, these problems are equivalent to dynamic rank with constant overhead.*

Proof. The reduction of Theorem 4.6(ii) uses the universe $\mathcal{U}(P)$, and by Theorem 5.25(i) its linear embedding can be computed in linear time and added to the instance. Conversely, dynamic β_p on every universe, embedded or not, reduces to dynamic rank by Theorem 4.6(i), and over $\mathbb{F}_2$ every matrix is a 0/1 matrix. □

Over other fields we do not know whether the universes $\mathcal{U}^{\pm}(P)$ with reversed tubes, which encode $\{0, \pm 1\}$-matrices, embed in $\mathbb{R}^4$.

5.8 Clique complexes and Vietoris–Rips complexes

Many complexes in topological data analysis are flag complexes, Vietoris–Rips complexes for example. The *clique complex* $\mathrm{Cl}(G)$ of a graph G has one simplex for every nonempty clique of G. The *Vietoris–Rips complex* $\mathrm{VR}_\varrho(X, \mathrm{dist})$ of a finite metric space (X, dist) at scale ϱ is the clique complex of the graph that joins two points at distance at most ϱ; for a finite point set $X \subset \mathbb{R}^d$ with the Euclidean metric we write $\mathrm{VR}_\varrho(X)$. Such a complex is determined by its graph, so the natural updates insert or delete a vertex together with its incident edges, or insert or delete a point. One such update may add or remove many simplices at once. We show that the lower bounds survive in these models.

Problem 5.27 (Dynamic clique-complex Betti numbers). Fix a field $\mathbb{F}$ and $p \geq 0$. The instance is a graph $G = (V, E)$ with n vertices. The data structure maintains a set $W \subseteq V$, initially empty, under insertion and deletion of single vertices, and on query returns $\beta_p(\mathrm{Cl}(G[W]); \mathbb{F})$, where $G[W]$ is the subgraph induced by W.

Problem 5.28 (Dynamic Vietoris–Rips Betti numbers). Fix a field $\mathbb{F}$ and an integer $p \geq 0$. The instance is a dimension $d \geq 1$ and a set $X_{\mathcal{U}} \subset \mathbb{Q}^d$ of n points. The data structure maintains a set $X \subseteq X_{\mathcal{U}}$, initially empty, under insertion and deletion of single points, and on query returns $\beta_p(\mathrm{VR}_1(X); \mathbb{F})$.

Theorem 5.29. *Assume the OMv conjecture, and let $\mathbb{F}$ be a field and $\varepsilon > 0$. For each of the following problems, no data structure with polynomial preprocessing time has amortized update time $O(n^{1/2-\varepsilon})$ and query time $O(n^{1-\varepsilon})$. In each problem, a query returns the first Betti number over $\mathbb{F}$ of the complex named.*

- *(i) The clique complex $\mathrm{Cl}(G[W])$, where G is a fixed graph with n vertices and maximum degree bounded by an absolute constant, and an update inserts a vertex into W or deletes one from it (Problem 5.27 with $p = 1$).*
- *(ii) The clique complex $\mathrm{Cl}(H)$, where H is a spanning subgraph of a fixed graph G as in (i), and an update inserts an edge of G into H or deletes one from it.*
- *(iii) The Vietoris–Rips complex $\mathrm{VR}_1(X, \mathrm{dist})$, where X is a fixed set of n points, dist is a metric on X with values in $\{1, 2\}$, and an update changes the distance between one pair of points.*
- *(iv) The Vietoris–Rips complex $\mathrm{VR}_1(X)$ of Problem 5.28 with $d = O(\log n)$, even if all coordinates of the points have $O(\log n)$ bits.*

The proof rests on barycentric subdivision. For a simplicial complex $\mathcal{U}$ let $G_{\mathcal{U}}$ be the *comparability graph* of its face poset. Its vertices are the simplices of $\mathcal{U}$, and two simplices are adjacent when one is a proper face of the other. For a subcomplex $K \subseteq \mathcal{U}$, the induced subgraph $G_{\mathcal{U}}[K]$ is the comparability graph of the face poset of K. Its cliques are exactly the chains $\sigma_0 \subsetneq \cdots \subsetneq \sigma_q$ of simplices of K. So $\mathrm{Cl}(G_{\mathcal{U}}[K])$ is the order complex of the face poset of K, which is the barycentric subdivision $\mathrm{sd}(K)$ [42]. Since $\mathrm{sd}(K)$ is homeomorphic to K,

$$\beta_p\big(\mathrm{Cl}(G_{\mathcal{U}}[K])\big) \;=\; \beta_p(K) \qquad \text{for every subcomplex } K \subseteq \mathcal{U} \text{ and every } p. \tag{7}$$

Proof of Theorem 5.29. Let $\mathcal{U}$ be the bounded-degree universe of Theorem 5.19. It has n simplices, every vertex lies in $O(1)$ simplices, and the lower bound of Theorem 5.19(i) for β_1 holds on it. In $G_{\mathcal{U}}$ a simplex is adjacent only to its faces and to the simplices that contain it, so $G_{\mathcal{U}}$ has maximum degree $O(1)$.

For (i) we maintain $W = K$. Inserting or deleting a simplex σ of K becomes inserting or deleting the vertex σ of W. By (7) the queries agree.

For (ii) we maintain the spanning subgraph H of $G_{\mathcal{U}}$ whose edges are the edges of $G_{\mathcal{U}}[K]$; the simplices of $\mathcal{U}$ that are not in K are isolated vertices of H. When σ is inserted into K, all its proper

faces lie in K and none of its cofaces does, so the new edges of H are the edges between σ and its proper faces; there are at most six of them, because $\dim \mathcal{U} = 2$. A deletion removes the same edges. So every simplex update becomes at most six edge updates, and we query only after all of them have been carried out. At that moment $\mathrm{Cl}(H)$ is the disjoint union of $\mathrm{Cl}(G_{\mathcal{U}}[K])$ and some isolated vertices, so $\beta_1(\mathrm{Cl}(H)) = \beta_1(K)$ by (7).

For (iii) we take X to be the vertex set of $G_{\mathcal{U}}$ and set $\mathrm{dist}(x, x') = 1$ if xx' is an edge of the graph H of (ii), and $\mathrm{dist}(x, x') = 2$ for all other pairs of distinct points. Every symmetric function with values in $\{1, 2\}$ satisfies the triangle inequality, because $2 \le 1 + 1$, so dist is a metric. By definition, $\mathrm{VR}_1(X, \mathrm{dist}) = \mathrm{Cl}(H)$, and an edge update of H is a single distance update.

For (iv) we realize $G_{\mathcal{U}}$ as a unit-ball graph: we map its vertices to points so that adjacent vertices are at distance less than 1 and nonadjacent vertices at distance greater than 1. Let A_G be the adjacency matrix of $G = G_{\mathcal{U}}$ and Δ its maximum degree. All eigenvalues of A_G lie in $[-\Delta, \Delta]$, so $A_G + \lambda I$ with $\lambda = \Delta + 1$ is positive definite. Write it as a Gram matrix with vectors $y_x \in \mathbb{R}^n$, one per vertex. Then $\|y_x - y_{x'}\|^2 = 2\lambda - 2$ if xx' is an edge and 2λ otherwise. By the Johnson–Lindenstrauss lemma [15], a random linear map $\Phi\colon \mathbb{R}^n \to \mathbb{R}^d$ with $d = O(\lambda^2 \log n)$ preserves all these squared distances up to a factor $1 \pm 1/(4\lambda)$ with high probability. It can be checked in polynomial time; we redraw it at most n times, and if all draws fail, which happens with probability $2^{-\Omega(n)}$, the reduction gives up, which only adds this probability to its error. Edges then have squared length below $2\lambda - 1$ and non-edges above it. Set $f(x) = \Phi(y_x)/\sqrt{2\lambda - 1}$. Then $\|f(x) - f(x')\| < 1$ for edges and > 1 for non-edges. This is the argument of Frankl and Maehara [31], who bound the sphericity of a graph in terms of its least eigenvalue. For every $W \subseteq V$, $\mathrm{VR}_1(f(W)) = \mathrm{Cl}(G_{\mathcal{U}}[W])$ as abstract complexes, because no pairwise distance equals 1. Inserting or deleting the vertex σ thus becomes inserting or deleting the point $f(\sigma)$. Since $\Delta = O(1)$, we get $d = O(\log n)$. Because the squared distances are separated from the threshold by a constant margin after scaling, rounding all coordinates to rationals with $O(\log n)$ bits does not change the graph, so the points have a polynomial-size description. □

A query of the form $\beta_1(\mathrm{VR}_\varrho(X))$ at a fixed scale ϱ is a persistent Betti number of the Vietoris–Rips filtration. Theorem 5.29 therefore also bounds every data structure that maintains the H_1-persistence of Vietoris–Rips filtrations of a dynamic point cloud, or of a finite metric space whose distances change one at a time, and can report the number of bars alive at a fixed scale. Applied to the bounded-degree universes of Theorem 5.19 for $p = 2$ and of Corollary 5.20 for $p \ge 3$, the same proof gives statements (i)–(iv) for β_p, for every fixed $p \ge 1$, with $2^{p+1} - 2$ in place of six in (ii); Theorem 5.30 transfers in the same way to statement (iii), and to statements (i) and (ii) without the degree bound, because the cone vertex of its universe lies in $\Theta(n)$ simplices. Two questions remain open: whether the Vietoris–Rips statement holds in a fixed dimension $d \ge 3$ (Section 7), and whether it holds under point *moves* at bounded speed rather than insertions and deletions.

5.9 Hardness with almost no homology

The complexes built in the proof of Theorem 5.5 carry a lot of homology, because they contain the whole 1-skeleton of the universe. Even without the unused edges the homology stays large. Let $A \in \{0, 1\}^{r \times k}$, and let $K^-(A)$ be the subcomplex of $K(A; [r], [k])$ that consists of its triangles, all their faces, and the row cycles. We call it the *lean* complex. In $K^-(A)$ all rows are active and all tethers are filled. The surface Σ_j together with the filling E_j of its tether is a sphere with one hole for every nonzero entry of column j, and the row cycles are circles, so

$$\chi(K^-(A)) \;=\; \sum_{j=1}^{k} \bigl(2 - \mathrm{nnz}(A_{*j})\bigr) \;=\; 2k - \mathrm{nnz}(A), \tag{8}$$

where $\mathrm{nnz}(\cdot)$ counts nonzero entries and A_{*j} is column j of A. Let G_A denote the *support graph* of A: the bipartite graph with a node ρ_i for every row, a node γ_j for every column, and an edge $\rho_i\gamma_j$ for every nonzero entry A_{ij}. Write $c(G_A)$ for its number of connected components. The components of $K^-(A)$ correspond to those of G_A, so $\beta_0(K^-(A)) = c(G_A)$. Lemma 4.4 applies verbatim to the lean complex, because it only concerns triangles, and gives $\beta_2(K^-(A)) = k - \operatorname{rank} A$. With (2) we get

$$\beta_1(K^-(A)) \;=\; (r - \operatorname{rank} A) \;+\; \big(\mathrm{nnz}(A) - r - k + c(G_A)\big). \tag{9}$$

The second summand is the number of independent cycles of G_A, which is typically of order n; Edelsbrunner and Parsa make the same observation for their static complex [26, Remark (2) after Theorem 3.1]. So one may hope for a data structure whose cost depends on the homology of K rather than on its size. Such a structure might be output-sensitive, with a running time that depends on the size of the answer. We now rule this out: small homology alone does not help. The lower bound survives on connected complexes with $\beta_1, \beta_2 \le 1$, and the question is only whether H_1 vanishes.

Theorem 5.30 (Deciding whether H_1 vanishes). *Assume the OMv conjecture, and let $\mathbb{F}$ be a field and $\varepsilon > 0$. No data structure with polynomial preprocessing time maintains a subcomplex K of a 2-dimensional universe with n simplices under simplex insertions and deletions and decides whether $H_1(K;\mathbb{F}) = 0$, with amortized update time $O(n^{1/2-\varepsilon})$ and query time $O(n^{1-\varepsilon})$. This holds even under the promise that at every query K is connected, $\chi(K) = 1$, and $\beta_1(K) = \beta_2(K) \in \{0, 1\}$.*

Under the promise, $H_1(K;\mathbb{F}) = 0$ holds exactly when K is $\mathbb{F}$-acyclic, so the same lower bound applies to deciding $\mathbb{F}$-acyclicity. It also applies to approximation: a data structure that returns a number in $[\beta_1(K), c\,\beta_1(K)]$ for some factor $c \ge 1$ returns 0 exactly when $\beta_1(K) = 0$, so it decides whether $H_1(K;\mathbb{F}) = 0$. Hence no multiplicative approximation of β_1, with any factor, can be maintained within the time bounds of Theorem 5.30.

The proof combines three ingredients. The first removes the cycles of the support graph without touching the rank information. The second is a bordered matrix whose rank deficiency is a single bit. The third is a standard isolation step that turns that bit into the Boolean answer of OuMv.

Coning off the support graph. Take $P = [r] \times [k]$ in Definition 4.2, so that column j has the triangle cycles $s_0^j, \dots, s_{r+1}^j$ and the socket between s_t^j and s_{t+1}^j belongs to row t for $t \in [r]$. Let a_t^j be the vertex 0 of s_t^j, and let c_i be the vertex 0 of the row cycle C_i. The *spine* $L(A) \subseteq K^-(A)$ is the graph on these vertices with two kinds of edges. The first kind consists of the edges $a_t^j a_{t+1}^j$ for $0 \le t \le r$ and $j \in [k]$. Such an edge is the edge $a_0 b_0$ of the cylinder of the socket between s_t^j and s_{t+1}^j, and it survives the removal of the hole, so it lies in $K^-(A)$ whether the socket is sealed or tubed. The second kind is an *attachment edge* $a_i^j c_i$ for every nonzero entry A_{ij}, namely the edge $h_0 c_0$ of the tube of position (i, j). Let z be a new vertex. The *coned complex* $\widehat{K}(A)$ is obtained from $K^-(A)$ by adding z, an edge zv for every vertex v of $L(A)$, and a triangle ze for every edge e of $L(A)$. In the coned complex a flip of A_{ij} costs 14 simplex updates. These are the 7 triangles of Lemma 4.5, the 6 edges of the tube that are not edges of its hole h or of C_i, and the cone triangle over the attachment edge. When the tube is inserted, we first delete the seal, then insert the six edges, the six tube triangles, and the cone triangle; a deletion of the tube proceeds in the reverse order, so every intermediate family is a subcomplex. Unlike in the proof of Theorem 5.5, these flips change the 1-skeleton.

Lemma 5.31. *For every field $\mathbb{F}$ and every matrix $A \in \{0,1\}^{r\times k}$,*

$$\beta_0(\widehat{K}(A)) = 1, \qquad \beta_1(\widehat{K}(A)) = r - \operatorname{rank} A, \qquad \beta_2(\widehat{K}(A)) = k - \operatorname{rank} A.$$

Proof. Write $K = K^-(A)$, $L = L(A)$, and $C = z * L$ for the cone, so that $\widehat{K}(A) = K \cup C$ and $K \cap C = L$. We first show that the inclusion $\iota\colon L \to K$ is injective on H_1 and bijective on $\widetilde{H}_0$. Define a vertex map φ from K to G_A that sends every vertex of C_i to ρ_i and every other vertex of column j to γ_j; the latter include the vertices of the tether Λ_j and the apices of the two cones of column j. Every simplex of K lies in one column surface together with its tether filling, or in one row cycle, or in the tube of a position (i, j) with $A_{ij} = 1$. In each case its image is a vertex or an edge of G_A, so φ is simplicial. On L, the map φ collapses the path $a_0^j a_1^j \cdots a_{r+1}^j$ of each column, which is a tree, to the node γ_j, and it maps each attachment edge onto the corresponding edge of G_A. Collapsing disjoint trees is a homotopy equivalence, so $(\varphi \circ \iota)_*$ is an isomorphism on all homology groups. Hence $\iota_*\colon H_1(L) \to H_1(K)$ is injective, with image of dimension $\operatorname{nnz}(A) - r - k + c(G_A)$. For the same reason $\iota_*\colon \widetilde{H}_0(L) \to \widetilde{H}_0(K)$ is injective, and it is an isomorphism because both groups have dimension $c(G_A) - 1$: the graph L has $c(G_A)$ components, being homotopy equivalent to G_A, and so does K.

The cone C is contractible and $H_2(L) = 0$, so the Mayer–Vietoris sequence of $K \cup C$ reads

$$\begin{aligned} 0 \to H_2(K) \to H_2(K \cup C) \to H_1(L) \xrightarrow{\iota_*} H_1(K) \to H_1(K \cup C) \\ \to \widetilde{H}_0(L) \xrightarrow{\iota_*} \widetilde{H}_0(K) \to \widetilde{H}_0(K \cup C) \to 0. \end{aligned}$$

Since ι_* is injective on H_1 and bijective on $\widetilde{H}_0$, we get $H_2(K \cup C) \cong H_2(K)$, $\widetilde{H}_0(K \cup C) = 0$, and $H_1(K \cup C) \cong H_1(K)/\iota_* H_1(L)$. Substituting $\beta_2(K) = k - \operatorname{rank} A$ and (9) proves the lemma. □

A bordered matrix with a one-bit rank deficiency. We use the Schur-complement identity of Lemma 5.14.

For $M \in \{0,1\}^{\ell \times \ell}$ and $u, v \in \{0,1\}^\ell$ we use the $(2\ell + 1) \times (2\ell + 1)$ matrix $\operatorname{Bord}(M; u, v)$ below, whose upper left $2\ell \times 2\ell$ block is the invertible matrix Y to which we apply Lemma 5.14:

$$\operatorname{Bord}(M; u, v) = \begin{pmatrix} I & M & 0 \\ 0 & I & v \\ u^\top & 0 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} I & M \\ 0 & I \end{pmatrix}, \quad Y^{-1} = \begin{pmatrix} I & -M \\ 0 & I \end{pmatrix}.$$

To keep the field product apart from the Boolean product $u^\top M v$ of OuMv, we write $\mu_{\mathbb{F}}(u, v) = \sum_{i,j} u_i M_{ij} v_j$, evaluated in $\mathbb{F}$. With $x = (u, 0)$ and $w = (0, v)$ we get $x^\top Y^{-1} w = -\mu_{\mathbb{F}}(u, v)$. By Lemma 5.14, the rank of $\operatorname{Bord}(M; u, v)$ is $2\ell + 1$ if $\mu_{\mathbb{F}}(u, v) \neq 0$ and 2ℓ otherwise. Lemma 5.31 therefore gives

$$\beta_0 = 1, \qquad \beta_1 = \beta_2 = \big[\mu_{\mathbb{F}}(u, v) = 0\big] \qquad \text{for } K = \widehat{K}\big(\operatorname{Bord}(M; u, v)\big). \tag{10}$$

So K is $\mathbb{F}$-acyclic exactly when $\mu_{\mathbb{F}}(u, v) \neq 0$.

From the field product to the Boolean product. The quantity $\mu_{\mathbb{F}}(u, v)$ in (10) is computed in $\mathbb{F}$, whereas OuMv asks whether the submatrix $S = M[\operatorname{supp} u, \operatorname{supp} v]$ is nonzero. These can differ; over $\mathbb{F}_2$, for instance, the entries of S may cancel in pairs. Random restrictions repair this.

Lemma 5.32. *Let $S \in \{0,1\}^{a \times b}$ be nonzero, and let $\xi \in \{0,1\}^a$ and $\eta \in \{0,1\}^b$ be independent and uniformly random. Then $\Pr[\xi^\top S \eta \neq 0$ in $\mathbb{F}] \geq 1/4$ for every field $\mathbb{F}$.*

Proof. The polynomial $f(\xi, \eta) = \sum_{i,j} S_{ij} \xi_i \eta_j$ is multilinear of degree 2, and it is nonzero because its coefficients are the entries of S. A nonzero multilinear polynomial of degree d does not vanish on at least a 2^{-d} fraction of the Boolean cube. To see this, fix a monomial of maximal degree; for every assignment to the other variables, the restriction to the variables of that monomial is a nonzero multilinear polynomial in d variables, which has a nonzero point among the 2^d assignments. □

Proof of Theorem 5.30. Let $M \in \{0,1\}^{\ell\times\ell}$ be an OuMv instance. The universe $\mathcal{U}$ is the union of the coned complexes $\widehat{K}(\mathrm{Bord}(M;u,v))$ over all $u,v \in \{0,1\}^{\ell}$. It contains the seal and the tube at every border position and has $n = \Theta(\ell^2)$ simplices. We insert the simplices of $\widehat{K}(\mathrm{Bord}(M;0,0))$, which costs $O(\ell^2)$ updates.

For a request (u,v), we perform $T = \lceil 8\ln\ell\rceil + 1$ independent probes. A probe draws uniform $\xi \in \{0,1\}^{\operatorname{supp} u}$ and $\eta \in \{0,1\}^{\operatorname{supp} v}$ and sets the border entries accordingly. It changes at most 2ℓ entries, each by a constant number of simplex updates. It then asks whether $H_1(K) = 0$. We answer 1 if and only if some probe finds $H_1(K) = 0$.

By (10), a probe finds $H_1(K) = 0$ exactly when $\xi^{\top} S \eta \neq 0$, where $S = M[\operatorname{supp} u, \operatorname{supp} v]$. If $S = 0$, no probe succeeds and the answer 0 is correct. If $S \neq 0$, each probe succeeds with probability at least $1/4$ by Lemma 5.32. All T probes then fail with probability at most $(3/4)^T < \ell^{-2}$. A union bound over the ℓ requests bounds the total error probability by $1/\ell$.

At every query, K is connected, $\beta_1(K) = \beta_2(K) \in \{0,1\}$, and $\chi(K) = \beta_0 - \beta_1 + \beta_2 = 1$, as promised. The ℓ requests use $O(\ell^2 \log \ell)$ updates and $O(\ell \log \ell)$ queries. A data structure with the stated bounds would therefore solve OuMv with polynomial preprocessing and computation time

$$O\big(\ell^2 \log \ell \cdot n^{1/2-\varepsilon} + \ell \log \ell \cdot n^{1-\varepsilon} + \ell^2 \log \ell\big) \;=\; O\big(\ell^{3-\delta}\big) \qquad \text{for every } \delta < \min\{2\varepsilon, 1\},$$

where the last term accounts for sampling the probes. This contradicts the OMv conjecture [33], which covers randomized algorithms with error probability at most $1/3$. The probes do not depend on the answers of the data structure, so the same argument applies to a randomized data structure that answers all queries of every fixed operation sequence correctly with probability at least $3/4$: the total error probability is then at most $1/4 + 1/\ell \le 1/3$ for $\ell \ge 12$. □

Remark 5.33. Over $\mathbb{F}_2$, setting the border entries to u and v themselves makes a single query compute the parity $\mu_{\mathbb{F}_2}(u,v)$. Under the analogous conjecture for matrix–vector products over $\mathbb{F}_2$, the reduction is deterministic and needs one query per request. The cone vertex z has degree $\Theta(n)$. We do not know whether bounded vertex degree, as in Theorem 5.19, can be combined with bounded homology.

5.10 Scope of the lower bounds

The bounds do not cover the following settings. First, in the vineyard model [12], an update transposes two simplices that are consecutive in a total order compatible with the filtration. A value update of Problem 5.8 can require up to $n-1$ such transpositions, so our bounds say nothing about the time per transposition. Second, for fully dynamic data structures with worst-case bounds, the only lower bounds that we obtain from the OMv conjecture itself are those implied by Theorem 5.5. Stronger worst-case bounds rest on the uMv-hinted uMv conjecture (Corollary 5.15), and their exponent depends on that of rectangular matrix multiplication. Third, the variants of the lower bound are proved in the degrees stated: bounded degree for every $p \ge 1$, linear embeddings in $\mathbb{R}^4$ for β_1 and β_2, clique and Vietoris–Rips complexes for every β_p with $p \ge 1$, and the acyclicity test for H_1. Finally, all these bounds are conditional; the only unconditional lower bound that we know is the $\Omega(\log n)$ cell-probe bound inherited from dynamic connectivity (Section 6.1).

6 Algorithms for special universes

This section contains our algorithms. Section 6.1 reduces β_0, and β_1 of graphs, to dynamic connectivity. Section 6.2 maintains all Betti numbers of a subcomplex of a fixed triangulation of the 3-sphere, and more generally of any fixed geometric complex in $\mathbb{R}^3$, in $O(\log^2 n)$ amortized time per update; together with the lower bound for complexes in $\mathbb{R}^4$ (Theorem 5.25), this separates three from four dimensions. Section 6.3 maintains every Betti number with polylogarithmically many

field operations per update in the worst case when the boundary matrices of the universe have small width or when the Hasse diagram of the universe has bounded treewidth.

Throughout, $c(G)$ denotes the number of connected components of a graph G. A *dynamic connectivity* structure maintains a graph on a fixed vertex set under edge insertions and deletions, and answers queries of the form "are u and v connected?" We use the deterministic structure of Holm, de Lichtenberg, and Thorup [36]. On a vertex set V it is initialized without edges in $O(|V| \log |V|)$ time. Afterwards, any sequence of s edge updates takes $O(s \log^2 |V|)$ time in total, and each query takes $O(\log |V| / \log \log |V|)$ time. Such a structure also maintains $c(G)$. Before inserting an edge uv we ask whether u and v are connected, and we decrease $c(G)$ by one if they are not. After deleting an edge uv we ask the same question, and we increase $c(G)$ by one if they are not.

6.1 Connected components

For a simplicial complex K, let $G_0(K)$ be the graph with vertex set K_0 and edge set K_1. For every field $\mathbb{F}$, $\beta_0(K;\mathbb{F}) = c(G_0(K))$ [49].

Proposition 6.1. *Let $\mathcal{U}$ be a universe with n simplices. There is a deterministic data structure that maintains a subcomplex $K \subseteq \mathcal{U}$ under the simplex updates of Problem 3.1 and returns $\beta_0(K;\mathbb{F})$, for every field $\mathbb{F}$, with preprocessing time $O(n \log n)$, amortized update time $O(\log^2 n)$, and query time $O(1)$. If $\dim \mathcal{U} \le 1$, the same data structure also returns $\beta_1(K;\mathbb{F})$ in $O(1)$ time.*

Proof. Let G be the graph with vertex set $\mathcal{U}_0$ and edge set K_1. Both endpoints of an edge of K lie in K, so G consists of $G_0(K)$ together with the $|\mathcal{U}_0| - n_0(K)$ vertices of $\mathcal{U}_0 \setminus K_0$, which are isolated. Hence

$$\beta_0(K;\mathbb{F}) = c(G_0(K)) = c(G) - |\mathcal{U}_0| + n_0(K).$$

We store $n_0(K)$, $n_1(K)$, and $c(G)$, and we keep G in the structure of [36] on the vertex set $\mathcal{U}_0$, initially without edges. Inserting or deleting a vertex changes only $n_0(K)$: the precondition of Delete ensures that a deleted vertex has no incident edge in K, so G does not change. Inserting or deleting an edge changes $n_1(K)$ and is one edge update of G, which also updates $c(G)$ with one connectivity query. Simplices of dimension at least two do not change G. So the preprocessing takes $O(n \log n)$ time, any sequence of t updates takes $O(t \log^2 n)$ time, and the formula above answers a query in $O(1)$ time.

If $\dim \mathcal{U} \le 1$, then $\partial_2 = 0$, and (1) gives $\operatorname{rank} \partial_1 = n_0(K) - \beta_0(K)$ and therefore

$$\beta_1(K) = n_1(K) - n_0(K) + \beta_0(K).$$

□

The proof uses only the interface of the connectivity structure. With the deterministic structure of Wulff-Nilsen [60], the amortized update time becomes $O(\log^2 n / \log \log n)$. With the randomized structure of Kapron, King, and Mountjoy [38], the update time becomes polylogarithmic in the worst case, and the answers inherit the Monte Carlo guarantee of [38]; with the deterministic structure of Chuzhoy et al. [11], it becomes $n^{o(1)}$ in the worst case. For graph filtrations, Dey, Hou, and Parsa [22] go further and update persistence diagrams, not only Betti numbers, under local changes in sublinear time.

Proposition 6.1 is optimal up to a logarithmic factor, without any conjecture. Pătraşcu and Demaine [50] proved that in the cell-probe model with cells of $\Theta(\log N)$ bits, every dynamic connectivity structure on N vertices has amortized update time or query time $\Omega(\log N)$. We take as universe the complete graph on these N vertices, with $n = N + \binom{N}{2}$ simplices, and insert all vertices during preprocessing. A connectivity query for two vertices u and v then reduces to two

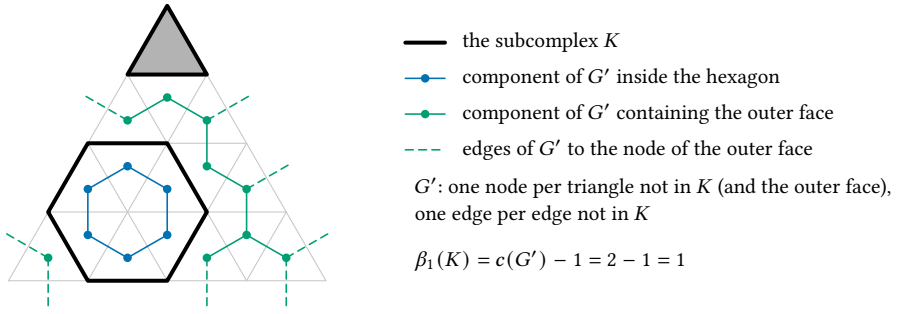


Fig. 8. The duality behind Theorem 6.4, one dimension lower. For a proper subcomplex K of a triangulated 2-sphere, of which only a triangulated disk is drawn (the triangles outside the disk are not in K and are merged into one node, the outer face), Alexander duality gives $\beta_1(K) = c - 1$, where c is the number of components of the complement of K, and these components are the components of the graph G' whose nodes are the triangles not in K (and the outer face) and whose edges are the edges not in K, as in the proof of Lemma 6.3. A simplex update changes G' by $O(1)$ nodes or edges, so dynamic connectivity maintains $\beta_1(K)$. For a triangulated 3-sphere, the same argument applies to β_2, with tetrahedra as nodes and triangles as edges.

updates and two queries of dynamic β_0: query β_0, insert the edge uv (if it is absent), query β_0 again, and delete the edge; the vertices are connected if and only if β_0 did not decrease. Since $\log n = \Theta(\log N)$, every data structure for dynamic β_0 also has amortized update time or query time $\Omega(\log n)$ in this model. By Lemma 4.8, the suspension shifts reduced homology by one degree, so $\beta_p(S^pK) = \beta_0(K) - 1$ for every nonempty complex K and every $p \geq 1$, where S^p is the p-fold suspension of Section 5.2. Since K always contains all vertices, it is nonempty, and each update of K costs 3^p updates of S^pK; hence the same unconditional bound holds for β_p for every fixed $p \geq 1$.

6.2 Subcomplexes of the 3-sphere

We write $\|K\|$ for the geometric realization of a simplicial complex K [49]; if K is a subcomplex of T, then $\|K\|$ is a subspace of $\|T\|$. A *triangulation* of a topological space Y is a simplicial complex T such that $\|T\|$ is homeomorphic to Y.

Let T be a triangulation of the 3-sphere S^3. We use three standard facts.

(T1) Every simplex of T is a face of a tetrahedron of T, and T has no simplex of dimension larger than three.
(T2) Every triangle of T is a facet of exactly two tetrahedra of T.
(T3) For every field $\mathbb{F}$, $\beta_0(T;\mathbb{F}) = \beta_3(T;\mathbb{F}) = 1$ and $\beta_1(T;\mathbb{F}) = \beta_2(T;\mathbb{F}) = 0$.

Facts (T1) and (T2) hold for every triangulation of a closed 3-manifold [49]. Fact (T3) holds because simplicial homology is a topological invariant [32, 49]. For a triangle t of T, let τ_t and τ'_t be the two tetrahedra that contain t. Figure 8 shows the idea of this subsection one dimension lower.

Definition 6.2 (Dual graph). For a subcomplex $K \subseteq T$, the *dual graph* $G_*(K)$ has the set T_3 of all tetrahedra of T as its vertex set and, for every triangle $t \in T_2 \setminus K$, an edge t^* that joins τ_t and τ'_t.

The graph $G_*(K)$ is simple, because two tetrahedra that share two triangles share all four vertices. A tetrahedron of K is an isolated vertex of $G_*(K)$, because all its triangles lie in K.

Lemma 6.3. *Let T be a triangulation of S^3, let $\mathbb{F}$ be a field, and let $K \subsetneq T$ be a subcomplex. Then*

$$\beta_3(K;\mathbb{F}) = 0 \qquad \text{and} \qquad \beta_2(K;\mathbb{F}) = c(G_*(K)) - n_3(K) - 1.$$

Proof. We fix an orientation of every tetrahedron and write $[\tau : t] \in \{1, -1\}$ for the coefficient of a triangle t in $\partial\tau$. By (T2), for every 3-chain $c = \sum_\tau c_\tau \tau$ and every triangle t,

$$\text{the coefficient of } t \text{ in } \partial c \text{ is } [\tau_t : t]\, c_{\tau_t} + [\tau'_t : t]\, c_{\tau'_t}. \tag{11}$$

Step 1: $G_*(\emptyset)$ is connected, and the orientations can be chosen so that $[\tau'_t : t] = -[\tau_t : t]$ for every triangle t. Over $\mathbb{F}_2$, (11) shows that a 3-chain is a cycle if and only if its coefficients agree at the two endpoints of every edge of $G_*(\emptyset)$. So $\dim Z_3(T;\mathbb{F}_2) = c(G_*(\emptyset))$. Since T has no 4-simplices by (T1), $\dim Z_3(T;\mathbb{F}_2) = \beta_3(T;\mathbb{F}_2)$, which is 1 by (T3). Hence $G_*(\emptyset)$ is connected. Over $\mathbb{Q}$, fact (T3) provides a cycle $z \neq 0$ in $Z_3(T;\mathbb{Q})$. By (11), $|z_{\tau_t}| = |z_{\tau'_t}|$ for every triangle t. Since $G_*(\emptyset)$ is connected, all coefficients of z have the same absolute value α, and $\alpha > 0$ because $z \neq 0$. We reverse the orientation of every tetrahedron τ with $z_\tau < 0$. Then $z = \alpha \sum_\tau \tau$, and (11) applied to z gives $[\tau_t : t] + [\tau'_t : t] = 0$ for every triangle t. These are integers, so the identity holds over every field. From now on we use these orientations. Then (11) says that the coefficient of t in ∂c is $[\tau_t : t]\,(c_{\tau_t} - c_{\tau'_t})$.

Step 2: $\beta_3(K) = 0$. By Step 1, every 3-cycle of T has equal coefficients at the two endpoints of every edge of the connected graph $G_*(\emptyset)$, so it has the same coefficient on all tetrahedra. If K contained every tetrahedron of T, then K would contain all their faces, and $K = T$ by (T1). So some tetrahedron is not in K. A 3-cycle of K is a 3-cycle of T with coefficient zero on that tetrahedron, so it is zero. Since K has no 4-simplices, $\beta_3(K) = \dim Z_3(K) = 0$.

Step 3: $\beta_2(K)$. The relative chain groups $C_q(T,K) = C_q(T)/C_q(K)$ have the simplices of $T_q \setminus K_q$ as a basis. The exact sequence of the pair (T,K) [49] contains the segment

$$H_3(K) \to H_3(T) \to H_3(T,K) \to H_2(K) \to H_2(T).$$

Here $H_3(K) = 0$ by Step 2, $H_3(T) \cong \mathbb{F}$ and $H_2(T) = 0$ by (T3), so $\beta_2(K) = \dim H_3(T,K) - 1$. Since T has no 4-simplices, $H_3(T,K)$ is the space of relative 3-cycles: chains c on the tetrahedra not in K such that ∂c has coefficient zero on every triangle $t \notin K$. For such a triangle, τ_t and τ'_t are not in K, because K is closed under taking faces. By Step 1, the condition for t says $c_{\tau_t} = c_{\tau'_t}$. Let G' be the subgraph of $G_*(K)$ induced by $T_3 \setminus K_3$; it contains every edge of $G_*(K)$. The relative 3-cycles are exactly the chains that are constant on every connected component of G', so $\dim H_3(T,K) = c(G')$. The tetrahedra of K are isolated vertices of $G_*(K)$, so $c(G') = c(G_*(K)) - n_3(K)$. □

Theorem 6.4. *Let T be a triangulation of S^3 with n simplices. There is a deterministic data structure that maintains a subcomplex $K \subseteq T$ under the simplex updates of Problem 3.1 with universe $\mathcal{U} = T$, and that returns $\beta_0(K;\mathbb{F}), \ldots, \beta_3(K;\mathbb{F})$, for every field $\mathbb{F}$, in $O(1)$ time. The preprocessing and any sequence of t updates take $O((n+t)\log^2 n)$ time in total.*

Proof. The data structure stores the counters $n_0(K), \ldots, n_3(K)$ and two dynamic connectivity structures [36] that maintain their numbers of components. The first, $\mathcal{D}_0$, stores the graph G with vertex set T_0 and edge set K_1, as in the proof of Proposition 6.1. The second, $\mathcal{D}_*$, stores the dual graph $G_*(K)$. In preprocessing we compute τ_t and τ'_t for every triangle t by sorting the triangles of all tetrahedra, in $O(n \log n)$ time. We initialize both structures, and we insert the $n_2(T) \leq n$ edges of $G_*(\emptyset)$ into $\mathcal{D}_*$.

An update of a vertex or of a tetrahedron changes only the corresponding counter. An update of an edge also inserts or deletes that edge in $\mathcal{D}_0$. Inserting a triangle into K deletes its dual edge from $\mathcal{D}_*$, and deleting a triangle from K inserts its dual edge. So every update costs $O(1)$ time plus at most one edge update and one connectivity query in one of the two structures. Both structures have at most n vertices. After their initialization, they receive at most $n + t$ edge updates and t queries in total, so the preprocessing and t updates take $O(n \log n + (n+t)\log^2 n + t \log n) = O((n+t)\log^2 n)$ time.

To answer a query, we distinguish two cases. If $n_3(K) = n_3(T)$, then K contains every tetrahedron, so $K = T$ by (T1), and we return $(1, 0, 0, 1)$, which is correct by (T3). Otherwise $K \subsetneq T$, and we return

$$\beta_0 = c(G) - n_0(T) + n_0(K), \quad \beta_2 = c(G_*(K)) - n_3(K) - 1, \quad \beta_3 = 0, \quad \beta_1 = \beta_0 + \beta_2 - \beta_3 - \chi(K),$$

where $\chi(K) = n_0(K) - n_1(K) + n_2(K) - n_3(K)$. The value β_0 is correct by the proof of Proposition 6.1, β_2 and β_3 are correct by Lemma 6.3, and β_1 is then correct by the Euler–Poincaré formula $\chi(K) = \sum_p (-1)^p \beta_p(K)$ of Section 3. None of these values depends on $\mathbb{F}$, and each is computed from stored numbers in $O(1)$ time. □

Thus, after preprocessing, any t updates take $O((n + t) \log^2 n)$ time, which is amortized update time $O(\log^2 n)$ in the sense of Section 3; the term $n \log^2 n$ pays for the potential that the $n_2(T)$ initial edges of $\mathcal{D}_*$ create in the analysis of [36]. With [60] the total time becomes $O((n+t) \log^2 n / \log\log n)$, with the randomized (Monte Carlo) structure of [38] every update takes polylogarithmic worst-case time, and with the deterministic structure of [11] every update takes $n^{o(1)}$ worst-case time. The formulas show that the Betti numbers of a subcomplex of T do not depend on the field.

Lemma 6.3 is a combinatorial form of Alexander duality. Identify $\|T\|$ with S^3 and let $\emptyset \neq K \subsetneq T$; then $\widetilde{H}_0(S^3 \setminus \|K\|; \mathbb{Z}) \cong \widetilde{H}^2(\|K\|; \mathbb{Z})$ [32, Section 3.3], the right-hand side has rank $\beta_2(K; \mathbb{Q})$ by the universal coefficient theorem, and so $c(G_*(K)) - n_3(K)$ is the number of path components of $S^3 \setminus \|K\|$. We do not use this fact. Delfinado and Edelsbrunner [16] use the same two graphs to compute the Betti numbers of a filtration of a subcomplex of a triangulated S^3 in $O(n\,\alpha(n))$ time, where α is the inverse Ackermann function. Since union–find cannot delete, they process the dual graph in reverse order. Fully dynamic connectivity removes this restriction, so in Theorem 6.4 insertions and deletions can arrive in any order. In their Section 6.2 they also drop the triangulation of the complement and handle every complex in $\mathbb{R}^3$ through a graph on the two sides of its triangles, built from the cyclic order of the triangles around each edge; Corollary 6.6 below obtains the dynamic version more simply, at the price of polynomial preprocessing.

The same algorithm applies to subcomplexes of a triangulated 3-ball.

Corollary 6.5. *Let B be a triangulation of the closed 3-ball with n_B simplices, and let the universe $\mathcal{U}$ be a subcomplex of B. There is a deterministic data structure that maintains a subcomplex $K \subseteq \mathcal{U}$ under the simplex updates of Problem 3.1 and returns $\beta_0(K; \mathbb{F}), \ldots, \beta_3(K; \mathbb{F})$, for every field $\mathbb{F}$, in $O(1)$ time. The preprocessing and any sequence of t updates take $O((n_B + t) \log^2 n_B)$ time in total.*

Proof. Let ∂B be the set of triangles of B that are facets of exactly one tetrahedron of B, together with all their faces. It is a subcomplex of B, and $\|\partial B\|$ is the boundary sphere of the ball $\|B\|$ [49]. Let o be a new vertex, let $C = \{\{o\}\} \cup \partial B \cup \{\sigma \cup \{o\} : \sigma \in \partial B\}$ be the cone on ∂B with apex o, and let $T = B \cup C$. Then T is a simplicial complex with at most $2n_B + 1$ simplices, $B \cap C = \partial B$, and $\|C\|$ is a cone on $\|\partial B\|$. So $\|T\|$ is the union of the ball $\|B\|$ and a cone on its boundary sphere, and the two meet exactly in that sphere. A homeomorphism from $\|B\|$ to the closed unit ball $D^3 \subset \mathbb{R}^3$ maps the boundary sphere $\|\partial B\|$ onto the unit sphere S^2, and it extends along the cone lines to a homeomorphism from $\|T\|$ to the space obtained from D^3 by attaching a cone on S^2 along S^2. The cone on S^2 is a closed 3-ball, and two closed 3-balls glued along their boundary spheres by a homeomorphism form S^3, so T is a triangulation of S^3. We compute ∂B and T in $O(n_B \log n_B)$ time. Every subcomplex of $\mathcal{U}$ is a subcomplex of T, and the preconditions of Insert and Delete depend only on K and on the updated simplex. So the data structure of Theorem 6.4 for T solves the problem for $\mathcal{U}$. □

A compact convex set in $\mathbb{R}^3$ with nonempty interior is homeomorphic to the closed 3-ball. So Corollary 6.5 applies to every subcomplex of a geometric simplicial complex in $\mathbb{R}^3$ whose underlying

space is a 3-dimensional convex polytope. An example is the Delaunay triangulation of a finite set of at least four points in general position in $\mathbb{R}^3$, whose subcomplexes include the alpha complexes of the point set [24].

In fact, polynomial preprocessing suffices to handle every fixed geometric complex in $\mathbb{R}^3$, with no assumption on its underlying space.

Corollary 6.6. *Let $\mathcal{U}$ be a geometric simplicial complex in $\mathbb{R}^3$ with n simplices, given by rational vertex coordinates. There is a deterministic data structure that maintains a subcomplex $K \subseteq \mathcal{U}$ under the simplex updates of Problem 3.1 and returns $\beta_0(K;\mathbb{F}), \ldots, \beta_3(K;\mathbb{F})$, for every field $\mathbb{F}$, in $O(1)$ time. Its preprocessing time is polynomial in the size of the input, and afterwards any sequence of t updates takes $O((n+t)\log^2 n)$ time in total.*

Proof. *A refinement.* Fix a closed tetrahedron $\Delta \subset \mathbb{R}^3$ whose interior contains $\|\mathcal{U}\|$. Cut Δ by the planes spanned by the triangles of $\mathcal{U}$ and, for every edge and every vertex of $\mathcal{U}$, by two or three planes through it whose intersection is exactly the line or point that it spans. Every simplex of $\mathcal{U}$ is then a union of closed cells, of all dimensions, of the resulting decomposition of Δ into convex polytopes, and the barycentric subdivision of this decomposition is a triangulation B of Δ in which every simplex of $\mathcal{U}$ is a union of simplices of B [49]. There are $O(n)$ planes, so B has polynomially many simplices and can be computed in polynomial time. Coning off ∂B as in the proof of Corollary 6.5 gives a triangulation T of S^3 with the same property. For a subcomplex $K \subseteq \mathcal{U}$, the simplices of T contained in $\|K\|$ form a subcomplex $K' \subsetneq T$ with $\|K'\| = \|K\|$, so K and K' have the same Betti numbers. By Lemma 6.3, $\beta_3(K) = 0$ and $\beta_2(K) = c(G_*(K')) - n_3(K') - 1$, where G_* is the dual graph of T.

A compressed dual graph. We call the connected components of $S^3 \setminus \|\mathcal{U}\|$ the *outer cells*. Near an interior point of a triangle t of $\mathcal{U}$, no simplex of $\mathcal{U}$ other than t and the tetrahedra that contain t is present, so each of the two sides of t lies in a single outer cell or in a single tetrahedron of $\mathcal{U}$ that contains t. Let $G(K)$ be the multigraph whose nodes are the outer cells and the tetrahedra of $\mathcal{U}$, with one edge for every triangle $t \in \mathcal{U}_2 \setminus K$, joining the nodes on the two sides of t. We claim that $c(G_*(K')) - n_3(K') = c(G(K)) - n_3(K)$. The interior of every tetrahedron of T lies in an outer cell or in the interior of a tetrahedron of $\mathcal{U}$. The tetrahedra of T in one outer cell are connected in $G_*(K')$: a path between two of them inside the open cell can be chosen to avoid the 1-skeleton of T and to cross triangles of T transversally, and the triangles that it crosses lie inside the cell, so they are not in K'. For a tetrahedron $\tau \in \mathcal{U}_3 \setminus K$, the tetrahedra of T in τ are connected through the triangles of T in the interior of τ, which are not in K'; for $\tau \in K$, each of them is an isolated vertex of $G_*(K')$, and it is also counted in $n_3(K')$. A triangle of T that is contained in a triangle t of $\mathcal{U}$ lies in K' if and only if $t \in K$, because $\|K\| \cap \|t\|$ is a union of faces of t; if $t \notin K$, it joins tetrahedra of T on the two sides of t. The relative interior of every other triangle of T lies in an outer cell or in the interior of a tetrahedron of $\mathcal{U}$. Contracting the groups of tetrahedra of T described above therefore turns $G_*(K')$ into $G(K)$, up to the multiplicity of edges (a triangle $t \notin K$ of $\mathcal{U}$ yields one edge for each triangle of T inside t, which does not affect connectivity), except that a tetrahedron $\tau \in K$ of $\mathcal{U}$ corresponds to its T-tetrahedra, which are isolated in $G_*(K')$ and counted in $n_3(K')$, while in $G(K)$ it is a single isolated node counted in $n_3(K)$. This proves the claim, so $\beta_2(K) = c(G(K)) - n_3(K) - 1$.

The data structure. In preprocessing we compute T, the outer cells, and the two sides of every triangle of $\mathcal{U}$. The multigraph $G(K)$ has at most $n_3(\mathcal{U}) + 2n_2(\mathcal{U}) + 1$ nodes and at most n edges. We then proceed as in the proof of Theorem 6.4, with a dynamic connectivity structure for $G(K)$ in place of $\mathcal{D}_*$; parallel edges and loops are handled by keeping a multiplicity for every pair of nodes. A simplex update of K changes at most one edge of $G(K)$ or of $G_0(K)$, so it takes $O(\log^2 n)$ amortized

time, and the query formulas are those of Theorem 6.4 with $c(G(K))$ in place of $c(G_*(K))$; the case $K = T$ does not occur. □

Three versus four dimensions. Theorem 6.4, Corollary 6.6, and Theorem 5.25 separate three from four dimensions. For subcomplexes of any fixed geometric complex in $\mathbb{R}^3$, all Betti numbers can be maintained in $O(\log^2 n)$ amortized time per update after polynomial preprocessing; preprocessing and t updates take $O((n+t)\log^2 n)$ time in total if the universe is a triangulation of the 3-sphere (Theorem 6.4), and $O((n_B+t)\log^2 n_B)$ time if it is a subcomplex of a given triangulated 3-ball with n_B simplices (Corollary 6.5). For subcomplexes of a fixed geometric 2-complex in $\mathbb{R}^4$, no data structure with polynomial preprocessing time maintains β_1 or β_2 with amortized update time $O(n^{1/2-\varepsilon})$ and query time $O(n^{1-\varepsilon})$, unless the OMv conjecture fails (Theorem 5.25). Duality explains where the boundary lies. The proof of Lemma 6.3 works verbatim for a triangulation T of the d-sphere with $d \geq 2$: for every proper subcomplex K, $\beta_d(K) = 0$ and $\beta_{d-1}(K) = c(G_*(K)) - n_d(K) - 1$, where $G_*(K)$ joins two d-simplices through every $(d-1)$-simplex not in K. With dynamic connectivity for β_0 and the Euler–Poincaré formula, this determines all Betti numbers if $d \leq 3$, because at most one Betti number, β_1, remains unknown. For $d = 4$, two Betti numbers remain, β_1 and β_2, tied by a single linear relation; these are exactly the Betti numbers for which Theorem 5.25 proves hardness in $\mathbb{R}^4$. This is the dynamic counterpart of the static picture: Betti numbers of subcomplexes of a triangulated S^3 can be computed in near-linear time [16], Dey and Guha [18] compute the homology groups of complexes in $\mathbb{R}^3$, and for 2-complexes that embed in $\mathbb{R}^4$ the problem is as hard as matrix rank [26]. By Corollary 6.6, both sides live in the same model: the universe is a fixed geometric complex, and preprocessing may take polynomial time. Whether the lower bound holds for subcomplexes of a triangulation of the 4-sphere with $O(n)$ simplices is open (Section 7).

6.3 Small width and bounded treewidth

Let $\mathbb{F}$ be a field and $M \in \mathbb{F}^{r\times k}$. We count field operations as unit cost. We consider dynamic rank under column switches only: the data structure maintains a set $J \subseteq [k]$ of active columns, initially empty, under Toggle(column j), and Query returns $\operatorname{rank} M[[r], J]$. By Lemma 4.1, this problem captures dynamic Betti numbers: the columns of a boundary matrix are the simplices of one dimension, and a simplex update switches one column. Zero rows and zero columns do not affect the rank. So we delete them in preprocessing, ignore switches of zero columns, and assume that every row and every column of M is nonzero. For a set S of rows, $\pi_S \colon \mathbb{F}^r \to \mathbb{F}^S$ denotes the coordinate projection, m_j denotes column j of M, and $\operatorname{nnz}(M)$ is the number of nonzero entries of M.

Column hierarchies. A *column hierarchy* for M is a rooted tree H whose leaves are the columns $1, \ldots, k$ and in which every inner node has one or two children. Its *height* is the largest number of edges on a path from the root to a leaf. For a node θ of H, let $L_\theta \subseteq [k]$ be the set of columns at the leaves below θ, so that $L_\theta = \{j\}$ if θ is the leaf j. Let $\operatorname{In}(\theta)$ be the set of rows all of whose nonzero entries lie in columns of L_θ, and let $\operatorname{Bd}(\theta)$ be the set of rows that have a nonzero entry in a column of L_θ and a nonzero entry in a column outside L_θ. The *width* of H is the largest size of a set $\operatorname{Bd}(\theta)$. Contracting an inner node that has a single child into that child changes neither the sets L_θ that occur nor the width, and it does not increase the height; so we may assume that every inner node has two children, and then H has fewer than $2k$ nodes.

Theorem 6.7. *Let $\mathbb{F}$ be a field, let $M \in \mathbb{F}^{r\times k}$, and let H be a column hierarchy for M of height h and width w. There is a deterministic data structure for dynamic rank of M under column switches with preprocessing time $O(r + k + \operatorname{nnz}(M)(h+1))$, worst-case update time $O((w+1)^3(h+1))$, and query time $O(1)$.*

For a node θ, let $V_\theta \subseteq \mathbb{F}^r$ be the span of the active columns in L_θ, let $r_\theta = \dim \pi_{\mathrm{In}(\theta)}(V_\theta)$, and let $W_\theta = \{v \in V_\theta : \pi_{\mathrm{In}(\theta)}(v) = 0\}$. If a column of L_θ has a nonzero entry in row i, then $i \in \mathrm{In}(\theta) \cup \mathrm{Bd}(\theta)$. So every vector of V_θ is supported on $\mathrm{In}(\theta) \cup \mathrm{Bd}(\theta)$, and every vector of W_θ is supported on $\mathrm{Bd}(\theta)$. By the rank–nullity theorem, $\dim V_\theta = r_\theta + \dim W_\theta$. At the root ρ, every row lies in $\mathrm{In}(\rho)$, so $W_\rho = 0$ and $\operatorname{rank} M[[r], J] = \dim V_\rho = r_\rho$.

Lemma 6.8. *Let θ be a node of H with children θ_1 and θ_2, and let $X = \mathrm{In}(\theta) \setminus (\mathrm{In}(\theta_1) \cup \mathrm{In}(\theta_2))$.*

(a) $X \subseteq \mathrm{Bd}(\theta_1) \cap \mathrm{Bd}(\theta_2)$; in particular $|X| \le w$.
(b) The set $\mathrm{Bd}(\theta_1) \cup \mathrm{Bd}(\theta_2)$ is the disjoint union of $\mathrm{Bd}(\theta)$ and X.
(c) $r_\theta = r_{\theta_1} + r_{\theta_2} + \dim \pi_X(W_{\theta_1} + W_{\theta_2})$ and $W_\theta = \{v \in W_{\theta_1} + W_{\theta_2} : \pi_X(v) = 0\}$.

Proof. (a) A row of X has all its nonzero entries in $L_\theta = L_{\theta_1} \sqcup L_{\theta_2}$, but neither all in L_{θ_1} nor all in L_{θ_2}. So it has a nonzero entry in each of L_{θ_1} and L_{θ_2}, and it lies in $\mathrm{Bd}(\theta_1)$ and in $\mathrm{Bd}(\theta_2)$.

(b) A row of $\mathrm{Bd}(\theta)$ has a nonzero entry in L_θ, hence in L_{θ_1} or in L_{θ_2}, and a nonzero entry outside L_θ, hence outside both; so it lies in $\mathrm{Bd}(\theta_1) \cup \mathrm{Bd}(\theta_2)$. By (a), so does every row of X. Conversely, let $i \in \mathrm{Bd}(\theta_1)$; the case of θ_2 is symmetric. If i has a nonzero entry outside L_θ, then $i \in \mathrm{Bd}(\theta)$. Otherwise $i \in \mathrm{In}(\theta)$; moreover $i \notin \mathrm{In}(\theta_1)$ because $i \in \mathrm{Bd}(\theta_1)$, and $i \notin \mathrm{In}(\theta_2)$ because i has a nonzero entry in L_{θ_1}; so $i \in X$. The sets $\mathrm{Bd}(\theta)$ and X are disjoint, because $X \subseteq \mathrm{In}(\theta)$.

(c) Let $Y = \mathrm{In}(\theta_1) \cup \mathrm{In}(\theta_2)$; this is a disjoint union, and $\mathrm{In}(\theta) = Y \cup X$. The rows of $\mathrm{In}(\theta_1)$ have all their nonzero entries in L_{θ_1}, so every vector of V_{θ_2} vanishes on them; symmetrically, every vector of V_{θ_1} vanishes on $\mathrm{In}(\theta_2)$. We have $V_\theta = V_{\theta_1} + V_{\theta_2}$. For $v_1 \in V_{\theta_1}$ and $v_2 \in V_{\theta_2}$, therefore, $\pi_{\mathrm{In}(\theta_1)}(v_1 + v_2) = \pi_{\mathrm{In}(\theta_1)}(v_1)$ and $\pi_{\mathrm{In}(\theta_2)}(v_1 + v_2) = \pi_{\mathrm{In}(\theta_2)}(v_2)$. Hence $\pi_Y(V_\theta)$ is the direct sum of $\pi_{\mathrm{In}(\theta_1)}(V_{\theta_1})$ and $\pi_{\mathrm{In}(\theta_2)}(V_{\theta_2})$, of dimension $r_{\theta_1} + r_{\theta_2}$. Moreover, $\pi_Y(v_1 + v_2) = 0$ if and only if $v_1 \in W_{\theta_1}$ and $v_2 \in W_{\theta_2}$, so the kernel of π_Y on V_θ is $W_{\theta_1} + W_{\theta_2}$. It follows that $\dim V_\theta = r_{\theta_1} + r_{\theta_2} + \dim(W_{\theta_1} + W_{\theta_2})$, and that W_θ, the kernel of $\pi_{\mathrm{In}(\theta)}$ on V_θ, consists of the vectors $v \in W_{\theta_1} + W_{\theta_2}$ with $\pi_X(v) = 0$. By rank–nullity for π_X on $W_{\theta_1} + W_{\theta_2}$, we get $r_\theta = \dim V_\theta - \dim W_\theta = r_{\theta_1} + r_{\theta_2} + \dim \pi_X(W_{\theta_1} + W_{\theta_2})$. □

Proof of Theorem 6.7. We assume that every inner node of H has two children. At each node θ we store the number r_θ and a basis of W_θ, written as vectors indexed by $\mathrm{Bd}(\theta)$; such a basis consists of at most w vectors, each with at most w coordinates.

For a leaf j, the set $\mathrm{In}(j)$ consists of the rows whose only nonzero entry lies in column j. If $j \notin J$, then $r_j = 0$ and $W_j = 0$. If $j \in J$ and $\pi_{\mathrm{In}(j)}(m_j) \neq 0$, then $r_j = 1$ and $W_j = 0$. If $j \in J$ and $\pi_{\mathrm{In}(j)}(m_j) = 0$, then $r_j = 0$ and W_j is spanned by m_j. For an inner node with children θ_1, θ_2, Lemma 6.8 computes the data of θ from that of its children. We write the at most $2w$ stored basis vectors of W_{θ_1} and W_{θ_2} as vectors indexed by $\mathrm{Bd}(\theta_1) \cup \mathrm{Bd}(\theta_2)$, which is $\mathrm{Bd}(\theta) \sqcup X$ by Lemma 6.8(b), and we bring them to row echelon form by Gaussian elimination, with the coordinates of X first. The nonzero echelon vectors form a basis of $W_{\theta_1} + W_{\theta_2}$. Those whose leading coordinate lies in X have linearly independent projections to $\mathbb{F}^X$, and their number is $\dim \pi_X(W_{\theta_1} + W_{\theta_2})$. The others vanish on X, and a vector of the span that vanishes on X has zero coefficients on the former, so the others form a basis of W_θ. This takes $O((w+1)^3)$ time. A switch of column j changes only the data of the leaf j and of its at most h ancestors, which we recompute bottom-up in $O((w+1)^3(h+1))$ time. A query returns r_ρ, which equals $\operatorname{rank} M[[r], J]$, in $O(1)$ time.

In preprocessing, we compute for every row i the lowest common ancestor λ_i of the leaves of its nonzero entries, by walking up from these leaves, and we record i at every node on the paths from these leaves to λ_i, excluding λ_i itself. These are exactly the nodes θ with $i \in \mathrm{Bd}(\theta)$, and λ_i is the unique node at which i belongs to the set X of Lemma 6.8 or, if λ_i is a leaf, to $\mathrm{In}(\lambda_i)$. This takes $O(r + k + \mathrm{nnz}(M)(h+1))$ time, and it yields the lists $\mathrm{Bd}(\theta)$, the sets X, and, for every column j, the restriction of m_j to $\mathrm{Bd}(j)$ and whether $\pi_{\mathrm{In}(j)}(m_j) \neq 0$. Initially $J = \emptyset$, so all stored ranks and bases are zero. □

Small width. For a nonzero row i, let a_i and b_i be the smallest and the largest index of a nonzero entry in row i. Row i *crosses* the cut $x \in [k-1]$ if $a_i \le x < b_i$. The *width* of M is the largest number of rows that cross a single cut, and 0 if $k = 1$. Deleting zero rows and zero columns, without reordering the remaining columns, does not increase the width.

Theorem 6.9. *Let $\mathbb{F}$ be a field, and let $M \in \mathbb{F}^{r\times k}$ have width w and $\operatorname{nnz}(M)$ nonzero entries. There is a deterministic data structure for dynamic rank of M under column switches with preprocessing time $O(r + \operatorname{nnz}(M) + k(w+1))$, worst-case update time $O((w+1)^3 \log(k+1))$, and query time $O(1)$.*

Proof. We may assume $k \ge 2$. Let H be a balanced binary tree whose leaves are the columns $1, \dots, k$ from left to right; its height is $\lceil \log_2 k \rceil$. For every node θ, the set L_θ is an interval $\{l, \dots, l'\}$. A row of $\operatorname{Bd}(\theta)$ has a nonzero entry in L_θ and one outside, so it crosses the cut $l-1$ or the cut l'; hence H has width at most $2w$. Theorem 6.7 gives the update and query times. For the preprocessing time, we compute all a_i and b_i in $O(r + \operatorname{nnz}(M))$ time and, for every cut, the list of rows that cross it; these lists have total length at most $(k-1)w$. For a node with $L_\theta = \{l, \dots, l'\}$, the set $\operatorname{Bd}(\theta)$ consists of the rows that cross the cut $l-1$ or the cut l' and have a nonzero entry in L_θ, where no row crosses the cuts 0 and k, and the set X of Lemma 6.8 consists of the rows that cross the cut between the two children but neither the cut $l-1$ nor the cut l'. For every row and every cut that it crosses, we record the nearest nonzero entries of the row on the two sides of the cut, in $O(\operatorname{nnz}(M) + kw)$ time in total; they tell in $O(1)$ time whether a row that crosses the cut $l-1$ or the cut l' has a nonzero entry in L_θ. These lists, and the data of the leaves, take $O(r + \operatorname{nnz}(M) + k(w+1))$ time in total. □

By Lemma 4.1, Theorem 6.9 applies to Betti numbers whenever the boundary matrices of the universe have small width. This is the case for universes of small bandwidth.

Corollary 6.10. *Let $\mathcal{U}$ be a universe with n simplices whose vertices are numbered so that the numbers of the two endpoints of every edge differ by at most b. Let $p \ge 0$, let $\mathbb{F}$ be a field, and let $w = \binom{b+2}{p+1}$. There is a deterministic data structure for Problem 3.1 on $\mathcal{U}$ with preprocessing time $O((p+1)^2 n \log n + wn)$, worst-case update time $O((w+1)^3 \log n)$, and query time $O(1)$.*

Proof. Every simplex σ of $\mathcal{U}$ satisfies $\max\sigma - \min\sigma \le b$, where we identify vertices with their numbers, because any two vertices of σ span an edge of $\mathcal{U}$. For $q \ge 1$, order the columns of the boundary matrix B_q of Section 4.1, that is, the q-simplices of $\mathcal{U}$, by their smallest vertex, breaking ties arbitrarily. Let μ_x be the smallest vertex of the q-simplex in position x. If a $(q-1)$-simplex ρ crosses the cut x, then ρ is a facet of a q-simplex σ in a position at most x and of a q-simplex σ' in a position larger than x. Then $\min\rho \ge \min\sigma' \ge \mu_x$ and $\max\rho \le \max\sigma \le \min\sigma + b \le \mu_x + b$. So ρ is a q-element subset of $\{\mu_x, \dots, \mu_x + b\}$, and the width of B_q is at most $\binom{b+1}{q}$.

By Lemma 4.1, $\beta_p(K) = n_p(K) - \operatorname{rank} B_p^K - \operatorname{rank} B_{p+1}^K$, where B_0^K is empty, and $\operatorname{rank} B_q^K$ is the rank of B_q restricted to the columns of K_q. We maintain $n_p(K)$ and, for $q \in \{p, p+1\}$ with $q \ge 1$, the data structure of Theorem 6.9 for B_q, with the columns of K_q active. A simplex update switches at most one column of one of these matrices. Both widths are at most $\binom{b+1}{p} + \binom{b+1}{p+1} = w$, and each matrix has at most n rows, at most n columns, and at most $(p+2)n$ nonzero entries. We build both matrices and sort their columns in $O((p+1)^2 n \log n)$ time, by sorting the simplices as vertex tuples. Theorem 6.9 then gives the stated bounds. □

Bounded treewidth. A *tree decomposition* of a graph G consists of a tree T and a *bag* $X_\mu \subseteq V(G)$ for every node μ of T, such that every vertex and every edge of G is contained in some bag, and for every vertex v the nodes whose bags contain v induce a connected subtree T_v of T. Its *width* is $\max_\mu |X_\mu| - 1$. For a matrix M, let G_M be the bipartite graph whose vertices are the rows and

the columns of M, with an edge between row i and column j whenever $M_{ij} \neq 0$. Static rank computation is fast when G_M has small treewidth [28]. We show that the same structure allows fast column switches, provided that the columns have few nonzero entries; boundary matrices have this property.

Corollary 6.11. *Let $\mathbb{F}$ be a field, and let $M \in \mathbb{F}^{r\times k}$ have at most $\Delta \geq 1$ nonzero entries in every column. Given a tree decomposition of G_M of width t, we can compute in polynomial time a column hierarchy for M of width $O(\Delta(t+1))$ and height $O(\log(r+k)\log(t+2))$. Consequently, there is a deterministic data structure for dynamic rank of M under column switches with polynomial preprocessing time, worst-case update time $O(\Delta^3(t+1)^3\log(r+k)\log(t+2))$, and query time $O(1)$.*

Proof. After first reducing the given decomposition in the standard way to at most $r+k$ nodes, by contracting tree edges $\mu\nu$ with $X_\mu \subseteq X_\nu$, the result of Bodlaender and Hagerup [4] transforms it in polynomial time into a tree decomposition $(T, (X_\mu)_\mu)$ of G_M of width $t' \leq 3t+2$ in which T is a rooted binary tree of height $h_T = O(\log(r+k))$. For a column j, let $\mathrm{home}(j)$ be the node of T_j closest to the root; all nodes of T_j are descendants of $\mathrm{home}(j)$. For a node μ, let $A_\mu = \{j : \mathrm{home}(j) = \mu\} \subseteq X_\mu$, and let C_μ be the set of columns whose home is μ or a descendant of μ. Then C_μ is the disjoint union of the *pieces* of μ: the sets $C_{\mu'}$ for the children μ' of μ, and the singletons $\{j\}$ for $j \in A_\mu$.

Claim. Let D be a union of pieces of a node μ. Every row with a nonzero entry in a column of D and a nonzero entry in a column not in D lies in X_μ or is adjacent in G_M to a column in X_μ. To see this, let i be such a row, with $M_{ic} \neq 0$ for some $c \in D$ and $M_{ic'} \neq 0$ for some $c' \notin D$, and suppose that neither c nor c' lies in X_μ. Then $c \notin A_\mu$, so the piece of D that contains c is C_{μ_1} for a child μ_1 of μ, and T_c lies in the subtree of μ_1. A bag that contains the edge ic shows that T_i meets the subtree of μ_1. Consider a node ν whose bag contains the edge ic'. If ν were in the subtree of μ_1, then $T_{c'}$ would contain ν and also $\mathrm{home}(c')$, which is not in that subtree because $c' \notin C_{\mu_1}$; being connected, $T_{c'}$ would contain μ, contradicting $c' \notin X_\mu$. So ν lies outside the subtree of μ_1, and the connected subtree T_i contains μ, that is, $i \in X_\mu$. This proves the claim. Since X_μ contains at most $t'+1$ rows and columns and every column has at most Δ nonzero entries, at most $\Delta(t'+1)$ rows satisfy its conclusion.

We build the hierarchy bottom-up along T. For a node μ, we take a balanced binary tree whose leaves are its at most $|A_\mu| + 2 \leq t'+3$ pieces, where a piece $C_{\mu'}$ stands for the hierarchy already built for μ', and a piece $\{j\}$ for the leaf j. We discard empty pieces and contract nodes with a single child. Every node of the resulting hierarchy H is created at some node μ of T, and its set of columns is a union of pieces of μ; by the claim, its set Bd has at most $\Delta(t'+1)$ rows. The height of H is at most $(h_T+1)\lceil\log_2(t'+3)\rceil = O(\log(r+k)\log(t+2))$. Theorem 6.7 now gives the data structure. □

The *Hasse diagram* of a universe $\mathcal{U}$ is the graph whose vertices are the simplices of $\mathcal{U}$, with an edge between every simplex and each of its facets.

Corollary 6.12. *Let $\mathbb{F}$ be a field, let $p \geq 0$, and let $\mathcal{U}$ be a universe with n simplices. Given a tree decomposition of the Hasse diagram of $\mathcal{U}$ of width t, there is a deterministic data structure for Problem 3.1 on $\mathcal{U}$ with polynomial preprocessing time, worst-case update time $O((p+2)^3(t+1)^3\log n\log(t+2))$, and query time $O(1)$, counting field operations.*

Proof. For $q \geq 1$, the graph G_{B_q} of the boundary matrix B_q of Section 4.1 is the subgraph of the Hasse diagram induced by the simplices of dimensions $q-1$ and q. Intersecting every bag with this vertex set gives a tree decomposition of G_{B_q} of width at most t, and every column of B_q has $q+1 \leq p+2$ nonzero entries. We maintain $n_p(K)$ and, for $q \in \{p, p+1\}$ with $q \geq 1$, the data structure of Corollary 6.11 for B_q, with the columns of K_q active. A simplex update switches at most one column of one of these matrices, and by Lemma 4.1, $\beta_p(K) = n_p(K) - \operatorname{rank} B_p^K - \operatorname{rank} B_{p+1}^K$. □

Bounded treewidth of the 1-skeleton suffices in fixed dimension. If the 1-skeleton of a d-dimensional universe $\mathcal{U}$ has a tree decomposition of width t, replace every bag by the set of simplices of $\mathcal{U}$ all of whose vertices lie in that bag. Every simplex σ spans a clique of the 1-skeleton, so the subtrees T_v of its vertices v pairwise intersect, and pairwise intersecting subtrees of a tree have a common node; hence σ lies in some new bag. A simplex and each of its facets lie together in every new bag that contains the simplex, and the nodes whose new bags contain σ form the intersection of the subtrees T_v, $v \in \sigma$, which is connected. So the Hasse diagram has a tree decomposition of width less than $\sum_{q=1}^{d+1} \binom{t+1}{q}$, and Corollary 6.12 maintains every Betti number in $O(\log n)$ field operations per update when t and d are constants. Since a numbering of bandwidth b yields a path decomposition of the 1-skeleton of width b, this also covers Corollary 6.10, with a worse dependence on b.

7 Conclusion and open problems

We showed that, for every $p \geq 1$, maintaining β_p under simplex insertions and deletions is dynamic matrix rank in disguise. Over $\mathbb{F}_2$, and over every field for matrices with entries in $\{0, \pm 1\}$, the two problems reduce to each other with constant overhead, through a gadget in which every matrix operation is a constant number of local changes to a 2-complex. Hence every lower bound for dynamic rank of 0/1 matrices, measured in the instance size, is a lower bound for dynamic Betti numbers, and every dynamic-rank algorithm is a dynamic-Betti algorithm. Under the OMv conjecture, this rules out data structures with polynomial preprocessing time, amortized update time $n^{1/2-\varepsilon}$, and query time $n^{1-\varepsilon}$, robustly across fields, dimensions, degree bounds, embeddings in $\mathbb{R}^4$, Vietoris–Rips complexes, zigzag persistence, and instances with almost no homology.

For the design of algorithms, the message is that, under the OMv conjecture, amortized update time $n^{1/2-\varepsilon}$ together with query time $n^{1-\varepsilon}$ is out of reach for homology in dimension at least one unless the universe has structure beyond its size; the uMv-hinted uMv conjecture extends this, over every field, to worst-case update and query time $n^{\alpha/2-\varepsilon}$, where $\alpha/2 \geq 5/8$, and $\alpha/2 \approx 0.703$ for the current bounds on matrix multiplication. Section 6 identifies two such structures, an embedding in $\mathbb{R}^3$ and a Hasse diagram of bounded treewidth, under which all Betti numbers can be maintained in polylogarithmic time per update. Without such structure, a data structure that maintains Betti numbers under insertions and deletions in general 2-complexes is, up to constant factors, a data structure for dynamic matrix rank. We close with open problems.

(1) *The exponent.* For amortized update time with query time $n^{1-\varepsilon}$, the best lower bound is $n^{1/2-o(1)}$ (Theorem 5.5), while the best algorithm, inherited from dynamic rank, takes $O(n^{1.407})$ worst-case time per update (Corollary 5.16). Our hard instances encode dense $\ell \times \ell$ matrices with $\ell = \Theta(\sqrt{n})$, so a lower bound proved with operation sequences of this kind, in which every complex has the form $K(M; R, J)$, cannot exceed $O(\ell^{1.407}) = O(n^{0.7035})$, because an algorithm that knows the encoding can run the algorithm of van den Brand, Nanongkai, and Saranurak [57] on the encoded matrix. Over $\mathbb{F}_2$, Theorem 4.6 shows that the gap coincides, up to constant factors, with the gap for dynamic rank of matrices with m admissible positions, measured in the instance size $N = r + k + m$. Closing it for either problem closes it for both. In particular, it is open whether any data structure maintains β_1 of 2-complexes, or equivalently the rank of a sparse matrix over $\mathbb{F}_2$, with update and query time $O(n^{1-\delta})$ for some $\delta > 0$.

(2) *Vineyards.* In the transposition model of vines and vineyards [12], the filtration order changes by swapping two consecutive simplices, and the data structure maintains the persistence pairing, which changes in at most two pairs. Our lower bounds do not apply: moving a simplex across the threshold of a sublevel set takes up to $n - 1$ transpositions. For graph filtrations of length L, Dey,

Hou, and Parsa [22] perform a transposition in $O(\log L)$ time. Is linear time per transposition optimal for 2-complexes?

(3) *Amortized partially dynamic bounds.* Theorem 5.12 needs worst-case update time, because rolling back operations defeats amortization. An insertion-only sequence is a filtration. Does incremental β_1 of 2-complexes admit amortized update time $n^{1/2-\varepsilon}$ with query time $n^{1-\varepsilon}$? Dahlgaard [14] proved amortized conditional lower bounds for some partially dynamic graph problems, such as maximum flow and bipartite matching; we do not know whether these techniques apply here.

(4) *Bounded degree and little homology together.* Our hard instances can have bounded degree (Theorem 5.19) or Betti numbers at most one (Theorem 5.30), but we do not know how to achieve both at once.

(5) *Fixed ambient dimension.* Theorem 5.29 gives hardness for Vietoris–Rips complexes of points in dimension $O(\log n)$. Does it hold for points in a fixed dimension $d \geq 3$? The static reduction of Kolbe and Mayr [41], which works for point sets of small doubling dimension, suggests a route. And does the lower bound hold for subcomplexes of a triangulation of the 4-sphere with $O(n)$ simplices? For their static construction, Edelsbrunner and Parsa remark that the whole space can be triangulated around their complex in linear time [26], which suggests that the answer to the last question may be positive. Finally, Corollary 6.6 maintains Betti numbers, not persistence: can the persistent Betti numbers of Problem 5.8 be maintained in polylogarithmic time per value update on a fixed universe in $\mathbb{R}^3$, for instance for alpha-complex filtrations?

(6) *Structure beyond size.* Small homology alone does not help (Theorem 5.30), and neither does bounded degree (Theorem 5.19), whereas an embedding in $\mathbb{R}^3$ (Corollary 6.6) and bounded treewidth of the Hasse diagram (Corollary 6.12) do. Which other structural parameters help? For example, does the algorithm for the 3-sphere extend to subcomplexes of other triangulated 3-manifolds, and can the treewidth structure support entry updates rather than only column switches?

Supplementary material. Code that checks the combinatorial identities used in our proofs by brute-force homology computations over $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{F}_5$, and $\mathbb{F}_7$, on random instances and exhaustively on small ones, is available in the folder `verification` of the branch `paper-v3` of the repository https://github.com/shawnray-research/dynamic-persistence-rank. No proof relies on it.

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